Momentum in large-batch training: Polyak enlarges the critical batch size, Nesterov improves data efficiency

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Abstract

We study how the benefits of momentum change with batch size in the one-pass training regime, using power-law kernel regression as a tractable multiscale model. We analyze two canonical momentum methods, Polyak and Nesterov, with a focus on the large-batch regime relevant to modern large-scale training. We first characterize mean-square stability and define the critical learning rate as the largest learning rate that preserves stable training. We show that $\eta^{\text{crit}}_{\text{polyak}} \eqsim \min\{1, B(1-\rho)\}$ and $\eta^{\text{crit}}_{\text{nesterov}} \eqsim \min\{1, B^{\beta}(1-\rho)\}$, where B is the batch size, $\rho \in [0, 1)$ is the momentum factor, and $\beta > 1$ is the capacity exponent. Within the stable regime, we derive scaling laws for the loss dynamics of both methods and obtain a complete batch-size phase diagram for their optimal data scaling. The results reveal two distinct benefits of momentum in large-batch training: Polyak substantially enlarges the critical batch size over SGD, enabling greater parallelism without sacrificing data efficiency, whereas Nesterov achieves a strictly better data-scaling rate in the large-batch regime. The additional advantage of Nesterov arises from its look-ahead mechanism, which induces curvature-dependent noise suppression. Numerical experiments validate the predicted stability boundaries, batch-size phase diagram, and scaling laws. The code is available at https://github.com/alexhuang13/Momentum_scaling.

1 Introduction

Momentum is a standard component of modern large-scale training, appearing in widely used optimizers such as Adam (Kingma & Ba, 2015) and Muon (Jordan et al., 2024; Liu et al., 2025). Classical momentum theories explain its benefit through accelerated optimization, typically measured by reduced iteration complexity (Nesterov, 1983; Su et al., 2016; Wibisono et al., 2016; Wilson et al., 2021; Gitman et al., 2019; Velikanov et al., 2023; Wang et al., 2024; Liu & Belkin, 2018; Gupta et al., 2024). However, these theories largely study optimization under fixed training configurations and do not address how the benefits of momentum change as data budget and batch size scale in modern large-scale training.

In this paper, we study momentum in the one-pass training regime that commonly arises in large language model (LLM) pre-training (Brown et al., 2020; Muennighoff et al., 2023). For a

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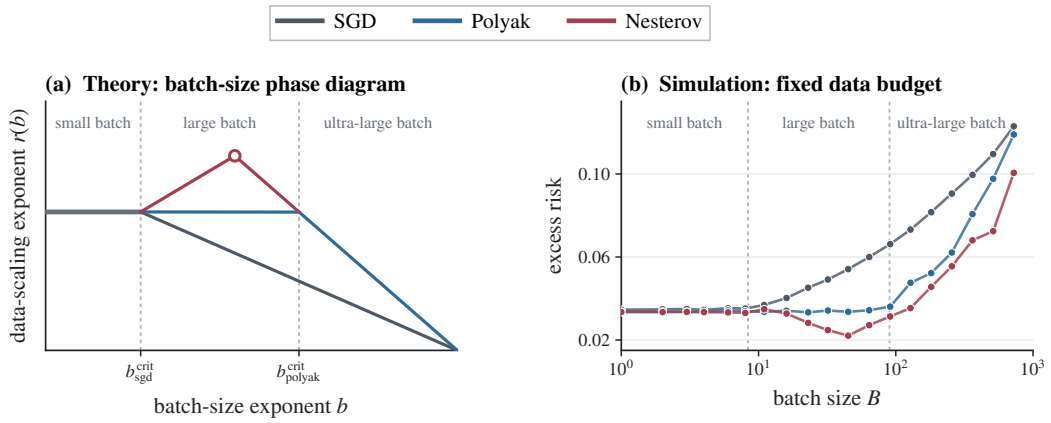


Figure 1: Polyak enlarges the critical batch size, while Nesterov improves data efficiency. (a) Batch-size phase diagram from Theorem 6.2; for $B \eqsim D^b$, $r(b)$ denotes the optimal data-scaling exponent defined by $\mathcal{E}^{\mathrm{opt}}(D; b) = D^{-r(b)+o(1)}$. The resulting phase diagram exhibits three regimes: small batch, Nesterov = Polyak = SGD; large batch, Nesterov > Polyak > SGD; ultra-large batch, Nesterov = Polyak > SGD. **(b)** Optimized excess risk versus batch size at fixed data budget $D = 2^{14}$ and $(s, \beta) = (0.3, 4)$, illustrating the same three-regime behavior. For each algorithm and batch size, the learning rate and momentum are optimally tuned.

given data budget D and batch size B, the number of training steps is

$$K = \frac{D}{B}.$$

Large-scale training often favors larger batches to better exploit hardware parallelism and increase training throughput. However, increasing B simultaneously reduces the number of training steps K, potentially degrading data efficiency. Understanding this batch-size trade-off is therefore central to large-scale training. We ask how momentum changes this trade-off:

Can momentum enable larger batches without sacrificing data efficiency, or even improve data efficiency itself?

We address this question for two canonical momentum methods, Polyak (Polyak, 1964) and Nesterov (Nesterov, 1983), in power-law kernel (PLK) regression. PLK regression is a tractable infinite-dimensional model with a broad spectrum of learning timescales, characterized by a capacity exponent $\beta > 1$ for the feature spectrum and a source exponent $s > 0$ for the target. This setting has recently been widely used to study scaling phenomena relevant to large-scale pre-training (Lin et al., 2024; Bahri et al., 2024; Bordelon et al., 2025; Li et al., 2026a; Wang et al., 2026; Li et al., 2026b). Within this framework, we characterize how momentum changes stability, loss dynamics, and data efficiency as the batch size increases.

Our contributions. Our main results are summarized as follows:

- **Critical learning rate.** We define the critical learning rate η^{crit} as the largest learning rate for which training remains mean-square stable. We derive its sharp scaling for SGD, Polyak, and Nesterov:

$$\eta^{\mathrm{crit}}_{\mathrm{sgd}} \eqsim 1, \qquad \eta^{\mathrm{crit}}_{\mathrm{polyak}} \eqsim \min\{1, B(1-\rho)\}, \qquad \eta^{\mathrm{crit}}_{\mathrm{nesterov}} \eqsim \min\{1, B^{\beta}(1-\rho)\},$$

  where $\rho \in [0, 1)$ is the momentum factor. These results reveal a direct coupling between momentum and batch size: high-momentum training requires larger batches to maintain a non-vanishing learning rate. This requirement is substantially weaker for Nesterov, which only requires $B^{\beta}(1-\rho) \gtrsim 1$ to sustain an order-one learning rate, compared with $B(1-\rho) \gtrsim 1$ for Polyak.

- **Scaling laws for loss dynamics.** Adopting the functional scaling law (FSL) framework of Li et al. (2026a), we derive sharp scaling laws for the loss dynamics of Polyak and Nesterov momentum within their stable regimes. The resulting decomposition separates signal learning from noise accumulation and reveals fundamentally different mechanisms behind the two methods. Polyak accelerates signal learning but amplifies noise accumulation at the same time, whereas Nesterov's look-ahead mechanism induces curvature-dependent noise suppression, allowing acceleration with less noise accumulation.
- **Batch-size phase diagram and data efficiency.** Lastly, we derive a complete batch-size phase diagram for the optimal data scaling by optimizing the momentum factor and learning rate at each prescribed batch size. The diagram reveals three regimes: (i) at *small batch*, SGD, Polyak, and Nesterov achieve the same data scaling; (ii) at *large batch*, Nesterov outperforms Polyak, which in turn outperforms SGD; and (iii) at *ultra-large batch*, Polyak and Nesterov achieve the same scaling, both outperforming SGD. Thus, momentum provides two distinct benefits in large-batch training: Polyak enlarges the critical batch size, allowing larger batches without sacrificing data efficiency, while Nesterov further improves data efficiency. Figure 1 provides a schematic illustration of the three-regime phase diagram together with numerical simulations.

The additional advantage of Nesterov over Polyak arises from the spectral adaptivity induced by its look-ahead mechanism: the effective momentum depends on curvature, leading to direction-dependent damping and noise forgetting.

2 Related work

Theory of momentum. Classical analyses of momentum have established acceleration through improved convergence rates and geometric interpretations (Nesterov, 1983; Su et al., 2016; Wibisono et al., 2016; Wilson et al., 2021). Subsequent works have studied momentum from various perspectives, including Lyapunov analyses, continuous-time limits, variational formulations, and stochastic convergence properties (Vidyasagar, 2025; Liu et al., 2020; Shi et al., 2022; Kovachki & Stuart, 2021; Yuan et al., 2016; Gitman et al., 2019; Velikanov et al., 2023). These theories primarily characterize momentum through iteration complexity or convergence under a fixed training configuration. By contrast, our focus is on how the benefit of momentum changes as batch size and data budget scale jointly in large-scale training.

Scaling-law theory. Scaling laws provide a framework for understanding how learning performance changes with training resources, including data, model, and compute (Hestness et al., 2017; Kaplan et al., 2020; Hoffmann et al., 2022; Grattafiori et al., 2024; DeepSeek-AI et al., 2026). A growing theoretical literature has derived scaling laws in linear, kernel, and related tractable models (Sharma & Kaplan, 2022; Hutter, 2021; Maloney et al., 2022; Wei et al., 2022; Jain et al., 2024; Michaud et al., 2023; Nam et al., 2024; Atanasov et al., 2026; Bahri et al., 2024; Dohmatob et al., 2024; Bordelon et al., 2024; Lin et al., 2024; Paquette et al., 2024; Bordelon et al., 2025; Yan et al., 2026; Lin et al., 2025; Sous & Winer, 2026; Li et al., 2026a; Zhang et al., 2025; Wang et al., 2026; Li et al., 2026b; Ding et al., 2025). Recent functional scaling laws (FSLs) further characterize the full loss dynamics through a decomposition into signal learning and noise accumulation (Li et al., 2026a). We build on this dynamical viewpoint and extend it to momentum methods. Our analysis shows that momentum not only changes the speed of signal learning, but also reshapes noise accumulation and, consequently, the batch-size dependence of data efficiency.

Stability of gradient-based optimization. Stability analysis characterizes the maximal learning rates under which training remains well behaved, and has been studied extensively for

gradient methods and SGD, with connections to implicit bias, generalization, and optimization geometry (Zhu et al., 2019; Wu et al., 2018; Nar & Sastry, 2018; Ma & Ying, 2021; Wu et al., 2022; Wu & Su, 2023; Chemnitz & Engel, 2025; Mulayoff et al., 2021; Nacson et al., 2023; Qiao et al., 2024; Kaur et al., 2023). For momentum methods, existing analyses have considered stability in various deterministic and stochastic settings, but how the stability boundary scales with batch size and momentum strength remains less understood. We provide sharp mean-square stability conditions for Polyak and Nesterov momentum under multiplicative gradient noise and show how these boundaries shape their large-batch scaling behavior.

Concurrent work. Morwani et al. (2026), which appeared while this manuscript was in preparation, independently study batch-size tradeoffs for stochastic momentum in linear regression. Their theoretical results provide lower bounds on the achievable contraction rates of heavy-ball and accelerated-SGD methods. While these bounds suggest that heavy-ball may sustain compute efficiency over a larger batch-size range, they do not establish a sharp scaling law: matching upper bounds are not proved and are explicitly left for future work. In contrast, we derive sharp scaling laws for the full loss dynamics of Polyak and Nesterov momentum. We then optimize the learning rate and momentum factor to characterize the achievable one-pass data scaling with exact piecewise exponents. This yields a complete batch-size phase diagram, including a distinct large-batch regime in which Nesterov strictly improves the data-scaling exponent through curvature-dependent noise suppression.

3 Preliminaries

Notation. In this paper, we use $\eqsim$ to denote equivalence up to a constant factor, and $\lesssim$ (resp. $\gtrsim$) indicates inequality up to a constant factor. For two non-negative functions f, g, we write $f(x) \lesssim g(x)$ (resp. $f(x) \gtrsim g(x)$) if there exists a constant $C > 0$ such that $f(x) \leqslant Cg(x)$ (resp. $g(x) \leqslant Cf(x)$) for all x under consideration. We write $f(x) \eqsim g(x)$ if both $f(x) \lesssim g(x)$ and $f(x) \gtrsim g(x)$ hold. Throughout the paper, $\langle \cdot, \cdot \rangle$ and $\|\cdot\|_2$ denote the standard inner product and the associated 2-norm, respectively.

3.1 Power-law kernel regression

We consider a power-law kernel (PLK) regression on a Hilbert space $\mathcal{H}$ with the input $\mathbf{x} \sim \mathcal{D}$, and the observed label is $y = \langle \boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\theta}^* \rangle + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is independent of $\mathbf{x}$. Here we assume the feature function $\boldsymbol{\phi}(\mathbf{x})$ to be Gaussian: $\boldsymbol{\phi}(\mathbf{x}) \sim \mathcal{N}(0, \mathbf{H})$, where $\mathbf{H} = \mathbb{E}_{\mathbf{x}\sim\mathcal{D}}[\boldsymbol{\phi}(\mathbf{x}) \otimes \boldsymbol{\phi}(\mathbf{x})]$ denotes the feature covariance operator. Let $\{\lambda_j\}_{j\geqslant 1}$ denote the eigenvalues of $\mathbf{H}$, arranged in decreasing order. We learn the target function using the linear student model $f(\mathbf{x}; \boldsymbol{\theta}) = \langle \boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\theta} \rangle$ by minimizing the population risk: $\mathcal{R}(\boldsymbol{\theta}) = \frac{1}{2}\mathbb{E}_{\mathbf{x},y}\left[(f(\mathbf{x}; \boldsymbol{\theta}) - y)^2\right]$. A direct calculation gives $\mathcal{R}(\boldsymbol{\theta}) = \mathcal{E}(\boldsymbol{\theta}) + \frac{\sigma^2}{2}$, where $\mathcal{E}(\boldsymbol{\theta}) = \frac{1}{2}\langle \boldsymbol{\theta} - \boldsymbol{\theta}^*, \mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\theta}^*) \rangle$ is the excess risk.

Assumption 3.1 (Diagonal covariance)**.** We work in an eigenbasis of the feature covariance, so that

$$\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots), \qquad \lambda_1 \geqslant \lambda_2 \geqslant \cdots > 0. \tag{1}$$

This is without loss of generality because SGD, Polyak, and Nesterov are equivariant under orthogonal transformations; see Appendix A.2.

Assumption 3.2 (Label-noise dominance)**.** Assume $\sigma \gtrsim 1$.

Assumption 3.3 (Source and capacity conditions)**.** There exist $\beta > 1$ and $s > 0$ such that

$$\lambda_j \eqsim j^{-\beta}, \quad \lambda_j |\theta_j^*|^2 \eqsim j^{-(1+s\beta)}.$$

Here β is the **capacity exponent**, which controls the decay rate of the feature spectrum, and s is the **source exponent**, which describes how the target energy decays across spectral directions. We also refer to s as the **relative difficulty** of the task: smaller s means that more target energy lies in low-curvature, small-eigenvalue directions, making the problem harder to learn. We remark that similar assumptions have been widely used in the analysis of kernel methods (Caponnetto & Vito, 2005; Caponnetto & De Vito, 2007; Spigler et al., 2020; Bordelon et al., 2020; Maloney et al., 2022). Our work builds upon and extends this line of research.

3.2 Optimization methods

At each step, we draw a fresh mini-batch sample $\mathcal{B}_k = \{(\mathbf{x}_{k,i}, y_{k,i})\}_{i=1}^{B}$, where B denotes the batch size. Let $\mathbf{g}(\boldsymbol{\theta}; \mathcal{B}_k) = \nabla_{\boldsymbol{\theta}}(\frac{1}{2B}\sum_{(\mathbf{x},y)\in\mathcal{B}_k}(f(\mathbf{x};\boldsymbol{\theta})-y)^2)$ denote the mini-batch gradient. SGD updates as follows

$$\text{SGD:} \qquad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \mathbf{g}(\boldsymbol{\theta}_k; \mathcal{B}_k), \tag{2}$$

where η is the learning rate. Polyak momentum, also known as the heavy-ball method, is given by

$$\text{Polyak:} \qquad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta \mathbf{g}(\boldsymbol{\theta}_k; \mathcal{B}_k) + \rho(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}), \tag{3}$$

where $\rho \in [0, 1)$ is the momentum factor. Nesterov momentum differs from Polyak momentum by evaluating the gradient at a look-ahead point:

$$\begin{aligned} \text{Nesterov:} \qquad \boldsymbol{\theta}_k^{\text{la}} &= \boldsymbol{\theta}_k + \rho(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}) \qquad \text{(look ahead)} \\ \boldsymbol{\theta}_{k+1} &= \boldsymbol{\theta}_k - \eta \mathbf{g}(\boldsymbol{\theta}_k^{\text{la}}; \mathcal{B}_k) + \rho(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}), \end{aligned} \tag{4}$$

where $\boldsymbol{\theta}_k^{\text{la}}$ is the look-ahead point obtained by extrapolating the current iterate along the previous update direction. This look-ahead mechanism distinguishes Nesterov from Polyak.

For clarity, we decompose $\mathbf{g}(\boldsymbol{\theta}; \mathcal{B}_k) = \nabla\mathcal{R}(\boldsymbol{\theta}) + \boldsymbol{\xi}_k(\boldsymbol{\theta})$, where $\nabla\mathcal{R}(\boldsymbol{\theta})$ denotes the population gradient and $\boldsymbol{\xi}_k(\boldsymbol{\theta})$ represents the stochastic gradient noise satisfying

$$\mathbb{E}[\boldsymbol{\xi}_k(\boldsymbol{\theta}) \mid \boldsymbol{\theta}] = \mathbf{0}, \qquad \mathbb{E}[\boldsymbol{\xi}_k(\boldsymbol{\theta}) \otimes \boldsymbol{\xi}_k(\boldsymbol{\theta}) \mid \boldsymbol{\theta}] = \frac{1}{B}\boldsymbol{\Sigma}(\boldsymbol{\theta}),$$

where $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ denotes the gradient-noise covariance of batch size 1. We next recall a noise–curvature alignment property that will be used throughout our analysis.

Proposition 3.4 (Noise–curvature alignment)**.** *For any $\boldsymbol{\nu} \in \mathbb{S}_{\mathcal{H}} := \{\boldsymbol{\nu} \in \mathcal{H} : \|\boldsymbol{\nu}\| = 1\}$, we have*

$$\mathbb{E}\Big[|\langle \boldsymbol{\xi}_k(\boldsymbol{\theta}_k), \boldsymbol{\nu}\rangle|^2 \mid \boldsymbol{\theta}_k\Big] \eqsim \frac{1}{B}\mathcal{R}(\boldsymbol{\theta}_k)\,\langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle\,. \tag{5}$$

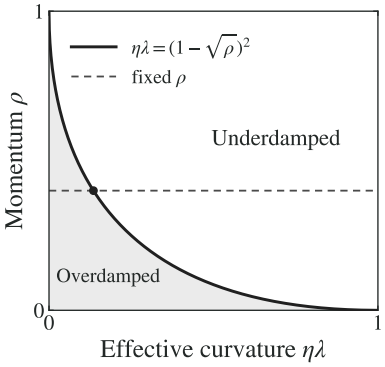
Here, $\langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle$ is the curvature along $\boldsymbol{\nu}$. Thus, Proposition 3.4 states that the current loss determines the overall scale of the gradient noise, while the curvature determines its directional profile. Variants of this alignment have been observed empirically in neural networks (Zhu et al., 2019; Wu et al., 2022) and established rigorously in related random-feature and linear-regression models (Wang & Wu, 2023). We restate it in the present PLK setting because it provides the constitutive relation used in our subsequent analysis. A proof is included in Appendix A.4 for completeness.

3.3 Regime-dependent damping in momentum methods

Consider the full-batch Polyak on the one-dimensional quadratic function $f(x) = \lambda x^2/2$, where $\lambda > 0$ denotes the curvature. Since $\nabla f(x) = \lambda x$, Polyak updates:

$$x_{k+1} = x_k - \eta\lambda x_k + \rho(x_k - x_{k-1}). \tag{6}$$

The corresponding characteristic polynomial is $r^2 - (1 + \rho - \eta\lambda)r + \rho = 0$. The relevant transition from real to complex roots occurs at $\eta\lambda = (1 - \sqrt{\rho})^2$.



For fixed η and ρ, different curvature directions can exhibit two qualitatively different damping behaviors.

- **Overdamped regime**: $\lambda\eta < (1 - \sqrt{\rho})^2$. In this regime, the characteristic roots r_+ and r_- are real with the dominant root satisfying $r_+ \approx 1 - \frac{\eta\lambda}{1-\rho}$ when $1 - \rho = o(1)$. Hence, for sufficiently large k, it holds that

  $$x_k = c_+ r_+^k + c_- r_-^k \approx c_+ r_+^k \approx c_+ \exp\left(-\frac{\eta\lambda}{1-\rho}k\right),$$

  where $c_\pm$ depend on the initialization. Hence, its update behaves like gradient descent with the *momentum-rescaled learning rate*

  $$\eta_\rho := \frac{\eta}{1-\rho}.$$

- **Underdamped regime**: $\lambda\eta > (1 - \sqrt{\rho})^2$. In this regime, the characteristic roots form a complex-conjugate pair with modulus $\sqrt{\rho}$. The iterate therefore has the form

  $$x_k = \rho^{k/2}\big[c_1 \cos(k\omega) + c_2 \sin(k\omega)\big].$$

  where c_1, c_2, and ω depend on the initialization and curvature. Thus, the iterate exhibits oscillatory convergence to zero, with the amplitude decaying as $\rho^{k/2}$.

The key difference between the two regimes lies in how curvature affects the dynamics. In the overdamped regime, the decay rate $\eta\lambda/(1-\rho)$ is proportional to the curvature, so flatter directions converge more slowly. In the underdamped regime, the oscillation frequency depends on the curvature, but the amplitude envelope $\rho^{k/2}$ does not.

Nesterov look-ahead induces curvature-adaptive damping. On the same quadratic objective, Nesterov updates using the gradient at the look-ahead point $x_k^{\text{la}} = x_k + \rho(x_k - x_{k-1})$:

$$\begin{aligned} x_{k+1} &= x_k - \eta\lambda[x_k + \rho(x_k - x_{k-1})] + \rho(x_k - x_{k-1}) \\ &= x_k - \eta\lambda x_k + \rho(1 - \eta\lambda)(x_k - x_{k-1}), \end{aligned} \tag{7}$$

which is equivalent to Polyak with the momentum factor $\rho(1 - \eta\lambda)$. For an underdamped direction, it accelerates convergence: $\rho^{k/2} \to [\rho(1 - \eta\lambda)]^{k/2}$, which damps sharper directions more rapidly. By contrast, in sufficiently flat directions where $\eta\lambda \ll 1$, we have $\rho(1 - \eta\lambda) \approx \rho$, and Nesterov retains approximately the same dynamics as Polyak.

Remark 3.5 (Spectral decomposition into damping regimes). For a fixed pair (η, ρ), the damping threshold $\lambda_{\text{cut}} := \frac{(1-\sqrt{\rho})^2}{\eta}$ partitions the spectrum into two parts. Directions with $\lambda_j > \lambda_{\text{cut}}$ are underdamped, whereas directions with $\lambda_j < \lambda_{\text{cut}}$ are overdamped. Hence, in PLK regression, the loss dynamics can be analyzed by treating the underdamped spectral head and overdamped spectral tail separately and then combining their contributions.

4 Mean-square stability and critical learning rates

In this section, we characterize the critical learning rates that guarantee training stability and study how they depend on the momentum factor and batch size. To make the notion of training stability precise, we view each optimization algorithm as a stochastic dynamical system and adopt the following notion of mean-square stability.

Definition 4.1 (Mean-square stability)**.** A stochastic iteration $\mathbf{z}_{k+1} = \mathcal{T}_k(\mathbf{z}_k)$ is said to be mean-square stable if for every initial state satisfying $\mathbb{E}\|\mathbf{z}_0\|^2 < \infty$, we have

$$\sup_k \mathbb{E}\|\mathbf{z}_k\|^2 < \infty.$$

Here $\mathbf{z}_k$ denotes the state variable of an algorithm. For SGD, $\mathbf{z}_k = \boldsymbol{\theta}_k$, while for Polyak and Nesterov, $\mathbf{z}_k = (\boldsymbol{\theta}_k, \boldsymbol{\theta}_{k-1})$.

Definition 4.2 (Critical learning rate)**.** Consider a stochastic optimization algorithm $\mathcal{A}$, parameterized by a learning rate η and a collection of additional parameters $\mathbf{h}$. For fixed $\mathbf{h}$, define its critical learning rate as

$$\eta_{\mathcal{A}}^{\mathrm{crit}}(\mathbf{h}) := \sup\left\{\eta > 0 : \mathbf{z}_{k+1} = \mathcal{A}(\mathbf{z}_k; \eta, \mathbf{h}) \text{ is mean-square stable}\right\}.$$

When the dependence on $\mathbf{h}$ is clear from context, we simply write $\eta_{\mathcal{A}}^{\mathrm{crit}}$. In our setting, $\mathbf{h} = B$ for SGD and $\mathbf{h} = (B, \rho)$ for Polyak and Nesterov momentum. The following theorem characterizes how the critical learning rate scales with batch size and momentum.

Theorem 4.3 (Critical learning-rate scaling)**.** Under Assumption 3.3, for sufficiently large B and sufficiently small $1-\rho$, the critical learning rates satisfy

$$\eta_{\mathrm{sgd}}^{\mathrm{crit}} \eqsim 1, \tag{8}$$

$$\eta_{\mathrm{polyak}}^{\mathrm{crit}} \eqsim \min\{1,\, B(1-\rho)\}, \tag{9}$$

$$\eta_{\mathrm{nesterov}}^{\mathrm{crit}} \eqsim \min\{1,\, B^{\beta}(1-\rho)\}, \tag{10}$$

where the implicit constants are independent of B and ρ.

Joint dependence on momentum and batch size. Theorem 4.3 reveals a direct coupling among the learning rate, momentum factor, and batch size in determining training stability. Increasing ρ toward one reduces the critical learning rate, whereas increasing the batch size enlarges it. In the high-momentum regime, $\eta_{\mathrm{polyak}}^{\mathrm{crit}} \eqsim B(1-\rho), \eta_{\mathrm{nesterov}}^{\mathrm{crit}} \eqsim B^{\beta}(1-\rho)$, before reaching their order-one ceilings. Thus, the batch-size effect on stability is substantially stronger for Nesterov, with a B^{β} dependence rather than the B dependence of Polyak. As shown later, this difference originates from the curvature adaptivity induced by Nesterov's look-ahead mechanism.

Proof idea and the distinct roles of multiplicative and additive noise. All three algorithms admit a unified random-state representation $\mathbf{z}_{k+1} = (\mathbf{T} + \mathbf{M}_k)\mathbf{z}_k + \sigma\boldsymbol{\xi}_k$, where $\mathbf{M}_k\mathbf{z}_k$ and $\sigma\boldsymbol{\xi}_k$ represent multiplicative and additive noise, respectively. Let $\mathbf{C}_k := \mathbb{E}[\mathbf{z}_k \otimes \mathbf{z}_k]$. Using the conditional mean-zero properties of the noise terms, the covariance dynamics take the form

$$\mathbf{C}_{k+1} = \mathcal{Q}(\mathbf{C}_k) + \sigma^2\mathbf{S},$$

where $\mathcal{Q}$ is determined by the deterministic transition $\mathbf{T}$ and the multiplicative noise $\mathbf{M}_k$, whereas the additive noise appears only through the forcing term $\sigma^2\mathbf{S}$. Consequently, the additive noise affects the magnitude of the covariance within the stable region but not the

stability threshold itself. Mean-square stability is determined by the homogeneous covariance dynamics $\mathbf{C}_{k+1} = \mathcal{Q}(\mathbf{C}_k)$. In finite dimensions, this is equivalently characterized by the spectral-radius condition $\mathrm{rad}(\mathcal{Q}) < 1$. Here for the infinite-dimensional case, the absence of a uniform spectral gap makes this criterion inapplicable. We instead reduce the covariance dynamics to a nonnegative renewal equation. Stability is then characterized by local mode stability and a global contraction condition for the resulting renewal feedback. The explicit forms of $\mathcal{Q}$ and the resulting stability characterizations are provided in Appendix B. Although additive noise does not affect stability, it plays a central role in the loss dynamics and data-scaling behavior studied in later sections.

5 Scaling law for excess-risk dynamics across damping regimes

In this section, we establish scaling laws for the excess-risk dynamics of Polyak and Nesterov, with explicit dependence on the learning rate η, batch size B, and momentum factor ρ. The key organizing principle is the mode-wise damping transition described in Section 3.3. Specifically, for fixed η and ρ, define the curvature cutoff and the spectral partition:

$$\lambda_{\mathrm{cut}} = \frac{(1-\sqrt{\rho})^2}{\eta}, \qquad \mathbb{I}_{\mathrm{over}} = \{j : \lambda_j \leqslant \lambda_{\mathrm{cut}}\}, \qquad \mathbb{I}_{\mathrm{under}} = \{j : \lambda_j > \lambda_{\mathrm{cut}}\},$$

where $\mathbb{I}_{\mathrm{over}}$ and $\mathbb{I}_{\mathrm{under}}$ consist of flat overdamped directions and sharp underdamped directions, respectively. Then, the parameter range $\rho \in [0, 1)$ can be divided into three regimes, illustrated in Figure 2:

- **Fully overdamped regime.** When $\rho = 0$, we have $\lambda_{\mathrm{cut}} = \frac{1}{\eta} \gtrsim \lambda_1$ under the assumption $\eta < 1$, and hence $\mathbb{I}_{\mathrm{under}} = \varnothing$. As ρ increases, λ_{cut} decreases, and the first underdamped direction appears when the cutoff reaches the largest curvature: $\lambda_{\mathrm{cut}} \eqsim \lambda_1$. Since $\lambda_1 \eqsim 1$, this condition is equivalent to $1 - \sqrt{\rho} \eqsim \sqrt{\eta}$. Thus, at the level of scaling, the transition from the fully overdamped to the mixed regime occurs at $\rho_1 := (1-\sqrt{\eta})^2$.
- **Mixed regime.** As ρ increases beyond ρ_1, λ_{cut} decreases and $\mathbb{I}_{\mathrm{under}}$ continues to expand. The dynamics enter a mixed regime consisting of both underdamped and overdamped modes. Since the PLK spectrum contains arbitrarily small eigenvalues, $\mathbb{I}_{\mathrm{over}}$ always remains nonempty, and the system never becomes fully underdamped.
- **Unstable regime.** Increasing ρ further eventually violates the mean-square stability condition derived in Section 4.



Figure 2: Damping and stability regimes of momentum The mixed regime consists of sharp underdamped modes and flat overdamped modes. Here $q = 1$ for Polyak and $q = \beta$ for Nesterov.

For comparison, we recall that the SGD scaling law for PLK regression is

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim (\eta k)^{-s} + \frac{\sigma^2}{B}\eta, \qquad k\eta \gtrsim 1. \tag{11}$$

The first term describes signal learning, whereas the second term captures the accumulation of additive noise under a constant learning rate. As discussed in Section 3.3, momentum rescales

the learning rate in overdamped directions by a factor $1/(1-\rho)$. This motivates defining the **momentum-rescaled learning rate**

$$\eta_\rho := \frac{\eta}{1-\rho}.$$

Hence, the momentum scaling laws below can be compared directly with the SGD scaling law.

Theorem 5.1 (Scaling law of Polyak)**.** Assume $\eta < 1$ and $\sigma \gtrsim 1$. For $k\eta_\rho \gtrsim 1$, we have:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \begin{cases} (\eta_\rho k)^{-s} + \frac{\sigma^2}{B}\eta_\rho, & 0 \leqslant \rho \lesssim (1-\sqrt{\eta})^2, \\ \rho^k + (\eta_\rho k)^{-s} + \frac{\sigma^2}{B}\eta_\rho, & (1-\sqrt{\eta})^2 \lesssim \rho \lesssim 1-\eta/B. \end{cases} \tag{12}$$

If $\rho \gtrsim 1-\eta/B$, the iteration is not mean-square stable.

The proof of Theorem 5.1 is deferred to Appendix D. In the fully overdamped regime, all spectral directions behave like SGD with learning rate η_ρ. Accordingly, the first branch of (12) coincides with the SGD scaling law (11) after replacing η by η_ρ.

In the mixed regime, the flat modes in $\mathbb{I}_{\text{over}}$ remain overdamped and produce the same power-law contribution $(\eta_\rho k)^{-s}$. The sharp modes in $\mathbb{I}_{\text{under}}$, by contrast, are underdamped. As shown in Section 3.3, their iterate amplitudes decay with the common envelope $\rho^{k/2}$, so their contribution to the loss produces the additional exponential transient ρ^k in the second branch of (12). SGD has no analogous underdamped component: its sharp modes decay rapidly and are negligible in the long-time scaling.

Theorem 5.2 (Scaling law of Nesterov)**.** Assume $\eta < 1$ and $\sigma \gtrsim 1$. For $\eta_\rho k \gtrsim 1$, we have:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \begin{cases} (\eta_\rho k)^{-s} + \frac{\sigma^2}{B}\eta_\rho, & 0 \leqslant \rho \lesssim (1-\sqrt{\eta})^2, \\ \min\{1, (\eta k)^{-s}\}\,\rho^k + (\eta_\rho k)^{-s} + \frac{\sigma^2}{B}\min\{\eta_\rho, \eta_\rho^{1/\beta}\}, & (1-\sqrt{\eta})^2 \lesssim \rho \lesssim 1-\eta/B^\beta. \end{cases}$$

If $\rho \gtrsim 1-\eta/B^\beta$, the iteration is not mean-square stable.

The proof of Theorem 5.2 is deferred to Appendix E.

How Nesterov differs from Polyak. The fully overdamped regime is the same for the two methods. Once underdamped modes appear, Nesterov's look-ahead changes the transient from

$$\rho^k \quad \longrightarrow \quad \left(\frac{(1-\rho)^2}{\eta}\right)^{s+1/\beta} p^k.$$

Since $p < \rho$, the underdamped contribution decays faster, reflecting the benefit of curvature adaptivity induced by look-ahead mechanism. In the high-momentum range $\eta_\rho = \eta/(1-\rho) > 1$, the same mechanism also reduces the noise-accumulation term:

$$\frac{\sigma^2}{B}\eta_\rho \quad \longrightarrow \quad \frac{\sigma^2}{B}\eta_\rho^{1/\beta}.$$

Moreover, Nesterov shifts the mean-square stability boundary from $\rho \eqsim 1-\eta/B$ to $\rho \eqsim 1-\eta/B^\beta$, allowing a wider stable high-momentum regime.

6 Optimal data scaling across batch-size regimes

In large-scale training, the batch size is often chosen according to hardware parallelism rather than optimized solely for data efficiency. For a fixed data budget D, increasing the batch

size B reduces the number of sequential training steps $K = D/B$, enabling greater hardware parallelism and higher training throughput. However, this may also degrade the achievable final-step loss, leading to a deterioration of data efficiency. We therefore ask:

For a prescribed batch size, what is the best final-step loss attainable by optimally tuning the remaining hyperparameters?

Consider the scaling regime $D \gg 1$ and prescribe the batch-size scaling:

$$B \eqsim D^b, \qquad K = \frac{D}{B} \eqsim D^{1-b}, \qquad 0 \leqslant b < 1,$$

where the exponent $b := \log_D B$ defines a relative batch size.

Definition 6.1 (Batch-size-dependent data-scaling exponent)**.** For a prescribed batch-size scaling $B \eqsim D^b$, define the optimal data-scaling exponent $r(b)$ through

$$\inf_{\eta,\rho} \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] = D^{-r(b)+o(1)},$$

where the infimum is taken over the stable parameter region and ρ is ignored for SGD.

A larger $r(b)$ corresponds to better data efficiency. When comparing different algorithms, we write $r_{\mathrm{sgd}}(b)$, $r_{\mathrm{polyak}}(b)$, and $r_{\mathrm{nesterov}}(b)$ for the corresponding exponents.

Define three transition exponents

$$b_1 = \frac{s}{s+1}, \quad b_2 = \frac{2s + 1/\beta}{2s + 1 + 1/\beta}, \quad b_3 = \frac{2s+1}{2s+2}. \tag{13}$$

Since $\beta > 1$ and $s > 0$, these exponents satisfy

$$0 < b_1 < b_2 < b_3 < 1.$$

The following theorem characterizes how the prescribed batch-size scaling affects the achievable data efficiency of the three methods. The proof is deferred to Appendix F.

Theorem 6.2 (Batch-size-dependent data scaling)**.** *For a prescribed batch-size scaling $B \eqsim D^b$, the corresponding data-scaling exponents of the three methods are given by*

$$r_{\mathrm{sgd}}(b) = \begin{cases} \frac{s}{s+1}, & 0 \leqslant b < b_1 \\ s(1-b), & b_1 \leqslant b < 1, \end{cases} \qquad r_{\mathrm{polyak}}(b) = \begin{cases} \frac{s}{s+1}, & 0 \leqslant b < b_3 \\ 2s(1-b), & b_3 \leqslant b < 1, \end{cases}$$

and

$$r_{\mathrm{nesterov}}(b) = \begin{cases} \frac{s}{s+1}, & 0 \leqslant b < b_1, \\ \frac{s(1+(\beta-1)b)}{s\beta+1}, & b_1 \leqslant b < b_2, \\ 2s(1-b), & b_2 \leqslant b < 1. \end{cases}$$

This theorem shows that the achievable data efficiency depends sharply on the batch-size scaling. Figure 1(left) provides a schematic illustration. For SGD, the optimal data scaling is preserved up to the critical batch-size exponent $b^{\mathrm{crit}}_{\mathrm{sgd}} = b_1$, beyond which data efficiency deteriorates as the number of training steps becomes too small. Polyak extends this threshold to $b^{\mathrm{crit}}_{\mathrm{polyak}} = b_3$, sustaining the optimal data scaling over a wider range of batch sizes. Nesterov behaves differently: its data-scaling exponent begins to improve beyond b_1 and reaches its maximum at $b^{\mathrm{opt}}_{\mathrm{nesterov}} = b_2$, yielding strictly better data efficiency than both SGD and Polyak in the large-batch regime.

To make this comparison explicit, we divide the batch-size scaling into three regimes: small batch, large batch, and ultra-large batch.

- **Small batch** ($b \leqslant b_1$). Momentum does not improve the optimal data-scaling exponent:

$$r_{\text{sgd}}(b) = r_{\text{polyak}}(b) = r_{\text{nesterov}}(b).$$

- **Large batch** ($b_1 < b < b_3$). The three methods separate:

$$r_{\text{sgd}}(b) < r_{\text{polyak}}(b) < r_{\text{nesterov}}(b).$$

  SGD deteriorates as the number of training steps decreases. Polyak preserves the optimal scaling over a larger batch-size range, while Nesterov further improves the data-scaling exponent. Within this regime, the Nesterov exponent reaches its maximum at $b = b_2$.

- **Ultra-large batch** ($b \geqslant b_3$). The additional advantage of Nesterov over Polyak disappears:

$$r_{\text{sgd}}(b) < r_{\text{polyak}}(b) = r_{\text{nesterov}}(b).$$

  Moreover, both momentum methods now deteriorate with increasing batch size, but remain strictly better than SGD.

These results reveal two distinct roles of momentum in large-batch training. Polyak enables larger batches while preserving the optimal data scaling of SGD, thereby supporting greater parallelism without sacrificing data efficiency. Nesterov goes further and improves the data-scaling exponent itself in the large-batch regime.

7 Numerical experiments

In this section, we empirically validate our theoretical predictions. The full experimental details and additional experiments can be found in Appendix G.

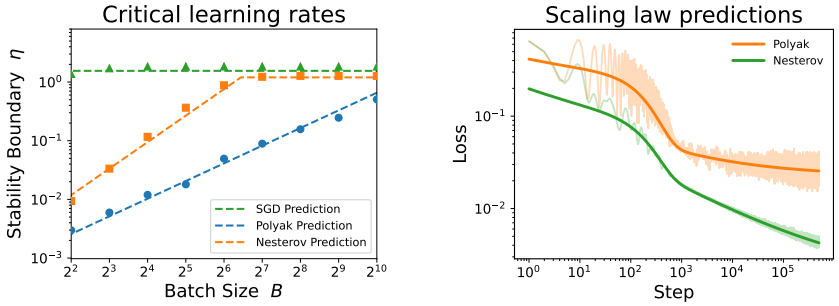


Figure 3: Numerical validation of the stability and scaling-law predictions. **Left:** Empirical mean-square stability boundaries for $(s, \beta) = (1.5, 2)$. Markers are numerical estimates and dashed curves are the predicted stability boundaries. **Right:** Loss trajectories of Polyak and Nesterov momentum for $(s, \beta) = (0.3, 4)$, with empirical results shown in light colors and theoretical predictions shown in dark colors. The scaling laws capture the transition from signal learning to noise-dominated long-time behavior and predict a smaller noise contribution of Nesterov.

Mean-square stability boundaries. We first test the stability conditions derived in Section 4. Fixing the momentum factor ρ, we vary the batch size B and scan the learning rate η for SGD, Polyak, and Nesterov. For each method and batch size, we estimate the empirical stability boundary as the largest learning rate for which the loss remains bounded over the prescribed training horizon. The precise stability criterion and experimental protocol are provided in Appendix G.1.

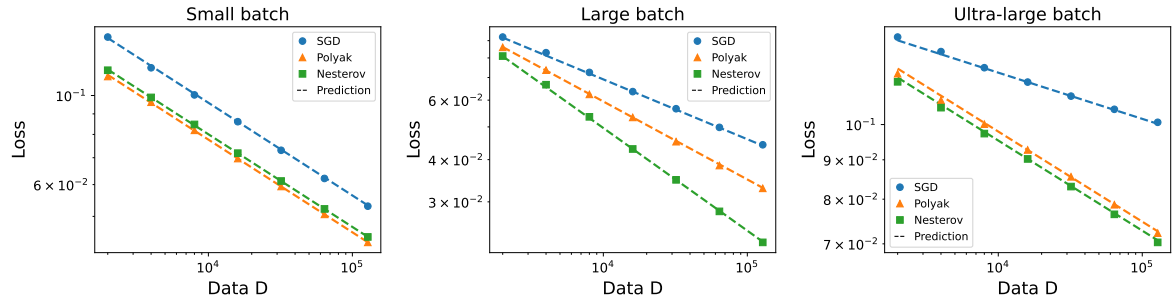


Figure 4: Data scaling at representative batch-size choices from the three regimes. For each panel, we choose one batch-size scaling from the corresponding regime and vary the data budget D. For each D, the learning rate and momentum factor are chosen according to their theoretically optimal scalings. Markers show the empirical final excess risks, while dashed lines show the rates predicted by Theorem 6.2. Across all three choices, the empirical data-scaling rates closely match the theoretical predictions. Consequently, the predicted ordering is also recovered: Nesterov = Polyak = SGD for the small batch choice (left); Nesterov > Polyak > SGD for the large batch choice (middle); and Nesterov = Polyak > SGD for the ultra-large batch choice (right).

Each marker in Figure 3 (left) reports the mean estimate over five independent random seeds. As predicted, the stability boundary of SGD remains of order one, while those of Polyak and Nesterov grow as $\eta^{\mathrm{crit}}_{\mathrm{polyak}} \asymp B$ and $\eta^{\mathrm{crit}}_{\mathrm{nesterov}} \asymp B^{\beta}$, respectively, before saturating at their order-one ceilings. These results confirm the distinct batch-size scalings predicted by Theorem 4.3. Additional results for different choices of s and β are reported in Appendix G.1.

The loss-dynamics scaling laws. We next test whether the predicted scaling laws capture the full loss trajectories of Polyak and Nesterov. At fixed B, η, and ρ, we simulate both methods over a long training horizon. Since Theorems 5.1 and 5.2 specify the transient, signal, and noise terms only up to constant factors, we keep their predicted forms fixed and fit a separate nonnegative prefactor for each term. The fitting procedure is detailed in Appendix G.2. In Figure 3 (right), the light curves show the empirical loss trajectories, while the dark curves show the corresponding theoretical predictions. The predictions capture both the initial signal-learning phase and the subsequent regime in which noise accumulation becomes significant. In particular, Polyak develops a larger noise contribution at long times, whereas the Nesterov loss continues to decrease to a lower level, consistent with the noise suppression induced by its look-ahead mechanism. Appendix G.2 shows that the same behavior persists across different (s, β) settings.

The three-regime batch-size phase diagram. We now validate the data-scaling predictions of Theorem 6.2. We set $(s, \beta) = (0.3, 4)$ and evaluate the three algorithms under three representative batch-size scalings: $B \asymp 1$, $B \asymp D^{0.4}$, and $B \asymp D^{0.8}$, corresponding respectively to the small, large, and ultra-large batch regimes. In each regime, the learning rate, and momentum factor vary with the data budget according to the theoretically prescribed scalings. See Appendix G.3 for the complete experimental setup and additional results. Each marker in Figure 4 reports final loss, and the dashed lines use the data-scaling exponents predicted by Theorem 6.2. Across the three panels of Figure 4, the empirical scaling rates closely follow the theoretical predictions. In the small-batch regime, SGD, Polyak, and Nesterov achieve the same data-scaling rate. In the large-batch regime, the three methods separate, with Nesterov achieving the fastest rate, followed by Polyak and then SGD. In the ultra-large-batch regime, Polyak and Nesterov again share the same scaling, both outperforming SGD. These experiments validate the three-regime ordering predicted by our batch-size phase diagram.

Critical batch size at a fixed data budget. Figure 1 (right) provides a complementary finite-budget view of the phase diagram. We fix the data budget $D = 2^{14}$ and, for each batch size, we optimally tune the learning rate and momentum factor and report the resulting excess risk. See Appendix G.4 for experimental details and additional results. At small batch sizes, all three methods achieve nearly the same loss. As the batch size increases, SGD first departs from this optimum, marking its critical batch size. Polyak maintains nearly constant loss over a substantially wider range, corresponding to its enlarged critical batch size. Nesterov exhibits a qualitatively different behavior: its loss continues to decrease in the large-batch regime before eventually increasing at ultra-large batch sizes. This finite-budget picture directly illustrates the two effects predicted by our theory: Polyak extends the data-efficient batch range, whereas Nesterov can further improve data efficiency.

8 Discussion and Conclusion

We developed a unified scaling-law analysis for Polyak and Nesterov momentum in power-law kernel regression, characterizing their mean-square stability, loss dynamics, and optimal data scaling as the training budget grows. A central conclusion is that the benefit of momentum depends critically on the batch size. At small batch sizes, neither method improves the data scaling of vanilla SGD. As the batch size increases, however, the two methods provide distinct benefits. Polyak enlarges the critical batch size, allowing training with fewer sequential steps while preserving the optimal data efficiency. Nesterov goes further: in the large-batch regime, it can improve the data-scaling exponent itself, although this advantage only emerges at sufficiently large batch sizes.

Note that our scaling analysis allows the batch size B, learning rate η, and momentum factor ρ to scale with the data budget D, while keeping them fixed within each training run. This captures how their joint scaling determines stability and data efficiency, but leaves open the effect of time-varying schedules. Such schedules may interact nontrivially with the stability and noise-accumulation mechanisms identified in this work.

A particularly instructive example is the momentum schedule. In deterministic optimization, increasing-momentum schedules such as $1 - \rho_k \eqsim k^{-1}$ are crucial to Nesterov acceleration (Nesterov, 1983). Substituting this schedule into Theorem 4.3 suggests that the training is stable only if

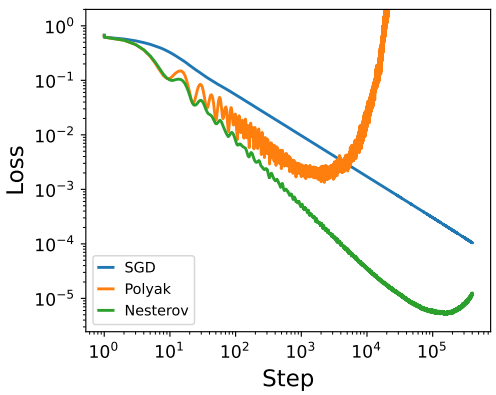
$$\eta \lesssim \min\left\{1, \frac{B}{k}\right\} \quad \text{for Polyak,} \qquad \eta \lesssim \min\left\{1, \frac{B^\beta}{k}\right\} \quad \text{for Nesterov.}$$

Thus, for fixed η and B, increasing momentum can accelerate the initial loss decay while eventually violating the stochastic stability condition. The corresponding instability times are predicted to scale as

$$k^*_{\text{polyak}} \eqsim \frac{B}{\eta}, \qquad k^*_{\text{nesterov}} \eqsim \frac{B^\beta}{\eta}.$$

The right panel illustrates this delayed-instability phenomenon for $(s, \beta) = (0.75, 2)$ under the momentum schedule $\rho_k = 1 - 1/(k+1)$; the experiment details are deferred to Appendix G.5. Both Polyak and Nesterov initially benefit from the growing momentum and exhibit accelerated convergence. As training proceeds, however, the stochastic stability constraint is eventually violated. Consistent with the predicted instability scales, Nesterov remains stable for substantially more

Fixed η; increasing momentum
$\rho_k = 1 - 1/(k+1)$



training steps than Polyak because of its stronger B^{β} dependence. This longer stable phase allows Nesterov to exploit the increasing-momentum schedule for a longer period and reach a lower loss before instability occurs. This example suggests that the practical benefit of a momentum schedule is determined not only by its deterministic acceleration effect, but also by how long the schedule can remain within the stable region.

A complete theory for learning-rate, momentum, and batch-size schedules remains open. A natural next step is to understand how schedules reshape stability, signal learning, and noise accumulation, and how these changes propagate to the resulting scaling behavior. Such a theory may in turn help explain why practical schedules work and provide principled guidance for their design. Extending the present framework to learning-rate schedules such as cosine decay (Hoffmann et al., 2022) and warmup–stable–decay (Li et al., 2026b), and ultimately to modern optimizers such as Adam and Muon, is therefore an important direction for future work.

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A Technical preliminaries

A.1 Kernel regression

In this section, we explain how our setup in Section 3.1 is equivalent to learning with kernels.

Definition A.1 (Positive semidefinite (PSD) kernel)**.** A function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is called a *positive semidefinite (PSD) kernel* if it satisfies:

- Symmetry: $K(\mathbf{x}, \mathbf{x}') = K(\mathbf{x}', \mathbf{x})$ for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$;
- Positive semidefiniteness: for any $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathcal{X}$ and $a_1, \ldots, a_n \in \mathbb{R}$,

$$\sum_{i,j=1}^{n} a_i a_j \, K(\mathbf{x}_i, \mathbf{x}_j) \geqslant 0.$$

Definition A.2 (Reproducing kernel Hilbert space (RKHS))**.** Given a kernel $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, the *reproducing kernel Hilbert space* $\mathcal{H}_K$ associated with K is a Hilbert space of functions over $\mathcal{X}$ such that

$$\langle f, K(\mathbf{x}, \cdot) \rangle_{\mathcal{H}_K} = f(\mathbf{x}), \quad \forall f \in \mathcal{H}_K, \ \mathbf{x} \in \mathcal{X}.$$

One can show that for a given kernel K, the associated RKHS is unique. Kernel methods consider the hypothesis space given by the associated RKHS. For instance, kernel ridge regression gives an estimator:

$$\widehat{f}_\lambda = \operatorname*{argmin}_{f \in \mathcal{H}_K} \frac{1}{n} \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}_K}^2.$$

Hence, model capacity is determined by the size of the RKHS $\mathcal{H}_K$.

Let $\mathcal{D}$ be the input distribution. We define the integral operator $\mathcal{T}_K : L^2(\mathcal{D}) \to L^2(\mathcal{D})$ by

$$\mathcal{T}_K f(\cdot) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[K(\cdot, \mathbf{x}) f(\mathbf{x})].$$

By assuming $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[K(\mathbf{x}, \mathbf{x})] < \infty$, the operator $\mathcal{T}_K$ is compact (Mercer's theorem) and consequently, the kernel admits the following eigenvalue decomposition

$$K(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^{\infty} \lambda_j e_j(\mathbf{x}) e_j(\mathbf{x}'),$$

where $\{\lambda_j\}_{j=1}^{\infty}$ and $\{e_j\}_{j=1}^{\infty}$ denote the eigenvalues and eigenfunctions, respectively. Moreover, $\langle e_i, e_j \rangle_{L^2(\mathcal{D})} = \delta_{i,j}$, i.e., the eigenfunctions form an orthonormal basis of $L^2(\mathcal{D})$.

Using the spectral decomposition, the RKHS admits the following representation:

$$\mathcal{H}_K = \left\{ \sum_{j=1}^{\infty} a_j e_j : \sum_{j=1}^{\infty} \frac{a_j^2}{\lambda_j} < \infty \right\}.$$

To better quantify the smoothness of functions, one often considers the interpolation space $\mathcal{H}_K^s$ with $s \geqslant 0$, defined as

$$\mathcal{H}_K^s = \left\{ \sum_{j=1}^{\infty} a_j e_j : \sum_{j=1}^{\infty} \frac{a_j^2}{\lambda_j^s} < \infty \right\}.$$

Clearly, $\mathcal{H}_K^1 = \mathcal{H}_K$, and

$$\mathcal{H}_K^{s_1} \subset \mathcal{H}_K^{s_2}, \qquad \forall \, s_1 > s_2 \geqslant 0.$$

Hence, the index s characterizes the smoothness of a function relative to the chosen kernel.

In the analysis of kernel methods, the following conditions are commonly used to describe the smoothness of the target function and the capacity of the kernel, respectively.

Assumption A.3 (Source condition)**.** There exists some $s > 0$ such that $f^* \in \mathcal{H}_K^s$.

Assumption A.4 (Capacity condition)**.** There exists some $\beta > 1$ such that $\lambda_j \eqsim j^{-\beta}$.

These conditions yield the following interpretation:

- A smaller s indicates that the target function f^* belongs to a larger space, corresponding to a more difficult learning problem.
- A smaller β implies a slower eigenvalue decay, meaning a richer hypothesis space $\mathcal{H}_K$ and thus higher model capacity.

Our formulation in Section 3.1 is equivalent to the above setting, but expressed in terms of the feature map $\boldsymbol{\phi}$. Under Assumption 3.1, we have

$$K_\phi(\mathbf{x}, \mathbf{x}') = \sum_{j=1}^{\infty} \phi_j(\mathbf{x})\phi_j(\mathbf{x}') = \sum_{j=1}^{\infty} \lambda_j \widehat{\phi}_j(\mathbf{x})\widehat{\phi}_j(\mathbf{x}'),$$

where $\widehat{\phi}_j = \phi_j / \lambda_j^{1/2}$. In this case, Assumption 3.3 corresponds exactly to the above capacity condition, while the task-difficulty assumption in Section 3.1 can be viewed as a power-law version of the source condition. Specifically, under Assumption 3.3,

$$f^* = \sum_{j=1}^{\infty} \theta_j^* \phi_j = \sum_{j=1}^{\infty} j^{-1/2} \lambda_j^s \, \widehat{\phi}_j =: \sum_{j=1}^{\infty} a_j^* \, \widehat{\phi}_j.$$

Hence, for any arbitrarily small $\delta \in (0, s)$, we have $f^* \in \mathcal{H}_{K_\phi}^{s-\delta}$, since

$$\sum_{j=1}^{\infty} \frac{|a_j^*|^2}{\lambda_j^{s-\delta}} = \sum_{j=1}^{\infty} j^{-1-\beta(s-\delta)} < \infty.$$

A.2 Rotational equivariance of SGD and momentum methods

In this section, we demonstrate that the optimization trajectories of SGD, Polyak, and Nesterov are equivariant under unitary transformations of the feature space. This rotational equivariance rigorously justifies our assumption that the population feature covariance matrix can be treated as diagonal without loss of generality.

Since the population covariance operator $\mathbf{H} = \mathbb{E}[\boldsymbol{\phi}(\mathbf{x}) \otimes \boldsymbol{\phi}(\mathbf{x})]$ is a self-adjoint positive trace-class operator on the Hilbert space $\mathcal{H}$, it admits the spectral decomposition $\mathbf{H} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^*$, where $\boldsymbol{U}$ is a unitary operator and $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots)$ is diagonal.

Consider the unitary change of basis for the parameters and features:

$$\boldsymbol{\theta}' = \boldsymbol{U}^*\boldsymbol{\theta}, \qquad \boldsymbol{\theta}^{*\prime} = \boldsymbol{U}^*\boldsymbol{\theta}^*, \qquad \boldsymbol{\phi}'(\mathbf{x}) = \boldsymbol{U}^*\boldsymbol{\phi}(\mathbf{x}).$$

For any input $\mathbf{x}$, the linear model prediction is invariant under this transformation:

$$f'(\mathbf{x}; \boldsymbol{\theta}') = \langle \boldsymbol{\phi}'(\mathbf{x}), \boldsymbol{\theta}' \rangle = \langle \boldsymbol{U}^*\boldsymbol{\phi}(\mathbf{x}), \boldsymbol{U}^*\boldsymbol{\theta} \rangle = \langle \boldsymbol{\phi}(\mathbf{x}), \boldsymbol{U}\boldsymbol{U}^*\boldsymbol{\theta} \rangle = \langle \boldsymbol{\phi}(\mathbf{x}), \boldsymbol{\theta} \rangle = f(\mathbf{x}; \boldsymbol{\theta}).$$

Consequently, the stochastic gradient over the mini-batch $\mathcal{B}_k$ in the transformed space is given by:

$$\begin{aligned}
\mathbf{g}_k'(\boldsymbol{\theta}') &= \nabla_{\boldsymbol{\theta}'} \left(\frac{1}{2B} \sum_{(\mathbf{x},y)\in\mathcal{B}_k} (f'(\mathbf{x}; \boldsymbol{\theta}') - y)^2 \right) \\
&= \frac{1}{B} \sum_{(\mathbf{x},y)\in\mathcal{B}_k} (f'(\mathbf{x}; \boldsymbol{\theta}') - y)\boldsymbol{\phi}'(\mathbf{x}) \\
&= \boldsymbol{U}^* \left(\frac{1}{B} \sum_{(\mathbf{x},y)\in\mathcal{B}_k} (f(\mathbf{x}; \boldsymbol{\theta}) - y)\boldsymbol{\phi}(\mathbf{x}) \right) \\
&= \boldsymbol{U}^* \mathbf{g}_k(\boldsymbol{\theta}).
\end{aligned}$$

Premultiplying the SGD update (2) by $\boldsymbol{U}^*$ yields:

$$\boldsymbol{U}^*\boldsymbol{\theta}_{k+1} = \boldsymbol{U}^*\boldsymbol{\theta}_k - \eta \boldsymbol{U}^*\mathbf{g}_k(\boldsymbol{\theta}_k) \quad \Longrightarrow \quad \boldsymbol{\theta}'_{k+1} = \boldsymbol{\theta}'_k - \eta \mathbf{g}'_k(\boldsymbol{\theta}'_k).$$

Premultiplying the Polyak update (3) by $\boldsymbol{U}^*$ yields:

$$\begin{aligned} &\boldsymbol{U}^*\boldsymbol{\theta}_{k+1} = \boldsymbol{U}^*\boldsymbol{\theta}_k - \eta \boldsymbol{U}^*\mathbf{g}_k(\boldsymbol{\theta}_k) + \rho \boldsymbol{U}^*(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1}) \\ \Longrightarrow \quad &\boldsymbol{\theta}'_{k+1} = \boldsymbol{\theta}'_k - \eta \mathbf{g}'_k(\boldsymbol{\theta}'_k) + \rho(\boldsymbol{\theta}'_k - \boldsymbol{\theta}'_{k-1}). \end{aligned}$$

Similarly, define the lookhead iterate $\widetilde{\boldsymbol{\theta}}_k := \boldsymbol{\theta}_k + \rho(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1})$, then applying $\boldsymbol{U}^*$ to the Nesterov update (4) yields:

$$\begin{aligned} \widetilde{\boldsymbol{\theta}}'_k &= \boldsymbol{U}^*\widetilde{\boldsymbol{\theta}}_k = \boldsymbol{\theta}'_k + \rho(\boldsymbol{\theta}'_k - \boldsymbol{\theta}'_{k-1}), \\ \boldsymbol{\theta}'_{k+1} &= \boldsymbol{U}^*\big(\widetilde{\boldsymbol{\theta}}_k - \eta \mathbf{g}_k(\widetilde{\boldsymbol{\theta}}_k)\big) = \widetilde{\boldsymbol{\theta}}'_k - \eta \mathbf{g}'_k(\widetilde{\boldsymbol{\theta}}'_k). \end{aligned}$$

Furthermore, the transformed feature covariance operator $\mathbf{H}'$ is strictly diagonal:

$$\mathbf{H}' = \mathbb{E}[\boldsymbol{\phi}'(\mathbf{x}) \otimes \boldsymbol{\phi}'(\mathbf{x})] = \mathbb{E}[(\boldsymbol{U}^*\boldsymbol{\phi}(\mathbf{x})) \otimes (\boldsymbol{U}^*\boldsymbol{\phi}(\mathbf{x}))] = \boldsymbol{U}^*\mathbf{H}\boldsymbol{U} = \boldsymbol{U}^*(\boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^*)\boldsymbol{U} = \boldsymbol{\Lambda}.$$

Since the transformed initialization remains $\boldsymbol{\theta}'_0 = \boldsymbol{U}^*\mathbf{0} = \mathbf{0}$, the optimization trajectories of SGD, Polyak, and Nesterov in the rotated eigenspace exactly coincide with their standard coordinate updates. The loss dynamics therefore depend intrinsically only on the eigenvalues of $\mathbf{H}$, establishing that Assumption 3.1 incurs no loss of generality.

A.3 Gaussian fourth-moment identity

We record a Gaussian fourth-moment identity that will be used throughout the stability analysis.

Lemma A.5 (Gaussian fourth-moment identity)**.** *Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{H})$ be a centered Gaussian random element in the separable Hilbert space $\mathcal{H}$, where $\mathbf{H}$ is a positive trace-class covariance operator. Then, for every self-adjoint trace-class operator $\mathbf{A}$ on $\mathcal{H}$,*

$$\mathbb{E}\left[(\mathbf{x} \otimes \mathbf{x})\mathbf{A}(\mathbf{x} \otimes \mathbf{x})\right] = 2\mathbf{H}\mathbf{A}\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{A}). \tag{14}$$

Moreover, let $\widehat{\mathbf{H}} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{x}_b \otimes \mathbf{x}_b, \mathbf{x}_b \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{H})$, and define the centered empirical-covariance fluctuation $\boldsymbol{\Sigma} := \widehat{\mathbf{H}} - \mathbf{H}$. Then

$$\mathbb{E}[\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}] = \frac{1}{B}\left(\mathbf{H}\mathbf{A}\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{A})\right). \tag{15}$$

Both identities hold in the trace-class sense.

Proof. All operator-valued expectations below are well defined in the trace-class sense, since $\mathbf{H}$ is trace class and $\mathbb{E}\|\mathbf{x}\|_2^4 < \infty$.

For arbitrary $\mathbf{u}, \mathbf{v} \in \mathcal{H}$, Wick's formula for Gaussian random elements gives

$$\mathbb{E}\left[\langle \mathbf{u}, \mathbf{x}\rangle\langle \mathbf{x}, \mathbf{A}\mathbf{x}\rangle\langle \mathbf{x}, \mathbf{v}\rangle\right] = \langle \mathbf{u}, \mathbf{H}\mathbf{v}\rangle \mathrm{Tr}(\mathbf{H}\mathbf{A}) + \langle \mathbf{u}, \mathbf{H}\mathbf{A}\mathbf{H}\mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{H}\mathbf{A}\mathbf{H}\mathbf{u}\rangle.$$

Since $\mathbf{A}$ and $\mathbf{H}$ are self-adjoint, $\mathbf{H}\mathbf{A}\mathbf{H}$ is self-adjoint. Therefore, $\langle \mathbf{v}, \mathbf{H}\mathbf{A}\mathbf{H}\mathbf{u}\rangle = \langle \mathbf{u}, \mathbf{H}\mathbf{A}\mathbf{H}\mathbf{v}\rangle$. Consequently,

$$\langle \mathbf{u}, \mathbb{E}\left[(\mathbf{x} \otimes \mathbf{x})\mathbf{A}(\mathbf{x} \otimes \mathbf{x})\right]\mathbf{v}\rangle = 2\langle \mathbf{u}, \mathbf{H}\mathbf{A}\mathbf{H}\mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{H}\mathbf{v}\rangle \mathrm{Tr}(\mathbf{H}\mathbf{A}).$$

Since this holds for every $\mathbf{u}, \mathbf{v} \in \mathcal{H}$, we obtain

$$\mathbb{E}\left[(\mathbf{x} \otimes \mathbf{x})\mathbf{A}(\mathbf{x} \otimes \mathbf{x})\right] = 2\mathbf{H}\mathbf{A}\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{A}).$$

We now prove (15). By independence of the samples,

$$\begin{aligned}\mathbb{E}[\widehat{\mathbf{H}}\mathbf{A}\widehat{\mathbf{H}}] &= \frac{1}{B^2}\sum_{b,b'=1}^{B}\mathbb{E}\left[(\mathbf{x}_b\otimes\mathbf{x}_b)\mathbf{A}(\mathbf{x}_{b'}\otimes\mathbf{x}_{b'})\right] \\ &= \frac{1}{B}\mathbb{E}\left[(\mathbf{x}\otimes\mathbf{x})\mathbf{A}(\mathbf{x}\otimes\mathbf{x})\right] + \frac{B-1}{B}\mathbf{H}\mathbf{A}\mathbf{H} \\ &= \left(1+\frac{1}{B}\right)\mathbf{H}\mathbf{A}\mathbf{H} + \frac{1}{B}\mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{A}).\end{aligned}$$

Since $\mathbb{E}[\widehat{\mathbf{H}}] = \mathbf{H}$, expanding $\boldsymbol{\Sigma} = \widehat{\mathbf{H}} - \mathbf{H}$ gives

$$\mathbb{E}[\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}] = \mathbb{E}[\widehat{\mathbf{H}}\mathbf{A}\widehat{\mathbf{H}}] - \mathbf{H}\mathbf{A}\mathbf{H} = \frac{1}{B}\left(\mathbf{H}\mathbf{A}\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{A})\right),$$

as claimed. □

A.4 Noise-curvature alignment

In this section, we prove Proposition 3.4. We restate it here for completeness.

Proposition A.6. *Under the Gaussian feature assumption, the stochastic-gradient noise in power-law kernel regression is curvature-aligned. More precisely, for any $\boldsymbol{\nu}$ in the unit sphere $\mathbb{S}(\mathcal{H}) := \{\boldsymbol{\nu}\in\mathcal{H} : \|\boldsymbol{\nu}\| = 1\}$, we have*

$$\mathbb{E}\left[|\langle\boldsymbol{\xi}_k,\boldsymbol{\nu}\rangle|^2\right] \eqsim \frac{1}{B}\left[\mathcal{E}(\boldsymbol{\theta}_k)+\sigma^2\right]\langle\boldsymbol{\nu},\mathbf{H}\boldsymbol{\nu}\rangle\,. \tag{16}$$

Proof of Proposition 3.4. For a mini-batch of size B, write $\boldsymbol{\xi}_k(\boldsymbol{\theta}) = \frac{1}{B}\sum_{i=1}^{B}\boldsymbol{\zeta}_{k,i}(\boldsymbol{\theta})$, where $\boldsymbol{\zeta}_{k,i}(\boldsymbol{\theta})$ is the gradient noise associated with a single data point. Conditional on $\boldsymbol{\theta}$, the random elements $\boldsymbol{\zeta}_{k,i}(\boldsymbol{\theta})$ are independent and have zero mean. Therefore, for any $\boldsymbol{\nu}\in\mathbb{S}(\mathcal{H})$,

$$\begin{aligned}\mathbb{E}\left[|\langle\boldsymbol{\xi}_k(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle|^2\middle|\boldsymbol{\theta}\right] &= \frac{1}{B^2}\sum_{i,j=1}^{B}\mathbb{E}\left[\langle\boldsymbol{\zeta}_{k,i}(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle\langle\boldsymbol{\zeta}_{k,j}(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle\middle|\boldsymbol{\theta}\right] \\ &= \frac{1}{B^2}\sum_{i=1}^{B}\mathbb{E}\left[|\langle\boldsymbol{\zeta}_{k,i}(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle|^2\middle|\boldsymbol{\theta}\right] \\ &= \frac{1}{B}\mathbb{E}\left[|\langle\boldsymbol{\zeta}(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle|^2\middle|\boldsymbol{\theta}\right].\end{aligned}$$

It remains to compute the directional second moment of the single-sample gradient noise. Consider a single data point $(\mathbf{x},y)$ and write $\boldsymbol{\delta} = \boldsymbol{\theta}-\boldsymbol{\theta}^*, \boldsymbol{\phi} = \boldsymbol{\phi}(\mathbf{x})$. Since $y = \langle\boldsymbol{\phi},\boldsymbol{\theta}^*\rangle + \epsilon$, the single-sample gradient is

$$\mathbf{g}_{\mathbf{x},y}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}\left[\frac{1}{2}\big(f(\mathbf{x};\boldsymbol{\theta})-y\big)^2\right] = \big(\langle\boldsymbol{\phi},\boldsymbol{\theta}\rangle - y\big)\boldsymbol{\phi} = \big(\langle\boldsymbol{\phi},\boldsymbol{\delta}\rangle-\epsilon\big)\boldsymbol{\phi}.$$

Using $\mathbb{E}[\epsilon]=0$ and $\mathbb{E}[\boldsymbol{\phi}\otimes\boldsymbol{\phi}] = \mathbf{H}$, we obtain

$$\nabla\mathcal{R}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{x},y}[\mathbf{g}_{\mathbf{x},y}(\boldsymbol{\theta})] = \mathbf{H}\boldsymbol{\delta}.$$

Thus, the single-sample gradient noise is

$$\boldsymbol{\zeta}(\boldsymbol{\theta}) = \mathbf{g}_{\mathbf{x},y}(\boldsymbol{\theta}) - \nabla\mathcal{R}(\boldsymbol{\theta}) = \langle\boldsymbol{\phi},\boldsymbol{\delta}\rangle\boldsymbol{\phi} - \mathbf{H}\boldsymbol{\delta} - \epsilon\boldsymbol{\phi}.$$

For any $\boldsymbol{\nu}\in\mathbb{S}(\mathcal{H})$,

$$\langle\boldsymbol{\zeta}(\boldsymbol{\theta}),\boldsymbol{\nu}\rangle = \langle\boldsymbol{\phi},\boldsymbol{\delta}\rangle\langle\boldsymbol{\phi},\boldsymbol{\nu}\rangle - \langle\boldsymbol{\delta},\mathbf{H}\boldsymbol{\nu}\rangle - \epsilon\langle\boldsymbol{\phi},\boldsymbol{\nu}\rangle.$$

The cross terms involving ϵ vanish because ϵ is independent of $\boldsymbol{\phi}$ and satisfies $\mathbb{E}[\epsilon] = 0$. Hence,

$$\mathbb{E}\left[|\langle \boldsymbol{\zeta}(\boldsymbol{\theta}), \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}\right] = \mathbb{E}\left[\langle \boldsymbol{\phi}, \boldsymbol{\delta}\rangle^2 \langle \boldsymbol{\phi}, \boldsymbol{\nu}\rangle^2\right] - \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\nu}\rangle^2 + \sigma^2 \mathbb{E}\left[\langle \boldsymbol{\phi}, \boldsymbol{\nu}\rangle^2\right].$$

Conditionally on $\boldsymbol{\theta}$, the vector $\boldsymbol{\delta}$ is fixed. Taking $\mathbf{A} = \boldsymbol{\delta} \otimes \boldsymbol{\delta}$ in Lemma A.5 and evaluating the resulting operator identity along $\boldsymbol{\nu}$, we obtain

$$\begin{aligned}\mathbb{E}\left[\langle \boldsymbol{\phi}, \boldsymbol{\delta}\rangle^2 \langle \boldsymbol{\phi}, \boldsymbol{\nu}\rangle^2\right] &= \langle \boldsymbol{\nu}, \mathbb{E}\left[(\boldsymbol{\phi} \otimes \boldsymbol{\phi})(\boldsymbol{\delta} \otimes \boldsymbol{\delta})(\boldsymbol{\phi} \otimes \boldsymbol{\phi})\right] \boldsymbol{\nu}\rangle \\ &= 2\left\langle \boldsymbol{\nu}, \mathbf{H}(\boldsymbol{\delta} \otimes \boldsymbol{\delta})\mathbf{H}\boldsymbol{\nu}\right\rangle + \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle \mathrm{Tr}(\mathbf{H}(\boldsymbol{\delta} \otimes \boldsymbol{\delta})).\end{aligned}$$

Since $\langle \boldsymbol{\nu}, \mathbf{H}(\boldsymbol{\delta} \otimes \boldsymbol{\delta})\mathbf{H}\boldsymbol{\nu}\rangle = \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\nu}\rangle^2$ and $\mathrm{Tr}(\mathbf{H}(\boldsymbol{\delta} \otimes \boldsymbol{\delta})) = \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle$, we have

$$\mathbb{E}\left[\langle \boldsymbol{\phi}, \boldsymbol{\delta}\rangle^2 \langle \boldsymbol{\phi}, \boldsymbol{\nu}\rangle^2\right] = \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle + 2\langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\nu}\rangle^2.$$

Moreover, $\mathbb{E}\left[\langle \boldsymbol{\phi}, \boldsymbol{\nu}\rangle^2\right] = \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle$. Therefore,

$$\mathbb{E}\left[|\langle \boldsymbol{\zeta}(\boldsymbol{\theta}), \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}\right] = \left(\sigma^2 + \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle\right) \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle + \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\nu}\rangle^2.$$

Dropping the nonnegative second term yields

$$\mathbb{E}\left[|\langle \boldsymbol{\zeta}(\boldsymbol{\theta}), \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}\right] \geqslant \left(\sigma^2 + \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle\right) \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle.$$

For the upper bound, the Cauchy–Schwarz inequality gives

$$\langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\nu}\rangle^2 \leqslant \langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle.$$

Consequently,

$$\mathbb{E}\left[|\langle \boldsymbol{\zeta}(\boldsymbol{\theta}), \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}\right] \leqslant \left(\sigma^2 + 2\langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle\right) \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle.$$

Using

$$\mathcal{E}(\boldsymbol{\theta}) = \frac{1}{2}\langle \boldsymbol{\delta}, \mathbf{H}\boldsymbol{\delta}\rangle,$$

we obtain

$$\left[\mathcal{E}(\boldsymbol{\theta}) + \sigma^2\right] \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle \leqslant \mathbb{E}\left[|\langle \boldsymbol{\zeta}(\boldsymbol{\theta}), \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}\right] \leqslant 4\left[\mathcal{E}(\boldsymbol{\theta}) + \sigma^2\right] \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle.$$

Combining this with the mini-batch identity and taking $\boldsymbol{\theta} = \boldsymbol{\theta}_k$ gives

$$\frac{1}{B}\left[\mathcal{E}(\boldsymbol{\theta}_k) + \sigma^2\right] \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle \leqslant \mathbb{E}\left[|\langle \boldsymbol{\xi}_k, \boldsymbol{\nu}\rangle|^2 \middle| \boldsymbol{\theta}_k\right] \leqslant \frac{4}{B}\left[\mathcal{E}(\boldsymbol{\theta}_k) + \sigma^2\right] \langle \boldsymbol{\nu}, \mathbf{H}\boldsymbol{\nu}\rangle.$$

This proves (5). □

B Mean square stability analysis

As explained in Section 4, additive label noise enters the second-moment dynamics only through an external forcing term and does not alter the homogeneous covariance transition that determines the critical learning-rate boundary.

To illustrate this clearly, consider a one-dimensional scalar linear regression with feature $x \sim \mathcal{N}(0, \lambda)$ ($\lambda > 0$) and label $y = \theta^* x + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$. Under full-batch gradient descent, the gradient at step k with fresh label noise is $g_k(\theta_k) = \lambda(\theta_k - \theta^*) - \xi_k$, where $\mathbb{E}[\xi_k] = 0$ and $\mathbb{E}[\xi_k^2] = \sigma^2\lambda$. Let $e_k = \theta_k - \theta^*$ be the parameter error. The error recursion obeys:

$$e_{k+1} = (1 - \eta\lambda)e_k + \eta\xi_k.$$

Taking the expected second moment $v_k := \mathbb{E}[e_k^2]$, the cross-term vanishes due to the independence of ξ_k, yielding the exact variance dynamics:

$$v_{k+1} = (1 - \eta\lambda)^2 v_k + \eta^2\sigma^2\lambda.$$

Mean-square stability requires the sequence $(v_k)_{k\geqslant 0}$ to converge to a finite limit. This holds if and only if the transition factor satisfies $(1 - \eta\lambda)^2 < 1$, which gives the classical stability boundary $\eta < 2/\lambda$. Crucially, the label noise variance σ^2 enters solely through the constant forcing term $\eta^2\sigma^2\lambda$ and is completely absent from the contraction factor $(1 - \eta\lambda)^2$. Therefore, label noise determines the asymptotic noise floor but exerts no influence on the stability threshold.

Guided by this exact decoupling, which holds identically in higher dimensions and for momentum algorithms, we set $\sigma = 0$ when determining the critical learning-rate boundary. Indeed, setting $\sigma = 0$ removes the accumulated forcing term and leaves

$$\mathbf{C}_k = \mathcal{Q}^k(\mathbf{C}_0).$$

B.1 Stability of SGD

With $\sigma = 0$, the SGD error recursion is

$$\mathbf{e}_{k+1} = (\mathbf{I} - \eta\widehat{\mathbf{H}}_k)\mathbf{e}_k. \tag{17}$$

Theorem B.1 (Mean-square stability of SGD)**.** *Let* $\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$ *be a positive trace-class operator, where* $\lambda_1 = \lambda_{\max} \geqslant \lambda_2 \geqslant \cdots > 0$*. Then the homogeneous SGD recursion* (17) *is mean-square stable if and only if*

$$\eta < \frac{2B}{(B+1)\lambda_{\max}} \tag{18}$$

and

$$\mathcal{R}_{\mathrm{sgd}}(\eta) := \sum_{i\geqslant 1} \frac{\eta\lambda_i}{2B - (B+1)\eta\lambda_i} < 1. \tag{19}$$

Proof. Define the covariance operator $\mathbf{C}_k := \mathbb{E}[\mathbf{e}_k \otimes \mathbf{e}_k]$. Since $\mathbf{e}_k$ is square-integrable, $\mathbf{C}_k$ is positive and trace class, with

$$\mathrm{Tr}(\mathbf{C}_k) = \mathbb{E}\|\mathbf{e}_k\|^2.$$

Because the mini-batch at iteration k is independent of the past and $\mathbb{E}[\mathbf{\Sigma}_k] = 0$, expanding (17) gives

$$\mathbf{C}_{k+1} = (\mathbf{I} - \eta\mathbf{H})\mathbf{C}_k(\mathbf{I} - \eta\mathbf{H}) + \eta^2\mathbb{E}[\mathbf{\Sigma}_k\mathbf{C}_k\mathbf{\Sigma}_k]. \tag{20}$$

Applying Lemma A.5 with $\mathbf{A} = \mathbf{C}_k$ gives

$$\mathbb{E}[\mathbf{\Sigma}_k\mathbf{C}_k\mathbf{\Sigma}_k] = \frac{1}{B}\left(\mathbf{H}\mathbf{C}_k\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{C}_k)\right).$$

Substituting this identity into (20), we obtain

$$\mathbf{C}_{k+1} = (\mathbf{I} - \eta\mathbf{H})\mathbf{C}_k(\mathbf{I} - \eta\mathbf{H}) + \frac{\eta^2}{B}\left(\mathbf{H}\mathbf{C}_k\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{C}_k)\right). \tag{21}$$

Strong mean-square stability is equivalent to $\mathrm{Tr}(\mathbf{C}_k) \to 0$. Since $\mathbf{C}_k$ is positive, its trace is completely determined by its diagonal entries. Moreover, because $\mathbf{H}$ is diagonal, the diagonal entries of (21) form a closed recursion that does not depend on the off-diagonal entries. Therefore, it is sufficient to analyze the diagonal covariance dynamics. Let $\{\mathbf{e}_i\}_{i\geqslant 1}$ denote the eigenbasis of $\mathbf{H}$, and define

$$u_{i,k} := \langle \mathbf{e}_i, \mathbf{C}_k\mathbf{e}_i\rangle, \qquad S_k := \mathrm{Tr}(\mathbf{H}\mathbf{C}_k) = \sum_{i\geqslant 1}\lambda_i u_{i,k}.$$

Since $\mathbf{C}_k$ is positive and trace class,

$$\mathbf{u}_k := (u_{1,k}, u_{2,k}, \ldots) \in \ell^1_+, \qquad \|\mathbf{u}_k\|_{\ell^1} = \mathrm{Tr}(\mathbf{C}_k).$$

Taking the diagonal entries of (21) gives

$$u_{i,k+1} = a_i u_{i,k} + \frac{\eta^2}{B}\lambda_i S_k, \qquad a_i := (1-\eta\lambda_i)^2 + \frac{\eta^2}{B}\lambda_i^2. \tag{22}$$

Equivalently,

$$\mathbf{u}_{k+1} = \mathbf{M}_\eta \mathbf{u}_k, \qquad \mathbf{M}_\eta = \mathbf{D}_\eta + \frac{\eta^2}{B}\boldsymbol{\lambda}\otimes\boldsymbol{\lambda},$$

where

$$\mathbf{D}_\eta = \mathrm{diag}(a_1, a_2, \ldots), \qquad \boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots)^\top.$$

We first determine the local stability condition. A necessary condition is $a_i < 1$ for every i. Since

$$1 - a_i = \eta\lambda_i\left[2 - \left(1 + \frac{1}{B}\right)\eta\lambda_i\right],$$

this is equivalent to

$$\eta < \frac{2B}{(B+1)\lambda_{\max}},$$

which proves (18). We now assume that (18) holds. Iterating (22) gives

$$u_{i,k} = a_i^k u_{i,0} + \frac{\eta^2}{B}\lambda_i \sum_{s=0}^{k-1} a_i^{k-1-s} S_s. \tag{23}$$

Multiplying by λ_i and summing over $i \geqslant 1$, we obtain the renewal equation

$$S_k = h_k + \sum_{s=0}^{k-1} K_{k-1-s} S_s, \tag{24}$$

where

$$h_k := \sum_{i\geqslant 1}\lambda_i a_i^k u_{i,0}, \qquad K_m := \frac{\eta^2}{B}\sum_{i\geqslant 1}\lambda_i^2 a_i^m.$$

The forcing sequence $(h_k)_{k\geqslant 0}$ is summable. Indeed,

$$\sum_{k\geqslant 0} h_k = \sum_{i\geqslant 1}\frac{\lambda_i u_{i,0}}{1-a_i} \leqslant \frac{\|\mathbf{u}_0\|_{\ell^1}}{\eta\left[2 - \left(1+\frac{1}{B}\right)\eta\lambda_{\max}\right]} < \infty.$$

Moreover,

$$\sum_{m\geqslant 0} K_m = \frac{\eta^2}{B}\sum_{i\geqslant 1}\frac{\lambda_i^2}{1-a_i} = \sum_{i\geqslant 1}\frac{\eta\lambda_i}{2B-(B+1)\eta\lambda_i} = \mathcal{R}_{\mathrm{sgd}}(\eta).$$

Suppose first that $\mathcal{R}_{\mathrm{sgd}}(\eta) < 1$. Since all terms in (24) are nonnegative, summing the renewal equation and applying Tonelli's theorem gives

$$\sum_{k\geqslant 0} S_k \leqslant \frac{\sum_{k\geqslant 0} h_k}{1-\mathcal{R}_{\mathrm{sgd}}(\eta)} < \infty.$$

Summing (23) over i gives

$$\|\mathbf{u}_k\|_{\ell^1} = \sum_{i\geqslant 1} a_i^k u_{i,0} + \frac{\eta^2}{B}\sum_{s=0}^{k-1} G_{k-1-s}S_s, \tag{25}$$

where

$$G_m := \sum_{i\geqslant 1}\lambda_i a_i^m.$$

For every fixed i, $a_i^k \to 0$. Since $\mathbf{u}_0 \in \ell^1$, dominated convergence gives

$$\sum_{i\geqslant 1} a_i^k u_{i,0} \longrightarrow 0.$$

Similarly, since $\lambda \in \ell^1$,

$$G_m \longrightarrow 0, \qquad \sup_{m\geqslant 0} G_m \leqslant \mathrm{Tr}(\mathbf{H}) < \infty.$$

Because $(S_k)_{k\geqslant 0} \in \ell^1$, the convolution term in (25) also converges to zero. Therefore,

$$\mathbb{E}\|\mathbf{e}_k\|^2 = \mathrm{Tr}(\mathbf{C}_k) = \|\mathbf{u}_k\|_{\ell^1} \longrightarrow 0.$$

This proves sufficiency. We next prove necessity. If (18) fails, then $a_i \geqslant 1$ for some i. Choosing an initialization supported on this eigendirection gives

$$u_{i,k+1} \geqslant a_i u_{i,k},$$

so the second moment does not converge to zero. Hence the local condition is necessary. Now assume that the local condition holds but $\mathcal{R}_{\mathrm{sgd}}(\eta) > 1$. For $\mu > 1$, define

$$F(\mu) := \frac{\eta^2}{B}\sum_{i\geqslant 1}\frac{\lambda_i^2}{\mu - a_i}.$$

The function F is continuous and strictly decreasing, with

$$\lim_{\mu\downarrow 1} F(\mu) = \mathcal{R}_{\mathrm{sgd}}(\eta) > 1, \qquad \lim_{\mu\to\infty} F(\mu) = 0.$$

Hence there exists $\mu > 1$ such that $F(\mu) = 1$. Define

$$v_i := \frac{\lambda_i}{\mu - a_i}.$$

Since $\mu > 1$ and $\lambda \in \ell^1$, we have $\mathbf{v} = (v_1, v_2, \ldots) \in \ell^1_+$. Moreover,

$$\mathbf{M}_\eta \mathbf{v} = \mu\mathbf{v}.$$

Choosing the deterministic initialization $(\mathbf{e}_0)_i = \sqrt{v_i}$ gives $\mathbf{u}_0 = \mathbf{v}$, and hence

$$\mathbf{u}_k = \mu^k \mathbf{v}.$$

Consequently,

$$\mathbb{E}\|\mathbf{e}_k\|^2 = \mu^k \|\mathbf{v}\|_{\ell^1} \longrightarrow \infty.$$

It remains to consider the boundary $\mathcal{R}_{\mathrm{sgd}}(\eta) = 1$. Define

$$w_i := \frac{\lambda_i}{1 - a_i} = \frac{1}{\eta\left[2 - \left(1 + \frac{1}{B}\right)\eta\lambda_i\right]}.$$

The local condition implies $\mathbf{w} = (w_1, w_2, \ldots) \in \ell^\infty_+$, and $\mathcal{R}_{\mathrm{sgd}}(\eta) = 1$ implies

$$\mathbf{M}^*_\eta \mathbf{w} = \mathbf{w}.$$

Thus, for every nonzero $\mathbf{u}_0 \in \ell^1_+$,

$$\langle \mathbf{w}, \mathbf{u}_k \rangle = \langle \mathbf{w}, \mathbf{u}_0 \rangle > 0 \qquad \text{for every } k.$$

If $\|\mathbf{u}_k\|_{\ell^1} \to 0$, then

$$\langle \mathbf{w}, \mathbf{u}_k \rangle \leqslant \|\mathbf{w}\|_{\ell^\infty} \|\mathbf{u}_k\|_{\ell^1} \longrightarrow 0,$$

which is a contradiction. Hence the critical boundary is not strongly mean-square asymptotically stable. □

Corollary B.2 (Critical learning rate of SGD)**.** *There exists a unique critical learning rate* $\eta^{\mathrm{crit}}_{\mathrm{sgd}}$ *satisfying*

$$0 < \eta^{\mathrm{crit}}_{\mathrm{sgd}} < \frac{2B}{(B+1)\lambda_{\max}}$$

and

$$\sum_{i \geqslant 1} \frac{\eta^{\mathrm{crit}}_{\mathrm{sgd}} \lambda_i}{2B - (B+1)\eta^{\mathrm{crit}}_{\mathrm{sgd}} \lambda_i} = 1. \tag{26}$$

The homogeneous SGD dynamics are mean-square stable if and only if

$$0 < \eta < \eta^{\mathrm{crit}}_{\mathrm{sgd}}.$$

Moreover,

$$\frac{2B}{(B+1)\lambda_{\max} + \mathrm{Tr}(\mathbf{H})} \leqslant \eta^{\mathrm{crit}}_{\mathrm{sgd}} \leqslant \min\left\{\frac{2B}{(B+1)\lambda_{\max}}, \frac{2B}{\mathrm{Tr}(\mathbf{H})}\right\}. \tag{27}$$

Consequently, under $\lambda_i \eqsim i^{-\beta}$, $\beta > 1$, *with* β *regarded as fixed,*

$$\eta^{\mathrm{crit}}_{\mathrm{sgd}} \eqsim 1.$$

Proof. The function $\mathcal{R}_{\mathrm{sgd}}(\eta)$ is continuous and strictly increasing on

$$0 < \eta < \frac{2B}{(B+1)\lambda_{\max}}.$$

Moreover, $\mathcal{R}_{\mathrm{sgd}}(0) = 0$, while $\mathcal{R}_{\mathrm{sgd}}(\eta) \to \infty$ as η approaches the local stability boundary. Hence the critical learning rate exists and is unique. At the critical learning rate, using

$$2B - (B+1)\eta^{\mathrm{crit}}_{\mathrm{sgd}} \lambda_i \leqslant 2B$$

in (26) gives

$$1 \geqslant \frac{\eta^{\mathrm{crit}}_{\mathrm{sgd}} \mathrm{Tr}(\mathbf{H})}{2B},$$

and hence

$$\eta_{\mathrm{sgd}}^{\mathrm{crit}} \leqslant \frac{2B}{\mathrm{Tr}(\mathbf{H})}.$$

The local condition gives the other upper bound in (27). Conversely,

$$2B - (B+1)\eta_{\mathrm{sgd}}^{\mathrm{crit}}\lambda_i \geqslant 2B - (B+1)\eta_{\mathrm{sgd}}^{\mathrm{crit}}\lambda_{\max}.$$

Therefore,

$$1 \leqslant \frac{\eta_{\mathrm{sgd}}^{\mathrm{crit}}\mathrm{Tr}(\mathbf{H})}{2B - (B+1)\eta_{\mathrm{sgd}}^{\mathrm{crit}}\lambda_{\max}},$$

which yields the lower bound in (27). Finally, when $\lambda_i \eqsim i^{-\beta}$ with fixed $\beta > 1$,

$$\lambda_{\max} \eqsim 1, \qquad \mathrm{Tr}(\mathbf{H}) = \sum_{i \geqslant 1} \lambda_i \eqsim 1.$$

The two-sided bound (27) therefore gives

$$\eta_{\mathrm{sgd}}^{\mathrm{crit}} \eqsim 1.$$

□

Remark B.3 (Absence of a uniform spectral gap). Although $a_i < 1$ for every fixed eigendirection in the stable regime, the power-law spectrum satisfies $\lambda_i \to 0$, and hence

$$a_i \longrightarrow 1 \qquad \text{as } i \to \infty.$$

Thus, the infinite-dimensional moment dynamics generally do not possess a uniform spectral gap. In particular, the spectral radius of the global covariance operator need not be strictly smaller than one even though every admissible covariance trajectory converges strongly to zero. Stability here is strong mean-square asymptotic stability: the convergence holds in trace norm for each admissible initial covariance, but need not be uniform or exponential over all spectral directions.

B.2 Stability of Polyak

With $\sigma = 0$, the Polyak error recursion is

$$\mathbf{e}_{k+1} = (\mathbf{I} - \eta\widehat{\mathbf{H}}_k)\mathbf{e}_k + \rho(\mathbf{e}_k - \mathbf{e}_{k-1}). \tag{28}$$

Writing

$$\widehat{\mathbf{H}}_k = \mathbf{H} + \mathbf{\Sigma}_k, \qquad \mathbf{D} = (1+\rho)\mathbf{I} - \eta\mathbf{H},$$

the recursion becomes

$$\mathbf{e}_{k+1} = \mathbf{D}\mathbf{e}_k - \rho\mathbf{e}_{k-1} - \eta\mathbf{\Sigma}_k\mathbf{e}_k.$$

Theorem B.4 (Mean-square stability of Polyak)**.** *Let* $\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$ *be a positive trace-class operator, where* $\lambda_1 = \lambda_{\max} \geqslant \lambda_2 \geqslant \cdots > 0$*, and let* $0 \leqslant \rho < 1$*. Then the homogeneous Polyak recursion* (28) *is mean-square stable if and only if*

$$\eta < \frac{2B(1-\rho^2)}{\lambda_{\max}\left[B(1-\rho) + (1+\rho)\right]} \tag{29}$$

and

$$\mathcal{R}_{\mathrm{polyak}}(\eta, \rho) := \sum_{i \geqslant 1} \frac{\eta\lambda_i(1+\rho)}{2B(1-\rho^2) - \eta\lambda_i\left[B(1-\rho) + (1+\rho)\right]} < 1. \tag{30}$$

Proof. Define

$$\mathbf{U}_k := \mathbb{E}[\mathbf{e}_k \otimes \mathbf{e}_k], \qquad \mathbf{P}_k := \mathbb{E}[\mathbf{e}_k \otimes \mathbf{e}_{k-1}], \qquad \mathbf{Q}_k := \mathbb{E}[\mathbf{e}_{k-1} \otimes \mathbf{e}_k], \qquad \mathbf{R}_k := \mathbb{E}[\mathbf{e}_{k-1} \otimes \mathbf{e}_{k-1}].$$

These operators are trace class whenever the augmented state has a finite second moment, and

$$\mathbb{E}\|\mathbf{z}_k\|^2 = \mathrm{Tr}(\mathbf{U}_k) + \mathrm{Tr}(\mathbf{R}_k).$$

Because the mini-batch at iteration k is independent of the past and $\mathbb{E}[\mathbf{\Sigma}_k] = 0$, expanding (28) and applying Lemma A.5 gives

$$\begin{cases} \mathbf{U}_{k+1} = \mathbf{D}\mathbf{U}_k\mathbf{D} - \rho\mathbf{D}\mathbf{P}_k - \rho\mathbf{Q}_k\mathbf{D} + \rho^2\mathbf{R}_k + \dfrac{\eta^2}{B}\left(\mathbf{H}\mathbf{U}_k\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{U}_k)\right), \\ \mathbf{P}_{k+1} = \mathbf{D}\mathbf{U}_k - \rho\mathbf{Q}_k, \\ \mathbf{Q}_{k+1} = \mathbf{U}_k\mathbf{D} - \rho\mathbf{P}_k, \\ \mathbf{R}_{k+1} = \mathbf{U}_k. \end{cases} \tag{31}$$

Strong mean-square stability is equivalent to

$$\mathrm{Tr}(\mathbf{U}_k) + \mathrm{Tr}(\mathbf{R}_k) \longrightarrow 0.$$

Since $\mathbf{R}_{k+1} = \mathbf{U}_k$, it is sufficient and necessary to prove $\mathrm{Tr}(\mathbf{U}_k) \to 0$. Moreover, because $\mathbf{H}$ and $\mathbf{D}$ are diagonal, the diagonal entries of (31) form a closed recursion. Let $\{\mathbf{e}_i\}_{i\geqslant 1}$ denote the eigenbasis of $\mathbf{H}$, and define

$$u_{i,k} := \langle \mathbf{e}_i, \mathbf{U}_k\mathbf{e}_i\rangle, \qquad v_{i,k} := \langle \mathbf{e}_i, \mathbf{P}_k\mathbf{e}_i\rangle, \qquad r_{i,k} := \langle \mathbf{e}_i, \mathbf{R}_k\mathbf{e}_i\rangle,$$

and

$$S_k := \mathrm{Tr}(\mathbf{H}\mathbf{U}_k) = \sum_{i\geqslant 1} \lambda_i u_{i,k}.$$

Since $\mathbf{Q}_k = \mathbf{P}_k^*$, the diagonal entries of $\mathbf{P}_k$ and $\mathbf{Q}_k$ coincide. Setting

$$d_i := 1 + \rho - \eta\lambda_i,$$

we obtain

$$\begin{cases} u_{i,k+1} = \left(d_i^2 + \dfrac{\eta^2}{B}\lambda_i^2\right) u_{i,k} - 2\rho d_i v_{i,k} + \rho^2 r_{i,k} + \dfrac{\eta^2}{B}\lambda_i S_k, \\ v_{i,k+1} = d_i u_{i,k} - \rho v_{i,k}, \\ r_{i,k+1} = u_{i,k}. \end{cases} \tag{32}$$

Define

$$\mathbf{X}_{i,k} := (u_{i,k}, v_{i,k}, r_{i,k})^\top, \qquad \mathbf{b} := (1,0,0)^\top,$$

and

$$\mathbf{M}_i := \begin{pmatrix} d_i^2 + \dfrac{\eta^2}{B}\lambda_i^2 & -2\rho d_i & \rho^2 \\ d_i & -\rho & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Then (32) becomes

$$\mathbf{X}_{i,k+1} = \mathbf{M}_i\mathbf{X}_{i,k} + \frac{\eta^2}{B}\lambda_i \mathbf{b} S_k. \tag{33}$$

We first determine the local stability condition. Ignoring the global trace-coupling term S_k, the characteristic polynomial of $\mathbf{M}_i$ is

$$p_i(\mu) = \mu^3 + \left(\rho - d_i^2 - \frac{\eta^2}{B}\lambda_i^2\right)\mu^2 + \rho\left(d_i^2 - \frac{\eta^2}{B}\lambda_i^2 - \rho\right)\mu - \rho^3.$$

The Jury stability criterion shows that all roots lie strictly inside the unit disk if and only if

$$2B(1-\rho^2) - \eta\lambda_i \left[B(1-\rho) + (1+\rho)\right] > 0.$$

Requiring this condition for every i is equivalent to (29). We now assume that the local condition holds. Define

$$g_{i,m} := \mathbf{b}^\top \mathbf{M}_i^m \mathbf{b}.$$

This quantity is nonnegative because it is the variance generated by the isolated i-th coordinate dynamics from the covariance state $\mathbf{b}$. For each fixed i, $g_{i,m} \to 0$. Moreover, the family of local blocks is uniformly power bounded under the strict local condition, so there exists $C < \infty$, independent of i and m, such that

$$0 \leqslant g_{i,m} \leqslant C.$$

Iterating (33), taking the first component, multiplying by λ_i, and summing over $i \geqslant 1$ gives

$$S_k = h_k + \sum_{s=0}^{k-1} K_{k-1-s} S_s, \tag{34}$$

where

$$h_k := \sum_{i \geqslant 1} \lambda_i \mathbf{b}^\top \mathbf{M}_i^k \mathbf{X}_{i,0}, \qquad K_m := \frac{\eta^2}{B} \sum_{i \geqslant 1} \lambda_i^2 g_{i,m}.$$

The sequence $(h_k)_{k \geqslant 0}$ is nonnegative and summable. Indeed,

$$\sum_{k \geqslant 0} h_k = \sum_{i \geqslant 1} \lambda_i \mathbf{b}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{X}_{i,0}.$$

A direct calculation gives

$$\mathbf{b}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} = \frac{B\big(1+\rho,\ -2\rho d_i,\ \rho^2(1+\rho)\big)}{\eta\lambda_i \left\{2B(1-\rho^2) - \eta\lambda_i \left[B(1-\rho) + (1+\rho)\right]\right\}}. \tag{35}$$

The initial coordinate covariance satisfies

$$\begin{pmatrix} u_{i,0} & v_{i,0} \\ v_{i,0} & r_{i,0} \end{pmatrix} \succeq 0.$$

Therefore,

$$v_{i,0}^2 \leqslant u_{i,0} r_{i,0}, \qquad |v_{i,0}| \leqslant \frac{u_{i,0} + r_{i,0}}{2}.$$

Define

$$\delta_{\text{polyak}} := 2B(1-\rho^2) - \eta\lambda_{\max} \left[B(1-\rho) + (1+\rho)\right] > 0.$$

Using (35) and $|d_i| \leqslant 1 + \rho + \eta\lambda_{\max}$, we obtain

$$\lambda_i \mathbf{b}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{X}_{i,0} \leqslant C(u_{i,0} + r_{i,0}),$$

where $C < \infty$ is independent of i. Hence

$$\sum_{k \geqslant 0} h_k \leqslant C \sum_{i \geqslant 1} (u_{i,0} + r_{i,0}) < \infty.$$

By Tonelli's theorem,

$$\sum_{m \geqslant 0} K_m = \frac{\eta^2}{B} \sum_{i \geqslant 1} \lambda_i^2 \mathbf{b}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{b}.$$

The first entry of (35) gives

$$\mathbf{b}^\top(\mathbf{I}-\mathbf{M}_i)^{-1}\mathbf{b}=\frac{B(1+\rho)}{\eta\lambda_i\left\{2B(1-\rho^2)-\eta\lambda_i\left[B(1-\rho)+(1+\rho)\right]\right\}},$$

and consequently

$$\sum_{m\geqslant 0}K_m=\mathcal{R}_{\text{polyak}}(\eta,\rho).$$

Suppose first that $\mathcal{R}_{\text{polyak}}(\eta,\rho)<1$. Summing (34) and applying Tonelli's theorem gives

$$\sum_{k\geqslant 0}S_k\leqslant\frac{\sum_{k\geqslant 0}h_k}{1-\mathcal{R}_{\text{polyak}}(\eta,\rho)}<\infty.$$

Taking the first component of the iterated block recursion and summing over i yields

$$\sum_{i\geqslant 1}u_{i,k}=H_k+\frac{\eta^2}{B}\sum_{s=0}^{k-1}G_{k-1-s}S_s,$$

where

$$H_k:=\sum_{i\geqslant 1}\mathbf{b}^\top\mathbf{M}_i^k\mathbf{X}_{i,0},\qquad G_m:=\sum_{i\geqslant 1}\lambda_i g_{i,m}.$$

For every fixed i, $\mathbf{b}^\top\mathbf{M}_i^k\mathbf{X}_{i,0}\to 0$. The uniform power bound and

$$\sum_{i\geqslant 1}(u_{i,0}+r_{i,0})<\infty$$

allow us to apply dominated convergence, giving $H_k\to 0$. Similarly,

$$G_m\longrightarrow 0,\qquad\sup_{m\geqslant 0}G_m\leqslant C\mathrm{Tr}(\mathbf{H})<\infty.$$

Since $(S_k)_{k\geqslant 0}\in\ell^1$, the convolution term also converges to zero. Therefore,

$$\mathrm{Tr}(\mathbf{U}_k)=\sum_{i\geqslant 1}u_{i,k}\longrightarrow 0.$$

Since $\mathbf{R}_{k+1}=\mathbf{U}_k$,

$$\mathbb{E}\|\mathbf{z}_k\|^2=\mathrm{Tr}(\mathbf{U}_k)+\mathrm{Tr}(\mathbf{R}_k)\longrightarrow 0.$$

This proves sufficiency. We next prove necessity. If (29) fails, at least one local block $\mathbf{M}_i$ is not asymptotically stable. The isolated moment map preserves the cone of positive semidefinite covariance matrices, so there exists a deterministic augmented initialization supported on this eigendirection whose second moment does not converge to zero. Hence the local condition is necessary. Now suppose that the local condition holds but $\mathcal{R}_{\text{polyak}}(\eta,\rho)>1$. For $\mu>1$, define

$$F(\mu):=\frac{\eta^2}{B}\sum_{i\geqslant 1}\lambda_i^2\mathbf{b}^\top(\mu\mathbf{I}-\mathbf{M}_i)^{-1}\mathbf{b}.$$

The Neumann-series representation shows that F is continuous and strictly decreasing, with

$$\lim_{\mu\downarrow 1}F(\mu)=\mathcal{R}_{\text{polyak}}(\eta,\rho)>1,\qquad\lim_{\mu\to\infty}F(\mu)=0.$$

Hence there exists $\mu>1$ such that $F(\mu)=1$. Define

$$\mathbf{W}_i:=\lambda_i(\mu\mathbf{I}-\mathbf{M}_i^*)^{-1}\mathbf{b}.$$

Then $\mathbf{W} = (\mathbf{W}_i)_{i\geqslant 1}$ is an eigenvector of the adjoint global moment operator with eigenvalue μ. Moreover, for every positive covariance state $\mathbf{X}_i$,

$$\langle \mathbf{W}_i, \mathbf{X}_i \rangle = \lambda_i \sum_{m\geqslant 0} \mu^{-m-1} \mathbf{b}^\top \mathbf{M}_i^m \mathbf{X}_i \geqslant 0.$$

Choose a deterministic initialization such that $\langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$. Then

$$\langle \mathbf{W}, \mathbf{X}_k \rangle = \mu^k \langle \mathbf{W}, \mathbf{X}_0 \rangle,$$

which cannot converge to zero. Hence the Polyak dynamics are not strongly mean-square asymptotically stable. It remains to consider $\mathcal{R}_{\text{polyak}}(\eta, \rho) = 1$. Define

$$\mathbf{W}_i := \lambda_i (\mathbf{I} - \mathbf{M}_i^*)^{-1} \mathbf{b}.$$

The local condition and (35) imply that $\mathbf{W}$ is bounded, while $\mathcal{R}_{\text{polyak}}(\eta, \rho) = 1$ gives

$$\mathbf{M}_{\text{polyak}}^* \mathbf{W} = \mathbf{W},$$

where $\mathbf{M}_{\text{polyak}}$ denotes the global diagonal moment operator. Choosing an initialization with $\langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$ gives

$$\langle \mathbf{W}, \mathbf{X}_k \rangle = \langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$$

for every k, contradicting convergence of the augmented covariance to zero. Hence the critical boundary is not strongly mean-square asymptotically stable. □

Corollary B.5 (Critical learning rate of Polyak)**.** *There exists a unique critical learning rate* $\eta_{\text{polyak}}^{\text{crit}}$ *satisfying*

$$0 < \eta_{\text{polyak}}^{\text{crit}} < \frac{2B(1-\rho^2)}{\lambda_{\max}\left[B(1-\rho) + (1+\rho)\right]}$$

and

$$\sum_{i\geqslant 1} \frac{\eta_{\text{polyak}}^{\text{crit}} \lambda_i (1+\rho)}{2B(1-\rho^2) - \eta_{\text{polyak}}^{\text{crit}} \lambda_i \left[B(1-\rho) + (1+\rho)\right]} = 1. \tag{36}$$

The homogeneous Polyak dynamics are mean-square stable if and only if

$$0 < \eta < \eta_{\text{polyak}}^{\text{crit}}.$$

Moreover,

$$\frac{2B(1-\rho^2)}{\lambda_{\max}\left[B(1-\rho) + (1+\rho)\right] + (1+\rho)\text{Tr}(\mathbf{H})} \leqslant \eta_{\text{polyak}}^{\text{crit}} \leqslant \min\left\{ \frac{2B(1-\rho^2)}{\lambda_{\max}\left[B(1-\rho) + (1+\rho)\right]}, \frac{2B(1-\rho)}{\text{Tr}(\mathbf{H})} \right\}. \tag{37}$$

Consequently, under $\lambda_i \eqsim i^{-\beta}$ *with fixed* $\beta > 1$*, in the scaling regime* $B \gg 1$ *and* $1 - \rho \ll 1$*,*

$$\eta_{\text{polyak}}^{\text{crit}} \eqsim \min\{1, B(1-\rho)\}.$$

Proof. The function $\mathcal{R}_{\text{polyak}}(\eta, \rho)$ is continuous and strictly increasing on the local stability interval. It vanishes at $\eta = 0$ and diverges as η approaches the local stability boundary. Hence the critical learning rate exists and is unique. At the critical learning rate, bounding each denominator in (36) from above by $2B(1-\rho^2)$ gives

$$\eta_{\text{polyak}}^{\text{crit}} \leqslant \frac{2B(1-\rho)}{\text{Tr}(\mathbf{H})}.$$

The local condition gives the other upper bound in (37). Conversely, bounding each denominator from below by

$$2B(1-\rho^2) - \eta_{\text{polyak}}^{\text{crit}} \lambda_{\max}\left[B(1-\rho) + (1+\rho)\right]$$

gives the lower bound in (37). Finally,

$$\lambda_{\max} \eqsim 1, \qquad \mathrm{Tr}(\mathbf{H}) \eqsim 1, \qquad 1-\rho^2 \eqsim 1-\rho.$$

Both sides of (37) therefore scale as

$$\frac{B(1-\rho)}{B(1-\rho)+1} \eqsim \min\{1, B(1-\rho)\}.$$

□

B.3 Stability of Nesterov

Define the look-ahead error

$$\mathbf{e}_k^{\mathrm{la}} := (1+\rho)\mathbf{e}_k - \rho\mathbf{e}_{k-1}.$$

With $\sigma = 0$, the Nesterov error recursion is

$$\mathbf{e}_{k+1} = (\mathbf{I} - \eta\widehat{\mathbf{H}}_k)\mathbf{e}_k^{\mathrm{la}}. \tag{38}$$

Writing

$$\widehat{\mathbf{H}}_k = \mathbf{H} + \mathbf{\Sigma}_k, \qquad \mathbf{D} = \mathbf{I} - \eta\mathbf{H},$$

the recursion becomes

$$\mathbf{e}_{k+1} = \mathbf{D}\mathbf{e}_k^{\mathrm{la}} - \eta\mathbf{\Sigma}_k\mathbf{e}_k^{\mathrm{la}}.$$

Theorem B.6 (Mean-square stability of Nesterov)**.** *Let* $\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \dots)$ *be a positive trace-class operator, where* $\lambda_1 = \lambda_{\max} \geqslant \lambda_2 \geqslant \cdots > 0$*, and let* $0 \leqslant \rho < 1$*. Define*

$$\Delta_{\mathrm{nesterov}}(x) := 2B(1-\rho^2) + \left[4B\rho^2 + (B-1)\rho - (B+1)\right]x - (B+1)\rho(1+2\rho)x^2.$$

Then the homogeneous Nesterov recursion (38) *is mean-square stable if and only if*

$$\Delta_{\mathrm{nesterov}}(\eta\lambda_{\max}) > 0 \tag{39}$$

and

$$\mathcal{R}_{\mathrm{nesterov}}(\eta, \rho) := \sum_{i\geqslant 1} \frac{\eta\lambda_i\left[1+\rho+\rho(1+2\rho)\eta\lambda_i\right]}{\Delta_{\mathrm{nesterov}}(\eta\lambda_i)} < 1. \tag{40}$$

Proof. Define

$$\mathbf{U}_k := \mathbb{E}[\mathbf{e}_k \otimes \mathbf{e}_k], \qquad \mathbf{V}_k := \mathbb{E}[\mathbf{e}_k^{\mathrm{la}} \otimes \mathbf{e}_k^{\mathrm{la}}], \qquad \mathbf{P}_k := \mathbb{E}[\mathbf{e}_k \otimes \mathbf{e}_{k-1}],$$

$$\mathbf{Q}_k := \mathbb{E}[\mathbf{e}_{k-1} \otimes \mathbf{e}_k], \qquad \mathbf{R}_k := \mathbb{E}[\mathbf{e}_{k-1} \otimes \mathbf{e}_{k-1}].$$

Because the mini-batch at iteration k is independent of the past and $\mathbb{E}[\mathbf{\Sigma}_k] = 0$, expanding (38) and applying Lemma A.5 gives

$$\begin{cases} \mathbf{U}_{k+1} = \mathbf{D}\mathbf{V}_k\mathbf{D} + \dfrac{\eta^2}{B}\left(\mathbf{H}\mathbf{V}_k\mathbf{H} + \mathbf{H}\mathrm{Tr}(\mathbf{H}\mathbf{V}_k)\right), \\ \quad \mathbf{V}_k = (1+\rho)^2\mathbf{U}_k - \rho(1+\rho)(\mathbf{P}_k + \mathbf{Q}_k) + \rho^2\mathbf{R}_k, \\ \mathbf{P}_{k+1} = \mathbf{D}\big((1+\rho)\mathbf{U}_k - \rho\mathbf{Q}_k\big), \\ \mathbf{Q}_{k+1} = \big((1+\rho)\mathbf{U}_k - \rho\mathbf{P}_k\big)\mathbf{D}, \\ \mathbf{R}_{k+1} = \mathbf{U}_k. \end{cases} \tag{41}$$

Since

$$\mathbb{E}\|\mathbf{z}_k\|^2 = \mathrm{Tr}(\mathbf{U}_k) + \mathrm{Tr}(\mathbf{R}_k),$$

strong mean-square stability is equivalent to

$$\mathrm{Tr}(\mathbf{U}_k) + \mathrm{Tr}(\mathbf{R}_k) \longrightarrow 0.$$

Because $\mathbf{R}_{k+1} = \mathbf{U}_k$, it is sufficient and necessary to prove $\mathrm{Tr}(\mathbf{U}_k) \to 0$. Moreover, since $\mathbf{H}$ and $\mathbf{D}$ are diagonal, the diagonal entries of (41) form a closed recursion. Let

$$u_{i,k} := \langle \mathbf{e}_i, \mathbf{U}_k \mathbf{e}_i \rangle, \qquad p_{i,k} := \langle \mathbf{e}_i, \mathbf{P}_k \mathbf{e}_i \rangle, \qquad r_{i,k} := \langle \mathbf{e}_i, \mathbf{R}_k \mathbf{e}_i \rangle,$$

and

$$v_{i,k} := \langle \mathbf{e}_i, \mathbf{V}_k \mathbf{e}_i \rangle.$$

Since $\mathbf{Q}_k = \mathbf{P}_k^*$,

$$v_{i,k} = (1+\rho)^2 u_{i,k} - 2\rho(1+\rho) p_{i,k} + \rho^2 r_{i,k}.$$

Set

$$S_k := \mathrm{Tr}(\mathbf{H}\mathbf{V}_k) = \sum_{i \geqslant 1} \lambda_i v_{i,k}, \qquad d_i := 1 - \eta\lambda_i, \qquad a_i := d_i^2 + \frac{\eta^2}{B}\lambda_i^2.$$

Taking the diagonal entries of (41) gives

$$\begin{cases} u_{i,k+1} = a_i \left[(1+\rho)^2 u_{i,k} - 2\rho(1+\rho) p_{i,k} + \rho^2 r_{i,k} \right] + \dfrac{\eta^2}{B} \lambda_i S_k, \\ p_{i,k+1} = d_i \left[(1+\rho) u_{i,k} - \rho p_{i,k} \right], \\ r_{i,k+1} = u_{i,k}. \end{cases} \tag{42}$$

Define

$$\mathbf{X}_{i,k} := (u_{i,k}, p_{i,k}, r_{i,k})^\top, \qquad \mathbf{b} := (1,0,0)^\top, \qquad \mathbf{c} := \left((1+\rho)^2, -2\rho(1+\rho), \rho^2\right)^\top,$$

and

$$\mathbf{M}_i := \begin{pmatrix} a_i(1+\rho)^2 & -2a_i\rho(1+\rho) & a_i\rho^2 \\ d_i(1+\rho) & -d_i\rho & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Then

$$v_{i,k} = \mathbf{c}^\top \mathbf{X}_{i,k}, \qquad S_k = \sum_{i \geqslant 1} \lambda_i \mathbf{c}^\top \mathbf{X}_{i,k},$$

and (42) becomes

$$\mathbf{X}_{i,k+1} = \mathbf{M}_i \mathbf{X}_{i,k} + \frac{\eta^2}{B} \lambda_i \mathbf{b} S_k. \tag{43}$$

We first determine the local stability condition. The characteristic polynomial of the isolated block $\mathbf{M}_i$ is

$$p_i(\mu) = \mu^3 - \left[a_i(1+\rho)^2 - d_i\rho\right] \mu^2 + a_i\rho \left[d_i(1+\rho)^2 - \rho\right] \mu - a_i d_i \rho^3.$$

The Jury stability criterion shows that all roots lie strictly inside the unit disk if and only if

$$\Delta_{\mathrm{nesterov}}(\eta\lambda_i) > 0.$$

Since $\Delta_{\mathrm{nesterov}}$ is a concave quadratic with

$$\Delta_{\mathrm{nesterov}}(0) = 2B(1-\rho^2) > 0$$

and exactly one positive root, requiring this condition for every i is equivalent to (39). We now assume that the local condition holds. Iterating (43) gives

$$\mathbf{X}_{i,k} = \mathbf{M}_i^k \mathbf{X}_{i,0} + \frac{\eta^2}{B} \lambda_i \sum_{s=0}^{k-1} \mathbf{M}_i^{k-1-s} \mathbf{b} S_s.$$

Multiplying by $\lambda_i \mathbf{c}^\top$ and summing over $i \geqslant 1$, we obtain

$$S_k = h_k + \sum_{s=0}^{k-1} K_{k-1-s} S_s, \tag{44}$$

where

$$h_k := \sum_{i \geqslant 1} \lambda_i \mathbf{c}^\top \mathbf{M}_i^k \mathbf{X}_{i,0}, \qquad K_m := \frac{\eta^2}{B} \sum_{i \geqslant 1} \lambda_i^2 \mathbf{c}^\top \mathbf{M}_i^m \mathbf{b}.$$

Both sequences are nonnegative: $\mathbf{c}^\top \mathbf{M}_i^k \mathbf{X}_{i,0}$ is the look-ahead variance generated by the isolated i-th coordinate dynamics, while $\mathbf{c}^\top \mathbf{M}_i^m \mathbf{b}$ is the look-ahead variance generated from the covariance state $\mathbf{b}$. We next show that $(h_k)_{k \geqslant 0}$ is summable. Since every local block is stable,

$$\sum_{k \geqslant 0} h_k = \sum_{i \geqslant 1} \lambda_i \mathbf{c}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{X}_{i,0}.$$

A direct calculation gives

$$\mathbf{c}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} = \frac{B\big(1 + \rho + \rho(1+2\rho)\eta\lambda_i,\ -2\rho(1+\rho),\ \rho^2[1 + \rho(1 - \eta\lambda_i)]\big)}{\eta\lambda_i \Delta_{\text{nesterov}}(\eta\lambda_i)}. \tag{45}$$

The initial coordinate covariance satisfies

$$\begin{pmatrix} u_{i,0} & p_{i,0} \\ p_{i,0} & r_{i,0} \end{pmatrix} \succeq 0,$$

and therefore

$$|p_{i,0}| \leqslant \frac{u_{i,0} + r_{i,0}}{2}.$$

Since Δ_{nesterov} is positive on $[0, \eta\lambda_{\max}]$, there exists $\delta > 0$ such that

$$\Delta_{\text{nesterov}}(\eta\lambda_i) \geqslant \delta \qquad \text{for every } i.$$

All coefficients in the numerator of (45) are uniformly bounded. Hence

$$\lambda_i \mathbf{c}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{X}_{i,0} \leqslant C(u_{i,0} + r_{i,0})$$

for some $C < \infty$ independent of i. Therefore,

$$\sum_{k \geqslant 0} h_k \leqslant C \sum_{i \geqslant 1} (u_{i,0} + r_{i,0}) < \infty.$$

By Tonelli's theorem,

$$\sum_{m \geqslant 0} K_m = \frac{\eta^2}{B} \sum_{i \geqslant 1} \lambda_i^2 \mathbf{c}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{b}.$$

The first entry of (45) gives

$$\mathbf{c}^\top (\mathbf{I} - \mathbf{M}_i)^{-1} \mathbf{b} = \frac{B\,[1 + \rho + \rho(1+2\rho)\eta\lambda_i]}{\eta\lambda_i \Delta_{\text{nesterov}}(\eta\lambda_i)},$$

and consequently

$$\sum_{m \geqslant 0} K_m = \mathcal{R}_{\text{nesterov}}(\eta, \rho).$$

Suppose first that $\mathcal{R}_{\text{nesterov}}(\eta, \rho) < 1$. Summing (44) and applying Tonelli's theorem gives

$$\sum_{k \geqslant 0} S_k \leqslant \frac{\sum_{k \geqslant 0} h_k}{1 - \mathcal{R}_{\text{nesterov}}(\eta, \rho)} < \infty.$$

Taking the first component of the iterated block recursion and summing over i gives

$$\sum_{i\geqslant 1} u_{i,k} = H_k + \frac{\eta^2}{B}\sum_{s=0}^{k-1} G_{k-1-s} S_s,$$

where

$$H_k := \sum_{i\geqslant 1} \mathbf{b}^\top \mathbf{M}_i^k \mathbf{X}_{i,0}, \qquad G_m := \sum_{i\geqslant 1} \lambda_i \mathbf{b}^\top \mathbf{M}_i^m \mathbf{b}.$$

For every fixed i, the local condition gives

$$\mathbf{b}^\top \mathbf{M}_i^k \mathbf{X}_{i,0} \longrightarrow 0, \qquad \mathbf{b}^\top \mathbf{M}_i^m \mathbf{b} \longrightarrow 0.$$

The family of local blocks is uniformly power bounded on

$$0 \leqslant \eta\lambda_i \leqslant \eta\lambda_{\max},$$

so dominated convergence gives

$$H_k \longrightarrow 0, \qquad G_m \longrightarrow 0, \qquad \sup_{m\geqslant 0} G_m < \infty.$$

Since $(S_k)_{k\geqslant 0} \in \ell^1$, the convolution term converges to zero. Therefore,

$$\mathrm{Tr}(\mathbf{U}_k) = \sum_{i\geqslant 1} u_{i,k} \longrightarrow 0.$$

Because $\mathbf{R}_{k+1} = \mathbf{U}_k$,

$$\mathbb{E}\|\mathbf{z}_k\|^2 \longrightarrow 0.$$

This proves sufficiency. We next prove necessity. If (39) fails, then at least one local block $\mathbf{M}_i$ is not asymptotically stable. Since the isolated moment map preserves the cone of positive semidefinite covariance matrices, there exists a positive semidefinite initial covariance whose second moment does not converge to zero. Decomposing this covariance into rank-one matrices shows that at least one deterministic augmented initialization supported on this eigendirection is not strongly mean-square asymptotically stable. Hence the local condition is necessary. Now assume that the local condition holds but $\mathcal{R}_{\mathrm{nesterov}}(\eta,\rho) > 1$. For $\mu > 1$, define

$$F(\mu) := \frac{\eta^2}{B}\sum_{i\geqslant 1} \lambda_i^2 \mathbf{c}^\top (\mu\mathbf{I} - \mathbf{M}_i)^{-1}\mathbf{b}.$$

Using

$$(\mu\mathbf{I} - \mathbf{M}_i)^{-1} = \sum_{m\geqslant 0} \mu^{-m-1}\mathbf{M}_i^m,$$

we see that F is continuous and strictly decreasing, with

$$\lim_{\mu\downarrow 1} F(\mu) = \mathcal{R}_{\mathrm{nesterov}}(\eta,\rho) > 1, \qquad \lim_{\mu\to\infty} F(\mu) = 0.$$

Hence there exists $\mu > 1$ such that $F(\mu) = 1$. For each i, define

$$\mathbf{W}_i := \lambda_i(\mu\mathbf{I} - \mathbf{M}_i^*)^{-1}\mathbf{c}.$$

Then $\mathbf{W} = (\mathbf{W}_i)_{i\geqslant 1}$ is an eigenvector of the adjoint global moment operator with eigenvalue μ. Moreover, for every positive semidefinite covariance state $\mathbf{X}_i$,

$$\langle \mathbf{W}_i, \mathbf{X}_i\rangle = \lambda_i \sum_{m\geqslant 0} \mu^{-m-1}\mathbf{c}^\top \mathbf{M}_i^m \mathbf{X}_i \geqslant 0.$$

Choosing a deterministic initialization such that $\langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$ gives

$$\langle \mathbf{W}, \mathbf{X}_k \rangle = \mu^k \langle \mathbf{W}, \mathbf{X}_0 \rangle,$$

which cannot converge to zero. Hence the dynamics are not strongly mean-square asymptotically stable. It remains to consider $\mathcal{R}_{\text{nesterov}}(\eta, \rho) = 1$. Define

$$\mathbf{W}_i := \lambda_i (\mathbf{I} - \mathbf{M}_i^*)^{-1} \mathbf{c}.$$

The local condition and (45) imply that $\mathbf{W}$ is bounded, while $\mathcal{R}_{\text{nesterov}}(\eta, \rho) = 1$ gives

$$\mathbf{M}_{\text{nesterov}}^* \mathbf{W} = \mathbf{W},$$

where $\mathbf{M}_{\text{nesterov}}$ denotes the global diagonal moment operator. Choosing an initialization with $\langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$ gives

$$\langle \mathbf{W}, \mathbf{X}_k \rangle = \langle \mathbf{W}, \mathbf{X}_0 \rangle > 0$$

for every k, contradicting convergence of the augmented covariance to zero. Hence the critical boundary is not strongly mean-square asymptotically stable. □

Corollary B.7 (Critical learning rate of Nesterov)**.** *There exists a unique critical learning rate* $\eta_{\text{nesterov}}^{\text{crit}}$ *satisfying*

$$\mathcal{R}_{\text{nesterov}}\big(\eta_{\text{nesterov}}^{\text{crit}}, \rho\big) = 1$$

before the local stability boundary. The homogeneous Nesterov dynamics are mean-square stable if and only if

$$0 < \eta < \eta_{\text{nesterov}}^{\text{crit}}.$$

Under $\lambda_i \eqsim i^{-\beta}$ *with fixed* $\beta > 1$*, in the scaling regime* $B \gg 1$ *and* $1 - \rho \ll 1$*,*

$$\eta_{\text{nesterov}}^{\text{crit}} \eqsim \min\{1, B^{\beta}(1-\rho)\}. \tag{46}$$

Proof. Since Δ_{nesterov} is a concave quadratic with $\Delta_{\text{nesterov}}(0) > 0$, the local condition

$$\Delta_{\text{nesterov}}(\eta \lambda_{\max}) > 0$$

is equivalent to

$$\eta \lambda_{\max} < x_+(B, \rho),$$

where $x_+(B, \rho)$ is the unique positive root of Δ_{nesterov}. In the regime $B \gg 1$ and $\rho \to 1^-$,

$$x_+(B, \rho) \eqsim 1.$$

Indeed,

$$x_+(B, 1) = \frac{4B - 2}{3(B+1)} \longrightarrow \frac{4}{3}.$$

Since $\lambda_{\max} \eqsim 1$, the local condition gives

$$\eta \lesssim 1.$$

On every fixed subinterval strictly inside the local stability region,

$$\mathcal{R}_{\text{nesterov}}(\eta, \rho) \eqsim \frac{1}{B} \sum_{i \geqslant 1} \frac{\eta \lambda_i}{1 - \rho + \eta \lambda_i}.$$

Under $\lambda_i \eqsim i^{-\beta}$,

$$\sum_{i \geqslant 1} \frac{\eta \lambda_i}{1 - \rho + \eta \lambda_i} \eqsim \begin{cases} \dfrac{\eta}{1-\rho}, & \eta \lesssim 1 - \rho, \\ \left(\dfrac{\eta}{1-\rho}\right)^{1/\beta}, & \eta \gtrsim 1 - \rho. \end{cases}$$

In the first regime, the critical equation $\mathcal{R}_{\text{nesterov}}(\eta, \rho) \eqsim 1$ would require

$$\frac{\eta}{1-\rho} \eqsim B,$$

which contradicts

$$\frac{\eta}{1-\rho} \lesssim 1$$

when $B \gg 1$. Hence the largest stable learning rate lies in the second regime, where the critical equation becomes

$$\left(\frac{\eta}{1-\rho}\right)^{1/\beta} \eqsim B.$$

Therefore,

$$\eta \eqsim B^{\beta}(1-\rho).$$

Combining this global condition with the local condition $\eta \lesssim 1$ gives

$$\eta^{\text{crit}}_{\text{nesterov}} \eqsim \min\{1, B^{\beta}(1-\rho)\}.$$

□

C SGD scaling law

We provide the complete proof of the SGD scaling law. Following Appendix D, we decompose the parameter error into deterministic signal and stochastic noise contributions, then bound the deterministic bias, convolution kernel, label-noise floor, and multiplicative noise over distinct temporal stages.

C.1 Signal–noise decomposition

Let

$$\mathbf{e}_k := \boldsymbol{\theta}_k - \boldsymbol{\theta}^*$$

denote the parameter error. Using the gradient decomposition

$$\mathbf{g}(\boldsymbol{\theta}_k; \mathcal{B}_k) = \mathbf{H}\mathbf{e}_k + \boldsymbol{\xi}_k(\boldsymbol{\theta}_k),$$

we can subtract $\boldsymbol{\theta}^*$ from both sides of the SGD update rule (2) to obtain the error recursion:

$$\mathbf{e}_{k+1} = (\mathbf{I} - \eta\mathbf{H})\mathbf{e}_k - \eta\boldsymbol{\xi}_k(\boldsymbol{\theta}_k).$$

In the eigenbasis of $\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$, the j-th coordinate $e_k(j)$ satisfies:

$$e_{k+1}(j) = (1 - \eta\lambda_j)e_k(j) - \eta\xi_k(j),$$

where $\xi_k(j)$ is the j-th coordinate of $\boldsymbol{\xi}_k(\boldsymbol{\theta}_k)$.

The discrete Green function is:

$$G_k(j) := (1 - \eta\lambda_j)^k, \qquad k \geqslant 0.$$

Unrolling from $e_0(j) = -\theta_j^*$ yields the exact impulse-response representation:

$$e_k(j) = e_k^{\mathrm{det}}(j) - \eta \sum_{m=0}^{k-1} G_{k-1-m}(j)\xi_m(j), \tag{47}$$

where $e_k^{\mathrm{det}}(j) = -(1 - \eta\lambda_j)^k\theta_j^*$ is the unperturbed deterministic trajectory. We define the deterministic bias and convolution kernel as:

$$S_{\mathrm{sgd}}(k) := \frac{1}{2}\sum_{j\geqslant 1} \lambda_j\, e_k^{\mathrm{det}}(j)^2, \qquad \mathcal{K}_{\mathrm{sgd}}(k) := \sum_{j\geqslant 1} \lambda_j^2\, G_k(j)^2.$$

Proposition C.1 (Risk recursion for SGD)**.** *Under the gradient-noise covariance assumption, the expected excess risk obeys the two-sided recursion:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\mathrm{sgd}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau) + \frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})]. \tag{48}$$

Proof. Substituting (47) into the excess risk, since the gradient noise is conditionally mean-zero and independent across time, cross-terms vanish:

$$\mathbb{E}[e_k(j)^2] = e_k^{\mathrm{det}}(j)^2 + \eta^2\sum_{m=0}^{k-1} G_{k-1-m}(j)^2\mathbb{E}[\xi_m(j)^2].$$

Using Proposition 3.4 for the variance $\mathbb{E}[\xi_m(j)^2] \eqsim \frac{1}{B}\lambda_j(\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] + \sigma^2)$, multiplying by $\frac{1}{2}\lambda_j$, and summing over j yields the stated recursion. □

C.2 Deterministic risk calculation

Proposition C.2 (Deterministic bias of SGD)**.** *Under Assumptions 3.1 and 3.3, assuming the local stability condition $\eta\lambda_1 \leqslant c_0$ for a sufficiently small constant $c_0 \in (0,1)$, we have:*

$$S_{\mathrm{sgd}}(k) \eqsim \begin{cases} 1, & k \lesssim \frac{1}{\eta}, \\ (\eta k)^{-s}, & k \gtrsim \frac{1}{\eta}. \end{cases}$$

Proof. By definition, $S_{\mathrm{sgd}}(k) = \frac{1}{2}\sum_{j\geqslant 1} \lambda_j(\theta_j^*)^2(1-\eta\lambda_j)^{2k}$. Using $\lambda_j(\theta_j^*)^2 \eqsim j^{-1-\beta s}$ and $\lambda_j \eqsim j^{-\beta}$, we evaluate the sum in two stages.

Stage 1: $k \lesssim \frac{1}{\eta}$.
For $j = \mathcal{O}(1)$, we have $\eta\lambda_j k \lesssim 1$. Thus,

$$(1-\eta\lambda_j)^{2k} \geqslant (1-\eta\lambda_1)^{2k} \eqsim \exp(-2\eta\lambda_1 k) \eqsim 1.$$

Since $(1-\eta\lambda_j)^{2k} \leqslant 1$, we have $(1-\eta\lambda_j)^{2k} \eqsim 1$. The bias is dominated by these modes:

$$S_{\mathrm{sgd}}(k) \eqsim \sum_{j\geqslant 1} j^{-1-\beta s} \eqsim 1,$$

which converges since $\beta s > 0$.

Stage 2: $k \gtrsim \frac{1}{\eta}$.
Using $(1-\eta\lambda_j)^{2k} \eqsim \exp(-2\eta k j^{-\beta})$:

$$S_{\mathrm{sgd}}(k) \eqsim \sum_{j\geqslant 1} j^{-1-\beta s}\exp(-2\eta k j^{-\beta}).$$

For the lower bound, restricting to $j \geqslant (\eta k)^{1/\beta}$ gives $\eta k j^{-\beta} \leqslant 1$, yielding:

$$\sum_{j\geqslant(\eta k)^{1/\beta}} j^{-1-\beta s} \eqsim (\eta k)^{-s}.$$

For the upper bound, letting $u = 2\eta k x^{-\beta}$:

$$\int_1^\infty x^{-1-\beta s}\exp(-2\eta k x^{-\beta})dx = \frac{1}{\beta}(2\eta k)^{-s}\int_0^{2\eta k} u^{s-1}e^{-u}du \lesssim (\eta k)^{-s}.$$

Thus, $S_{\mathrm{sgd}}(k) \eqsim (\eta k)^{-s}$. □

C.3 Uniform boundedness and total risk

As long as the additive label noise is strictly positive ($\sigma > 0$), we can prove that the expected risk is uniformly bounded, which allows the multiplicative gradient noise to be completely absorbed by the additive label noise floor.

Lemma C.3 (Label-noise contribution)**.** *For $k \gtrsim \frac{1}{\eta}$, the accumulated label-noise contribution establishes the additive noise floor:*

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau) \eqsim \frac{\eta\sigma^2}{B}.$$

Proof. The saturated sum for each coordinate is:

$$\sum_{\tau=0}^{\infty} G_\tau(j)^2 = \frac{1}{1-(1-\eta\lambda_j)^2} = \frac{1}{2\eta\lambda_j - \eta^2\lambda_j^2} \eqsim \frac{1}{\eta\lambda_j},$$

assuming the stability condition $\eta\lambda_j \ll 1$. Summing over all coordinates gives:

$$\frac{\eta^2\sigma^2}{B}\sum_{j\geqslant 1}\lambda_j^2\sum_{\tau=0}^{\infty}G_\tau(j)^2 \eqsim \frac{\eta^2\sigma^2}{B}\sum_{j\geqslant 1}\frac{\lambda_j^2}{\eta\lambda_j} = \frac{\eta\sigma^2}{B}\sum_{j\geqslant 1}\lambda_j \eqsim \frac{\eta\sigma^2}{B},$$

since $\sum\lambda_j \eqsim 1$ for $\beta > 1$. For $k \gtrsim 1/\eta$, the finite sum captures a constant fraction of this infinite series, yielding $\frac{\eta\sigma^2}{B}$. □

Lemma C.4 (Uniform boundedness of expected risk)**.** *Suppose $\sigma > 0$. Under the stability condition $\eta \leqslant c$ for a sufficiently small universal constant $c > 0$, the expected excess risk of SGD is uniformly bounded for all $k \geqslant 0$:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M,$$

where $M > 0$ is a constant independent of k, η, and B.

Proof. We proceed by induction on k. By Proposition C.1, there exist universal constants $C_1, C_2, C_3 > 0$ such that for any $k \geqslant 1$:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant C_1 S_{\mathrm{sgd}}(k) + C_2\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau) + C_3\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})]. \tag{49}$$

First, by Proposition C.2, the deterministic bias monotonically decays and is uniformly bounded by its initialization: $S_{\mathrm{sgd}}(k) \leqslant \mathcal{E}(\boldsymbol{\theta}_0) =: M_1$.

Second, as shown in the derivation of Lemma C.3, the infinite sum of the convolution kernel is bounded by $\sum_{\tau=0}^{\infty}\mathcal{K}_{\mathrm{sgd}}(\tau) \leqslant \frac{C_K}{\eta}$. Therefore, the label noise contribution is universally bounded:

$$C_2\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau) \leqslant C_2\sigma^2\frac{C_K\eta}{B} \leqslant C_2\sigma^2 C_K c =: M_2,$$

where we used the stability condition $\eta \leqslant c$ and the fact that $B \geqslant 1 \implies \eta/B \leqslant c$.

Assuming inductively $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] \leqslant M$ for $m < k$, the multiplicative noise term yields:

$$C_3\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})] \leqslant C_3\frac{\eta^2}{B}\left(\frac{C_K}{\eta}\right)M = C_3C_K\frac{\eta}{B}M \leqslant C_3C_K cM.$$

Choosing $c \leqslant \frac{1}{2C_3C_K}$ limits this term to $M/2$. Substituting into (49) yields:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M_1 + M_2 + \frac{M}{2}.$$

Setting $M = 2(M_1 + M_2)$ guarantees $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M$. The base case $k = 0$ holds trivially since $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_0)] = S_{\mathrm{sgd}}(0) \leqslant M_1 \leqslant M$. This completes the induction. □

Corollary C.5 (Absorption of multiplicative noise)**.** *For $\sigma > 0$, under the stability condition $\eta \lesssim 1$, the multiplicative gradient noise is strictly bounded by the additive label noise up to a constant factor:*

$$\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})] \leqslant \frac{M}{\sigma^2}\left(\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau)\right).$$

Consequently, the expected excess risk is asymptotically equivalent to the sum of the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\mathrm{sgd}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau).$$

Theorem C.6 (SGD scaling law)**.** *Assume that the primary structural stability condition* $\eta \lesssim 1$ *holds for normal SGD execution. For temporal parameter mapping extending across long-timescale horizons where* $k \gtrsim 1/\eta$*, we rigorously establish:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \underbrace{(\eta k)^{-s}}_{\textit{signal}} + \underbrace{\frac{\eta\sigma^2}{B}}_{\textit{additive noise}} . \tag{50}$$

Proof. By Corollary C.5, the expected risk is completely determined by the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\mathrm{sgd}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau).$$

For $k \gtrsim \frac{1}{\eta}$, we substitute the deterministic bias from Proposition C.2:

$$S_{\mathrm{sgd}}(k) \eqsim (\eta k)^{-s}.$$

We substitute the label noise floor from Lemma C.3:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\mathrm{sgd}}(\tau) \eqsim \frac{\eta\sigma^2}{B}.$$

Summing these two terms directly yields the exact scaling law. □

D Polyak scaling law

In this section, we prove Theorem 5.1. We first decompose the loss into signal and noise contributions, then classify the spectral modes into overdamped and underdamped modes. Finally, we analyze the loss separately in the fully overdamped regime and in the mixed overdamped–underdamped regime.

D.1 Signal–noise decomposition and risk recursion

To analyze the Polyak scaling law, we first decompose the parameter error into a deterministic signal trajectory and a stochastic noise accumulation term. Recall the parameter error

$$\mathbf{e}_k := \boldsymbol{\theta}_k - \boldsymbol{\theta}^*.$$

Substituting the gradient decomposition into the Polyak update (3), we obtain the error recursion:

$$\mathbf{e}_{k+1} = \big((1+\rho)\mathbf{I} - \eta\mathbf{H}\big)\mathbf{e}_k - \rho\mathbf{e}_{k-1} - \eta\boldsymbol{\xi}_k(\boldsymbol{\theta}_k).$$

In the diagonal eigenbasis of the population Hessian $\mathbf{H} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$, the j-th coordinate of the parameter error, denoted as $e_k(j)$, obeys the following second-order scalar difference equation:

$$e_{k+1}(j) = (1+\rho-\eta\lambda_j)e_k(j) - \rho e_{k-1}(j) - \eta\xi_k(j),$$

where $\xi_k(j)$ is the j-th coordinate of the stochastic gradient noise vector $\boldsymbol{\xi}_k(\boldsymbol{\theta}_k)$.

The homogeneous characteristic polynomial associated with the deterministic part of this coordinate-wise recursion is

$$y^2 - (1+\rho-\eta\lambda_j)y + \rho = 0.$$

For each coordinate j, let $y_{j,+}$ and $y_{j,-}$ denote the two roots of this polynomial.

We define the discrete Green function (impulse response) for the j-th coordinate as

$$G_k(j) := \frac{y_{j,+}^{k+1} - y_{j,-}^{k+1}}{y_{j,+} - y_{j,-}}, \qquad k \geqslant 0,$$

with the natural continuous limiting definition when $y_{j,+} = y_{j,-}$ (i.e., critical damping).

By viewing the stochastic noise sequence $-\eta\xi_k(j)$ as an external forcing term, the stochastic parameter error trajectory can be decomposed exactly into a deterministic signal trajectory and a discrete convolution of the gradient noise. This exact impulse-response representation is formalized in the following proposition.

Proposition D.1 (Impulse-response representation)**.** *For any step $k \geqslant 0$, the coordinate-wise error trajectory of Polyak satisfies:*

$$e_k(j) = e_k^{\mathrm{det}}(j) - \eta\sum_{m=0}^{k-1} G_{k-1-m}(j)\xi_m(j), \tag{51}$$

where $e_k^{\mathrm{det}}(j)$ is the unperturbed deterministic trajectory solving:

$$e_{k+1}^{\mathrm{det}}(j) = (1+\rho-\eta\lambda_j)e_k^{\mathrm{det}}(j) - \rho e_{k-1}^{\mathrm{det}}(j),$$

with initializations $e_0^{\mathrm{det}}(j) = e_0(j) = -\theta_j^*$ and $e_{-1}^{\mathrm{det}}(j) = e_{-1}(j) = -\theta_j^*$ (assuming $\boldsymbol{\theta}_0 = \boldsymbol{\theta}_{-1} = \mathbf{0}$).

Proof. Define the stochastic residual error $\tilde{e}_k(j) := e_k(j) - e_k^{\text{det}}(j)$. Subtracting the deterministic trajectory from the stochastic parameter error yields the inhomogeneous system:

$$\tilde{e}_{k+1}(j) - (1 + \rho - \eta\lambda_j)\tilde{e}_k(j) + \rho\tilde{e}_{k-1}(j) = -\eta\xi_k(j), \tag{52}$$

subject to the initial conditions:

$$\tilde{e}_0(j) = \tilde{e}_{-1}(j) = 0.$$

We introduce the discrete linear difference operator $\mathcal{L}_j$:

$$(\mathcal{L}_j u)_k := u_{k+1} - (1 + \rho - \eta\lambda_j)u_k + \rho u_{k-1}.$$

By the definition of the discrete Green function $G_k(j)$:

$$(\mathcal{L}_j G(j))_k = 0 \quad (\forall k \geqslant 0), \qquad \text{with} \quad G_{-1}(j) = 0, \quad G_0(j) = 1.$$

Let the impulse response to a point source at step $m \geqslant 0$ be defined as $H_k(j; m) := G_{k-1-m}(j)$, which satisfies:

$$(\mathcal{L}_j H(j; m))_k = \delta_{k,m}, \qquad \text{where} \quad \delta_{k,m} = \begin{cases} 1, & k = m, \\ 0, & k \neq m. \end{cases}$$

Applying the superposition principle to the linear operator $\mathcal{L}_j$ under the forcing term $-\eta\xi_k(j)$ in (52) yields:

$$\begin{aligned} \tilde{e}_k(j) &= \sum_{m=0}^{k-1} \big(-\eta\xi_m(j) \big) H_k(j; m) \\ &= -\eta \sum_{m=0}^{k-1} G_{k-1-m}(j)\xi_m(j). \end{aligned}$$

Substituting the definition of $\tilde{e}_k(j)$ back into the equation completes the proof:

$$e_k(j) = e_k^{\text{det}}(j) - \eta \sum_{m=0}^{k-1} G_{k-1-m}(j)\xi_m(j).$$

□

To quantify the excess risk, we define the deterministic bias (the *signal* component) and the convolution kernel as follows:

$$S_{\text{polyak}}(k) := \frac{1}{2} \sum_{j \geqslant 1} \lambda_j \, e_k^{\text{det}}(j)^2 = \mathcal{E}(\boldsymbol{\theta}_k^{\text{det}}), \qquad \mathcal{K}_{\text{polyak}}(k) := \sum_{j \geqslant 1} \lambda_j^2 \, G_k(j)^2.$$

The following proposition establishes that the expected excess risk strictly decouples into the deterministic signal bias, the accumulated additive label noise, and the accumulated multiplicative gradient noise.

Proposition D.2 (Risk recursion for Polyak)**.** *Under the gradient-noise covariance assumption, the expected excess risk obeys the two-sided recursion:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{polyak}}(k) + \frac{\eta^2\sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) + \frac{\eta^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})]. \tag{53}$$

Equivalently, the right-hand side provides both rigorous upper and lower bounds for the expected excess risk up to universal constant factors.

Proof. By substituting the Green function representation (51) into the definition of the excess risk, we analyze the second moment for each error coordinate:

$$e_k(j)^2 = \left(e_k^{\det}(j) - \eta \sum_{m=0}^{k-1} G_{k-1-m}(j)\xi_m(j) \right)^2 .$$

Because the stochastic gradient noise is conditionally mean-zero ($\mathbb{E}[\xi_m(j) \mid \boldsymbol{\theta}_m] = 0$) and independent across distinct time steps, all cross-terms between the deterministic trajectory and the noise, as well as cross-time noise products, vanish upon taking the expectation. Hence,

$$\mathbb{E}[e_k(j)^2] = e_k^{\det}(j)^2 + \eta^2 \sum_{m=0}^{k-1} G_{k-1-m}(j)^2 \mathbb{E}[\xi_m(j)^2].$$

Multiplying by $\frac{1}{2}\lambda_j$ and summing over all coordinates $j \geqslant 1$ gives the expected excess risk:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \frac{1}{2} \sum_{j \geqslant 1} \lambda_j e_k^{\det}(j)^2 + \frac{\eta^2}{2} \sum_{m=0}^{k-1} \sum_{j \geqslant 1} \lambda_j G_{k-1-m}(j)^2 \mathbb{E}[\xi_m(j)^2].$$

According to the property of curvature-aligned noise defined in Eq. (5), for the coordinate basis $\boldsymbol{u}_j$, the variance satisfies the following:

$$\mathbb{E}[\xi_m(j)^2] \eqsim \frac{1}{B} \lambda_j \big(\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] + \sigma^2 \big),$$

where the factor $1/B$ accounts for the mini-batching variance reduction.

Substituting this into the previous equation yields:

$$\begin{aligned}
\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] &\eqsim \frac{1}{2} \sum_{j \geqslant 1} \lambda_j e_k^{\det}(j)^2 + \frac{\eta^2}{B} \sum_{m=0}^{k-1} \big(\sigma^2 + \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] \big) \sum_{j \geqslant 1} \lambda_j^2 G_{k-1-m}(j)^2 \\
&= S_{\text{polyak}}(k) + \frac{\eta^2 \sigma^2}{B} \sum_{m=0}^{k-1} \mathcal{K}_{\text{polyak}}(k-1-m) + \frac{\eta^2}{B} \sum_{m=0}^{k-1} \mathcal{K}_{\text{polyak}}(k-1-m) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)].
\end{aligned}$$

Finally, performing a change of variables by setting $\tau = k - 1 - m$ yields:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{polyak}}(k) + \frac{\eta^2 \sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) + \frac{\eta^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})].$$

The two-sided equivalence follows directly from the two-sided comparability in the noise covariance assumption and the non-negativity of $S_{\text{polyak}}(k)$ and $\mathcal{K}_{\text{polyak}}(k)$. □

D.2 Overdamped and underdamped Modes decomposition

Since both the deterministic error signal $e_k^{\det}(j)$ and the discrete Green function $G_k(j)$ obey the exact same homogeneous difference equation, their dynamic behaviors are simultaneously dictated by the same characteristic roots. Depending on the sign of the corresponding discriminant, these roots transition from real to complex conjugate pairs, driving the system from monotonic decay to oscillatory transients. The following proposition rigorously establishes the damping threshold, defines the crossover index $J_{\max}$, and partitions the spectrum into two distinct global regimes.

Proposition D.3 (Exact spectral decomposition and crossover boundary)**.** *For each coordinate j, the characteristic roots of the Polyak recursion are given by*

$$y_{j,\pm} = \frac{(1+\rho-\eta\lambda_j) \pm \sqrt{\Delta_j}}{2},$$

where the discriminant is $\Delta_j := (1+\rho-\eta\lambda_j)^2 - 4\rho$.

The critical transition $\Delta_j = 0$ corresponds exactly to the spectral threshold $\eta\lambda_j = (1-\sqrt{\rho})^2$, which is asymptotically equivalent to $\eta\lambda_j \eqsim (1-\rho)^2$. This threshold classifies the individual eigen-directions into:

1. ***Overdamped Modes*** *($\eta\lambda_j < (1-\sqrt{\rho})^2$, i.e., $\Delta_j > 0$): The roots are strictly real.*
2. ***Underdamped Modes*** *($\eta\lambda_j > (1-\sqrt{\rho})^2$, i.e., $\Delta_j < 0$): The roots are complex conjugates.*

In the Mixed Overdamped–Underdamped Regime, the boundary index $J_{\max}$ separating the underdamped modes from the overdamped modes is defined as the crossover index satisfying $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$:

$$J_{\max} \eqsim \left(\frac{\eta}{(1-\rho)^2}\right)^{1/\beta}. \tag{54}$$

Consequently, based on the maximum eigenvalue $\lambda_{\max}$ of the population Hessian $\mathbf{H}$, the algorithm globally operates in one of two regimes:

- ***Fully Overdamped Regime*** *($\eta\lambda_{\max} \lesssim (1-\rho)^2$): All spectral modes are overdamped.*
- ***Mixed Overdamped–Underdamped Regime*** *($\eta\lambda_{\max} \gtrsim (1-\rho)^2$): The spectrum bifurcates into high-curvature underdamped modes ($j \leqslant J_{\max}$) and flat overdamped modes ($j > J_{\max}$).*

Proof. The phase transition between real and complex roots occurs exactly at the boundary where the discriminant vanishes ($\Delta_j = 0$). Solving the quadratic equation $(1+\rho-\eta\lambda_j)^2 = 4\rho$ for the relevant stable branch yields the exact threshold:

$$\eta\lambda_j = 1+\rho-2\sqrt{\rho} = (1-\sqrt{\rho})^2.$$

By multiplying the numerator and denominator by $(1+\sqrt{\rho})^2$, we can rewrite this exact threshold as: $\eta\lambda_j = \frac{(1-\rho)^2}{(1+\sqrt{\rho})^2}$. Since $1+\sqrt{\rho} \eqsim 2$ for all $\rho \in (0,1)$, we obtain the fundamental scaling equivalence for the damping boundary:

$$\eta\lambda_j \eqsim (1-\rho)^2.$$

To locate the exact crossover boundary in the Mixed Overdamped–Underdamped Regime, we apply the power-law spectrum $\lambda_j \eqsim j^{-\beta}$. Setting the critical damping boundary condition $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$, we obtain: $\eta J_{\max}^{-\beta} \eqsim (1-\rho)^2 \iff J_{\max}^{-\beta} \eqsim \frac{(1-\rho)^2}{\eta}$. Taking the $-1/\beta$-th power of both sides yields:

$$J_{\max} \eqsim \left(\frac{\eta}{(1-\rho)^2}\right)^{1/\beta},$$

which establishes that the underdamped modes are exactly indexed by $j \leqslant J_{\max}$ and the overdamped modes by $j > J_{\max}$.

For an overdamped mode where $\eta\lambda_j < (1-\sqrt{\rho})^2$, the discriminant is strictly positive ($\Delta_j > 0$), and the roots $y_{j,+}$ and $y_{j,-}$ are real and distinct.

For an underdamped mode where $\eta\lambda_j > (1-\sqrt{\rho})^2$, the discriminant is strictly negative ($\Delta_j < 0$). The roots become a complex conjugate pair $y_{j,\pm} = \sqrt{\rho}e^{\pm i\omega_j}$, where the exact phase angle $\omega_j \in (0,\pi)$ satisfies: $\cos\omega_j = \frac{1+\rho-\eta\lambda_j}{2\sqrt{\rho}}$.

Finally, to determine the global behavior of the system, we examine the maximum eigenvalue $\lambda_{\max}$. If $\eta\lambda_{\max} \lesssim (1-\rho)^2$, then every coordinate satisfies $\Delta_j \geqslant 0$ in the scaling sense, placing the system entirely in the Fully Overdamped Regime. If $\eta\lambda_{\max} \gtrsim (1-\rho)^2$, the spectral modes with $\lambda_j \gtrsim (1-\rho)^2/\eta$ form a non-empty underdamped subset, while the remaining tail $\lambda_j \lesssim (1-\rho)^2/\eta$ remains overdamped, placing the system globally in the Mixed Overdamped–Underdamped Regime. □

D.3 Label noise calculation

Before bounding the full multiplicative risk, we must isolate the purely additive label-noise contribution, which establishes the noise floor for both regimes.

Lemma D.4 (Label-noise contribution)**.** *There exist constants $C, c_-, c_+ > 0$ such that if*

$$k \geqslant C \max\left\{\frac{1}{1-\rho}, \frac{1-\rho}{\eta}\right\}, \tag{55}$$

then the finite-time label-noise contribution satisfies

$$c_- \frac{\sigma^2\eta}{B(1-\rho)} \leqslant \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) \leqslant c_+ \frac{\sigma^2\eta}{B(1-\rho)}. \tag{56}$$

In particular,

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) \eqsim \frac{\sigma^2}{B}\frac{\eta}{1-\rho}.$$

Proof. We first compute the saturated coordinate Green energy exactly. Fix a coordinate j and write

$$x_j := \eta\lambda_j, \qquad A_j := 1+\rho-x_j.$$

The Green function satisfies

$$G_{k+1}(j) = A_j G_k(j) - \rho G_{k-1}(j), \qquad G_0(j) = 1, \qquad G_{-1}(j) = 0.$$

Assume the coordinate is stable. Define

$$S_j := \sum_{\tau=0}^{\infty} G_\tau(j)^2, \qquad C_j := \sum_{\tau=0}^{\infty} G_\tau(j)G_{\tau-1}(j),$$

where $G_{-1}(j) = 0$. Multiplying the recursion by $G_k(j)$ and summing over $k \geqslant 0$ gives $C_j = A_j S_j - \rho C_j$, hence $C_j = \frac{A_j}{1+\rho} S_j$.

Next square the recursion and sum over $k \geqslant 0$:

$$S_j - 1 = \left(A_j^2 + \rho^2\right) S_j - 2A_j\rho C_j.$$

Substituting $C_j = A_j S_j/(1+\rho)$ gives

$$\begin{aligned} 1 &= S_j\left[1 - A_j^2 - \rho^2 + \frac{2A_j^2\rho}{1+\rho}\right] \\ &= S_j\frac{1-\rho}{1+\rho}\left[(1+\rho)^2 - (1+\rho-x_j)^2\right] \\ &= S_j\frac{1-\rho}{1+\rho}x_j\left(2(1+\rho) - x_j\right). \end{aligned}$$

Therefore

$$\sum_{\tau=0}^{\infty} G_\tau(j)^2 = \frac{1+\rho}{(1-\rho)\eta\lambda_j\,(2(1+\rho)-\eta\lambda_j)}. \tag{57}$$

We now pass from the saturated sum to the finite-time sum. The root formula for $G_k(j)$ implies the following tail bound: there exist universal constants $C_G, c_G > 0$ such that

$$\sum_{\tau=k}^{\infty} G_\tau(j)^2 \leqslant C_G \exp\left\{-c_G k \min\left(1-\rho, \frac{\eta\lambda_j}{1-\rho}\right)\right\} \sum_{\tau=0}^{\infty} G_\tau(j)^2. \tag{58}$$

This is the standard Polyak relaxation estimate.

Because $\lambda_j \eqsim j^{-\beta}$ and $\beta > 1$, the spectral mass is finite: $\sum_{j\geqslant 1} \lambda_j < \infty$.
Choose a fixed integer J_*, depending only on the spectrum constants and β, such that $\sum_{j=1}^{J_*} \lambda_j \geqslant \frac{1}{2}\sum_{j\geqslant 1} \lambda_j$.
Since J_* is fixed, λ_{J_*} is a positive constant independent of η, ρ, k, B. Thus, after increasing the universal constant C in (55) if necessary, condition (55) implies that for every $1 \leqslant j \leqslant J_*$,

$$k \min\left(1-\rho, \frac{\eta\lambda_j}{1-\rho}\right) \geqslant \frac{\log(2C_G)}{c_G}.$$

By (58), for all $j \leqslant J_*$,

$$\sum_{\tau=k}^{\infty} G_\tau(j)^2 \leqslant \frac{1}{2}\sum_{\tau=0}^{\infty} G_\tau(j)^2.$$

Therefore

$$\sum_{\tau=0}^{k-1} G_\tau(j)^2 \geqslant \frac{1}{2}\sum_{\tau=0}^{\infty} G_\tau(j)^2, \qquad 1 \leqslant j \leqslant J_*. \tag{59}$$

For all j, the trivial upper bound

$$\sum_{\tau=0}^{k-1} G_\tau(j)^2 \leqslant \sum_{\tau=0}^{\infty} G_\tau(j)^2 \tag{60}$$

holds.

We now insert these bounds into the finite-time label-noise term. Since all terms are nonnegative, we can exchange the τ- and j-sums:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) = \frac{\eta^2\sigma^2}{B}\sum_{j\geqslant 1}\lambda_j^2 \sum_{\tau=0}^{k-1} G_\tau(j)^2.$$

For the lower bound, use (59) and (57) on the first J_* modes:

$$\begin{aligned}\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) &\geqslant \frac{1}{2}\frac{\eta^2\sigma^2}{B}\sum_{j=1}^{J_*}\lambda_j^2 \frac{1+\rho}{(1-\rho)\eta\lambda_j(2(1+\rho)-\eta\lambda_j)} \\ &= \frac{1}{2}\frac{\eta\sigma^2(1+\rho)}{B(1-\rho)}\sum_{j=1}^{J_*}\frac{\lambda_j}{2(1+\rho)-\eta\lambda_j}.\end{aligned}$$

Since $2(1+\rho)-\eta\lambda_j \leqslant 2(1+\rho)$,

$$\frac{1+\rho}{2(1+\rho)-\eta\lambda_j} \geqslant \frac{1}{2}.$$

Thus

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) \geqslant \frac{1}{4}\frac{\eta\sigma^2}{B(1-\rho)}\sum_{j=1}^{J_*}\lambda_j.$$

The finite sum $\sum_{j=1}^{J_*}\lambda_j$ is a positive constant, so this gives the lower bound in (56) for a suitable constant $c_- > 0$.

For the upper bound, use (60) and (57) over all modes:

$$\begin{aligned}\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) &\leqslant \frac{\eta\sigma^2(1+\rho)}{B(1-\rho)}\sum_{j\geqslant 1}\frac{\lambda_j}{2(1+\rho)-\eta\lambda_j}\\ &\leqslant \frac{\eta\sigma^2(1+\rho)}{B(1-\rho)}\frac{1}{2(1+\rho)-\eta\lambda_1}\sum_{j\geqslant 1}\lambda_j.\end{aligned}$$

The last factor is finite, and the denominator is bounded below by the stability margin. Hence the upper bound in (56) holds for a suitable constant $c_+ > 0$. □

D.4 Uniform boundedness of expected risk

As long as the additive label noise is strictly positive ($\sigma > 0$), the scaling law derivation can be vastly simplified. We can prove that the expected risk is uniformly bounded, which allows the multiplicative gradient noise to be completely absorbed by the additive label noise floor.

Lemma D.5 (Uniform boundedness of expected risk)**.** *Suppose $\sigma > 0$. Under the stability condition $\eta \leqslant cB(1-\rho)$ for a sufficiently small universal constant $c > 0$, the expected excess risk of Polyak is uniformly bounded for all $k \geqslant 0$:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M,$$

where $M > 0$ is a constant independent of k, η, B, and ρ.

Proof. We proceed by induction on k. By Proposition D.2, there exist universal constants $C_1, C_2, C_3 > 0$ such that for any $k \geqslant 1$:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant C_1 S_{\text{polyak}}(k) + C_2\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) + C_3\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})]. \tag{61}$$

First, Propositions D.7 and D.9 in following subsections establish that the deterministic bias monotonically decays or is uniformly bounded by its initialization across all regimes. Thus:

$$S_{\text{polyak}}(k) \leqslant C_h^2\mathcal{E}(\boldsymbol{\theta}_0) =: M_1.$$

Second, applying $\sum_{\tau=0}^{\infty}\mathcal{K}_{\text{polyak}}(\tau) \leqslant \frac{C_K}{\eta(1-\rho)}$ from Lemma D.4 bounds the label noise:

$$C_2\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) \leqslant C_2\sigma^2\frac{C_K\eta}{B(1-\rho)}.$$

Under $\eta \leqslant cB(1-\rho)$, this is uniformly bounded by $C_2\sigma^2 C_K c =: M_2$.

Assuming inductively $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] \leqslant M$ for $m < k$, the multiplicative noise term yields:

$$C_3\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})] \leqslant C_3\frac{\eta^2}{B}\left(\frac{C_K}{\eta(1-\rho)}\right)M = C_3C_K\frac{\eta}{B(1-\rho)}M \leqslant C_3C_K cM.$$

Choosing $c \leqslant \frac{1}{2C_3C_K}$ limits this term to $M/2$. Substituting into (61) yields:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M_1 + M_2 + \frac{M}{2}.$$

Setting $M = 2(M_1 + M_2)$ guarantees $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M$. The base case $k = 0$ holds trivially since $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_0)] = S_{\text{polyak}}(0) \leqslant M_1 \leqslant M$. This completes the induction. □

Corollary D.6 (Absorption of multiplicative noise)**.** *For $\sigma > 0$, under the stability condition $\eta \lesssim B(1-\rho)$, the multiplicative gradient noise is strictly bounded by the additive label noise up to a constant factor:*

$$\frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau)\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-1-\tau})] \leqslant \frac{M}{\sigma^2}\left(\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau)\right).$$

Consequently, the multiplicative noise can be safely absorbed, and the expected excess risk is asymptotically equivalent to the sum of the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{polyak}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau).$$

D.5 Risk dynamics in the fully overdamped regime

In the fully overdamped regime, the deterministic error trajectory decays monotonically, yielding a polynomial bias.

Proposition D.7 (Deterministic loss, fully overdamped regime)**.** *Under Assumptions 3.1 and 3.3, in the fully overdamped regime $\eta \lesssim (1-\rho)^2$, we have:*

$$S_{\text{polyak}}(k) \eqsim \begin{cases} 1, & k \lesssim \frac{1-\rho}{\eta}, \\ \left(\frac{1-\rho}{\eta k}\right)^s, & k \gtrsim \frac{1-\rho}{\eta}. \end{cases}$$

Proof. In the fully overdamped regime, we can assume: $\eta\lambda_1 \leqslant c_0(1-\rho)^2$, with a sufficiently small universal constant $c_0 > 0$.

Let the deterministic coordinate multiplier $h_k(j) := e_k^{\text{det}}(j)/e_0^{\text{det}}(j) = -e_k^{\text{det}}(j)/\theta_j^*$ satisfy

$$h_{k+1}(j) = (1+\rho-\eta\lambda_j)h_k(j) - \rho h_{k-1}(j), \qquad h_0(j) = 1, \qquad h_1(j) = 1-\eta\lambda_j.$$

The characteristic roots are

$$r_{j,+}, r_{j,-} = \frac{1+\rho-\eta\lambda_j \pm \sqrt{(1+\rho-\eta\lambda_j)^2 - 4\rho}}{2}.$$

We first give explicit root bounds. Since $\eta \leqslant \frac{c_0}{\lambda_1}(1-\rho)^2$ and $\lambda_j \leqslant \lambda_1$, we have $0 \leqslant \eta\lambda_j \leqslant c_0(1-\rho)^2$ for all j. Put $r = 1-u$. The characteristic polynomial evaluated at $1-u$ is

$$(1-u)^2 - (1+\rho-\eta\lambda_j)(1-u) + \rho = \eta\lambda_j - (1-\rho+\eta\lambda_j)u + u^2.$$

Choosing $c_0 > 0$ sufficiently small, the two evaluations

$$\eta\lambda_j - (1-\rho+\eta\lambda_j)\frac{\eta\lambda_j}{2(1-\rho)} + \left(\frac{\eta\lambda_j}{2(1-\rho)}\right)^2 > 0,$$

and

$$\eta\lambda_j - (1-\rho+\eta\lambda_j)\frac{2\eta\lambda_j}{1-\rho} + \left(\frac{2\eta\lambda_j}{1-\rho}\right)^2 < 0$$

hold uniformly for $0 < \eta\lambda_j \leqslant c_0(1-\rho)^2$. Therefore the slow root satisfies

$$1 - \frac{2\eta\lambda_j}{1-\rho} \leqslant r_{j,+} \leqslant 1 - \frac{\eta\lambda_j}{2(1-\rho)}. \tag{62}$$

The fast root satisfies $r_{j,+}r_{j,-} = \rho = 1-(1-\rho)$. Since $r_{j,+} \geqslant 1 - 2c_0(1-\rho)$, after decreasing c_0 if needed,

$$0 \leqslant r_{j,-} \leqslant 1 - \frac{1-\rho}{2}. \tag{63}$$

Now write the solution as

$$h_k(j) = A_j r_{j,+}^k + B_j r_{j,-}^k.$$

From $h_0 = 1$ and $h_1 = 1 - \eta\lambda_j$, we obtain the coefficients

$$A_j = \frac{1 - \eta\lambda_j - r_{j,-}}{r_{j,+} - r_{j,-}}, \qquad B_j = 1 - A_j = \frac{r_{j,+} - (1-\eta\lambda_j)}{r_{j,+} - r_{j,-}}.$$

Using (62)–(63), and taking c_0 sufficiently small, these coefficients obey uniform bounds:

$$\frac{1}{2} \leqslant A_j \leqslant 2, \qquad |B_j| \leqslant 2. \tag{64}$$

Equivalently, applying the initial conditions, the exact solution can be expressed in the fractional form:

$$h_k(j) = \frac{r_{j,+}r_{j,-}\left(-r_{j,+}^{k-1} + r_{j,-}^{k-1}\right) + (1-\eta\lambda_j)\left(r_{j,+}^k - r_{j,-}^k\right)}{r_{j,+} - r_{j,-}}. \tag{65}$$

To evaluate the deterministic bias $S_{\text{polyak}}(k) = \sum_{j\geqslant 1} \lambda_j(\theta_j^*)^2 h_k(j)^2$, we divide the time k into three successive stages.

Stage 1: $k \lesssim \frac{1}{1-\rho}$.

In this extremely short-time regime, taking the limits $r_{j,+} \approx r_{j,-} \approx r \approx 1$ and using $r^2 \approx \rho$ in (65) yields the heuristic trajectory:

$$h_k(j) \simeq -\rho(k-1)r^{k-2} + (1-\eta\lambda_j)kr^{k-1} = k\left((1-\eta\lambda_j)r - \rho\right)r^{k-2} + \rho r^{k-2}.$$

Since $(1-\eta\lambda_j)r - \rho = \mathcal{O}(1-\rho)$, the first term is bounded by $\mathcal{O}(k(1-\rho)) \lesssim 1$, yielding $h_k(j) \simeq 1$. Rigorously, for the leading modes ($j = \mathcal{O}(1)$), since $\eta\lambda_j \leqslant c_0(1-\rho)^2$, the slow root satisfies $r_{j,+} \geqslant 1 - 2c_0(1-\rho)$. For $k \lesssim \frac{1}{1-\rho}$, we have

$$r_{j,+}^k \geqslant (1 - 2c_0(1-\rho))^{\frac{C}{1-\rho}} \asymp 1.$$

Since the dynamics are unconditionally stable ($h_k(j) \leqslant 1$), it follows that $h_k(j) \asymp 1$ for the leading modes. Consequently, the bias is dominated by these non-decayed modes:

$$S_{\text{polyak}}(k) \asymp \sum_{j\geqslant 1} \lambda_j(\theta_j^*)^2 \asymp 1.$$

Stage 2: $\frac{1}{1-\rho} \lesssim k \lesssim \frac{1-\rho}{\eta}$.

For $k \gtrsim \frac{1}{1-\rho}$, the fast root decays exponentially since $r_{j,-}^k \leqslant \left(1 - \frac{1-\rho}{2}\right)^k \ll 1$. The multiplier is overwhelmingly dominated by the slow root:

$$h_k(j) \simeq \frac{1 - \eta\lambda_j - r_{j,-}}{r_{j,+} - r_{j,-}} r_{j,+}^k \simeq r_{j,+}^k.$$

For the leading modes with $\lambda_j = \Theta(1)$, within the time window $k \lesssim \frac{1-\rho}{\eta}$, the exponent is bounded by:

$$r_{j,+}^k \eqsim \left(1 - c\frac{\eta\lambda_j}{1-\rho}\right)^k \eqsim \exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right) \eqsim 1,$$

where we used the condition $\frac{\eta\lambda_j}{1-\rho}k \lesssim \lambda_j \leqslant \lambda_1 = \mathcal{O}(1)$. Thus, the leading modes have not yet experienced significant exponential decay, preserving $h_k(j) \eqsim 1$. As a result, the deterministic bias remains unchanged:

$$S_{\text{polyak}}(k) \eqsim 1.$$

Stage 3: $k \gtrsim \frac{1-\rho}{\eta}$.
In this long-time regime, the exponential decay of the slow root becomes significant across the spectrum. Combining (62), (63), and (64), there are universal constants $c_1, c_2, c_3, C_1 > 0$ such that, for all $k \geqslant C_1(1-\rho)^{-1}$,

$$c_1 \exp\left(-c_2\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant |h_k(j)| \leqslant C_1 \exp\left(-c_3\frac{\eta\lambda_j}{1-\rho}k\right). \tag{66}$$

Substituting (66) into the deterministic bias gives

$$c\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2 \exp\left(-C\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant S_{\text{polyak}}(k) \leqslant C\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2 \exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right). \tag{67}$$

By Assumption 3.3,

$$\lambda_j(\theta_j^*)^2 \eqsim j^{-1-\beta s}, \qquad \lambda_j \eqsim j^{-\beta}.$$

Thus it remains to bound sums of the form

$$\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho}j^{-\beta}\right).$$

The condition on k implies $\frac{\eta k}{1-\rho} \geqslant C$.

For the lower bound, set $J := \left\lceil \left(\frac{\eta k}{1-\rho}\right)^{1/\beta} \right\rceil$. For $j \geqslant J$, we have $\frac{\eta k}{1-\rho}j^{-\beta} \leqslant 1$, so $\exp(-C\frac{\eta k}{1-\rho}j^{-\beta}) \geqslant e^{-C}$. Therefore,

$$\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-C\frac{\eta k}{1-\rho}j^{-\beta}\right) \geqslant e^{-C}\sum_{j\geqslant J} j^{-1-\beta s} \geqslant c\left(\frac{\eta k}{1-\rho}\right)^{-s}.$$

For the upper bound, the integral comparison with $u = c\frac{\eta k}{1-\rho}x^{-\beta}$ gives

$$\begin{aligned}\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho}j^{-\beta}\right) &\leqslant C\int_1^\infty x^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho}x^{-\beta}\right) dx \\ &= C\left(\frac{\eta k}{1-\rho}\right)^{-s}\int_0^{c\frac{\eta k}{1-\rho}} u^{s-1}e^{-u}\,du \leqslant C\left(\frac{\eta k}{1-\rho}\right)^{-s}.\end{aligned}$$

Combining the upper and lower bounds in (67) yields

$$S_{\text{polyak}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s.$$

Synthesizing the three stages completes the proof. □

Theorem D.8 (Scaling law, fully overdamped regime)**.** *Under Assumptions 3.1, 3.3, the stability condition $\eta \lesssim B(1-\rho)$, and $\eta \lesssim (1-\rho)^2$, for $k \gtrsim \frac{1-\rho}{\eta}$, the expected excess risk satisfies:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \left(\frac{1-\rho}{\eta k}\right)^s + \frac{\sigma^2\eta}{B(1-\rho)}.$$

Proof. By Corollary D.6, the expected risk is completely determined by the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{polyak}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau).$$

For $k \gtrsim \frac{1-\rho}{\eta}$, we substitute the deterministic bias from Proposition D.7:

$$S_{\text{polyak}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s.$$

We substitute the label noise floor from Lemma D.4:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{polyak}}(\tau) \eqsim \frac{\sigma^2\eta}{B(1-\rho)}.$$

Summing these two terms directly yields the scaling law. □

D.6 Risk dynamics in the mixed regime

When underdamped modes are present, the deterministic bias acquires an exponentially decaying oscillatory transient alongside the polynomial tail.

Proposition D.9 (Deterministic loss, mixed regime)**.** *Under Assumptions 3.1 and 3.3, assume the stability condition $\eta \lesssim \min\{1, B(1-\rho)\}$. For the mixed regime $\eta \gtrsim (1-\rho)^2$, we have:*

$$S_{\text{polyak}}(k) \eqsim \begin{cases} \rho^k + \left(\frac{(1-\rho)^2}{\eta}\right)^s, & k \lesssim \frac{1}{1-\rho}, \\ \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s, & k \gtrsim \frac{1}{1-\rho}. \end{cases}$$

Proof. Let the coordinate multiplier $h_k(j) := -e_k^{\text{det}}(j)/\theta_j^*$. Then it satisfies

$$h_{k+1}(j) = (1+\rho-\eta\lambda_j)h_k(j) - \rho h_{k-1}(j), \qquad h_0(j) = 1, \qquad h_1(j) = 1-\eta\lambda_j.$$

The characteristic discriminant is negative precisely when

$$(1-\sqrt{\rho})^2 < \eta\lambda_j < (1+\sqrt{\rho})^2.$$

Thus the comparison between η and $(1-\rho)^2$ determines whether a coordinate is overdamped or underdamped. In the mixed regime $\eta \gtrsim (1-\rho)^2$, the crossover index is defined by $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$. We decompose the deterministic bias into the overdamped tail ($j > J_{\max}$) and the underdamped leading block ($j \leqslant J_{\max}$).

For overdamped coordinates in the stable small-step range, the root comparison used in Proposition D.7 gives constants $c_1, C_1 > 0$ such that

$$|h_k(j)| \leqslant C_1 \exp\left(-c_1 \frac{\eta\lambda_j}{1-\rho}k\right). \tag{68}$$

Moreover, if $\eta\lambda_j \leqslant c_1(1-\rho)^2$ and $k \geqslant C_1(1-\rho)^{-1}$, the slow root dominates the fast root and

$$|h_k(j)| \geqslant c_1 \exp\left(-C_1 \frac{\eta\lambda_j}{1-\rho}k\right).$$

To evaluate the overdamped contribution to the bias, we discuss two cases for k:

Case 1: $k \lesssim \frac{1}{1-\rho}$.

In this regime, for all overdamped modes $j > J_{\max}$, we have $\eta\lambda_j \lesssim (1-\rho)^2$. Consequently, the exponent satisfies $\frac{\eta\lambda_j}{1-\rho}k \lesssim \frac{(1-\rho)^2}{1-\rho}\frac{1}{1-\rho} = 1$. The exponential decay is bounded away from zero, so $|h_k(j)| \eqsim 1$. Therefore, the sum is strictly dominated by its lower limit:

$$\sum_{j>J_{\max}} \lambda_j(\theta_j^*)^2 h_k(j)^2 \eqsim \sum_{j>J_{\max}} j^{-1-\beta s} \eqsim (J_{\max})^{-\beta s} \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^s.$$

Case 2: $k \gtrsim \frac{1}{1-\rho}$.

Since $\lambda_j \eqsim j^{-\beta}$, the time condition implies that every $j \geqslant \left(\frac{C\eta k}{1-\rho}\right)^{1/\beta}$ satisfies $\eta\lambda_j \leqslant c_1(1-\rho)^2$, after increasing C if necessary.
Using $\lambda_j(e_0(j))^2 \eqsim j^{-1-\beta s}$, we get

$$\sum_{j>J_{\max}} \lambda_j(\theta_j^*)^2 h_k(j)^2 \geqslant c \sum_{j\geqslant (C\eta k/(1-\rho))^{1/\beta}} j^{-1-\beta s} \geqslant c\left(\frac{1-\rho}{\eta k}\right)^s. \tag{69}$$

Conversely, by (68) and the same integral comparison,

$$\sum_{j>J_{\max}} \lambda_j(\theta_j^*)^2 h_k(j)^2 \leqslant C\sum_{j\geqslant 1} j^{-1-\beta s}\exp\left(-c_1\frac{\eta k}{1-\rho}j^{-\beta}\right) \leqslant C\left(\frac{1-\rho}{\eta k}\right)^s.$$

Therefore the overdamped tail contributes exactly the polynomial term, up to constants.

It remains to estimate the underdamped contribution. For $j \leqslant J_{\max}$, write the two roots as

$$\sqrt{\rho}e^{i\omega_j}, \quad \sqrt{\rho}e^{-i\omega_j}, \qquad \text{where} \quad \cos\omega_j = \frac{1+\rho-\eta\lambda_j}{2\sqrt{\rho}}.$$

Solving the two initial conditions gives the exact identity

$$h_k(j) = \rho^{k/2}\left[\cos(k\omega_j) + \frac{1-\rho-\eta\lambda_j}{2\sqrt{\rho}\sin\omega_j}\sin(k\omega_j)\right]. \tag{70}$$

The sine term is retained in the oscillatory estimates below. Set

$$A_j := \frac{1-\rho-\eta\lambda_j}{2\sqrt{\rho}\sin\omega_j}.$$

On the separated underdamped band

$$C_{\mathrm{mom}}(1-\rho)^2 \leqslant \eta\lambda_j \leqslant c_{\mathrm{st}}/2,$$

one has

$$4\rho\sin^2\omega_j = \left(\eta\lambda_j - (1-\sqrt{\rho})^2\right)\left((1+\sqrt{\rho})^2 - \eta\lambda_j\right) \geqslant c\eta\lambda_j,$$

and

$$(1-\rho-\eta\lambda_j)^2 \leqslant (1-\rho)^2,$$

hence $|A_j| \leqslant C$. Thus (70) immediately gives the upper bound

$$\sum_{j\leqslant J_{\max}} \lambda_j(\theta_j^*)^2 h_k(j)^2 \leqslant C\rho^k,$$

because $\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2 < \infty$.

For the lower bound of the underdamped contribution, we evaluate the sum over the underdamped modes $j \leqslant J_{\max}$, which is given by:

$$S_{\text{under}}(k) = \rho^k \sum_{j \leqslant J_{\max}} \lambda_j (\theta_j^*)^2 (1 + A_j^2) \cos^2(k\omega_j - \psi_j).$$

To isolate the oscillatory behavior, we define the positive spectral weights w_j and decompose the cosine-squared terms as follows:

$$S_{\text{under}}(k) = \sum_{j \leqslant J_{\max}} w_j \cos^2(k\omega_j - \psi_j) = \frac{1}{2} \sum_{j \leqslant J_{\max}} w_j + \frac{1}{2} \sum_{j \leqslant J_{\max}} w_j \cos(2k\omega_j - 2\psi_j),$$

where $w_j = \lambda_j (\theta_j^*)^2 (1 + A_j^2)$.

The first sum represents the stable static mass, which satisfies $\sum_{j \leqslant J_{\max}} w_j \geqslant M_w > 0$. To analyze the oscillatory second sum, we view the indices as a continuous variable $x \in [1, J_{\max}]$. For the continuous oscillatory integral $I(k)$, we define:

$$I(k) = \int_1^{J_{\max}} w(x) \cos(2k\omega(x) - 2\psi(x))\, dx.$$

We introduce the substitution $u = \omega(x)$ to map the continuous domain $[1, J_{\max}]$ to $[u_{\min}, u_{\max}]$, denoting the inverse function as $x(u)$. Under this substitution, the oscillatory integral is represented by:

$$I(k) = \int_{u_{\min}}^{u_{\max}} g(u) \cos(2ku - \phi(u))\, du, \qquad \text{where } g(u) = w(x(u)) \left|x'(u)\right|, \phi(u) = 2\psi(x(u)).$$

To bound the integral, we perform integration by parts using the exact differential form:

$$d\left(\sin(2ku - \phi(u))\right) = (2k - \phi'(u)) \cos(2ku - \phi(u))\, du$$

Applying this integration yields:

$$I(k) = \int_{u_{\min}}^{u_{\max}} \frac{g(u)}{2k - \phi'(u)}\, d\left(\sin(2ku - \phi(u))\right).$$

Let $v_k(u) = \frac{g(u)}{2k - \phi'(u)}$, and then evaluating this expression completely gives the boundary terms and the residual integral:

$$I(k) = \left[v_k(u) \sin(2ku - \phi(u))\right]_{u_{\min}}^{u_{\max}} - \int_{u_{\min}}^{u_{\max}} v_k'(u) \sin(2ku - \phi(u))\, du.$$

Assuming that $g(u)$, $\phi(u)$, and their derivatives are continuous and bounded on $[u_{\min}, u_{\max}]$, we define the maximum values:

$$M_0 = \max |g(u)|, \quad M_1 = \max |g'(u)|, \quad L_1 = \max |\phi'(u)|, \quad L_2 = \max |\phi''(u)|$$

For a sufficiently large step k satisfying $k \geqslant L_1$, we have $2k - \phi'(u) \geqslant k$, which allows us to write:

$$k \geqslant L_1, \quad 2k - \phi'(u) \geqslant k$$

$$|v_k(u)| \leqslant \frac{M_0}{k}, \qquad |v_k'(u)| = \left| \frac{g'(u)}{2k - \phi'(u)} + \frac{g(u)\phi''(u)}{(2k - \phi'(u))^2} \right| \leqslant \frac{M_1 + M_0 L_2}{k}.$$

Combining these bounds on the boundary terms and the integral, we bound the absolute value of $I(k)$ as:

$$|I(k)| \leqslant \frac{2M_0 + (u_{\max} - u_{\min})(M_1 + M_0 L_2)}{k} =: \frac{C_1}{k}.$$

This establishes the temporal decay of the continuous oscillatory component. Thus, $I(k) = O(1/k)$ as $k \to \infty$.

We find that for any step k satisfying:

$$k \geqslant C \max\left\{\frac{1}{1-\rho}, \frac{1-\rho}{\eta}\right\}$$

The oscillatory term is strictly dominated by a small fraction of the static mass:

$$|I(k)| \leqslant \frac{1}{4}\int_1^{J_{\max}} w(x)\,dx$$

This guarantees that the oscillatory interference cannot cancel the static component, yielding a strictly positive lower bound for the sum:

$$\sum_{j \leqslant J_{\max}} \lambda_j (\theta_j^*)^2 (1 + A_j^2) \cos^2(k\omega_j - \psi_j) \geqslant c > 0.$$

Thus, the discrete sum is strictly bounded away from zero, yielding $S_{\text{under}}(k) \geqslant c\rho^k$.

Synthesizing the overdamped and underdamped contributions, we obtain the full deterministic bias $S_{\text{polyak}}(k) = S_{\text{over}}(k) + S_{\text{under}}(k)$:

- When $k \lesssim \frac{1-\rho}{\eta}$, combining the short-time overdamped term with the underdamped bound gives:
$$S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{(1-\rho)^2}{\eta}\right)^s.$$
- When $k \gtrsim \frac{1-\rho}{\eta}$, combining the long-time overdamped polynomial tail with the underdamped bound gives:
$$S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

This concludes the proof. □

Theorem D.10 (Two-sided scaling law, mixed regime)**.** *Under Assumptions 3.1 and 3.3, the stability condition $\eta \lesssim \min\{1, B(1-\rho)\}$, and for the mixed regime $\eta \gtrsim (1-\rho)^2$ with $k \gtrsim \frac{1-\rho}{\eta}$, the expected excess risk satisfies:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s + \frac{\sigma^2\eta}{B(1-\rho)}.$$

Proof. By Corollary D.6, the multiplicative gradient noise is strictly absorbed by the additive label noise floor, yielding the asymptotic equivalence:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{polyak}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau). \tag{71}$$

By Lemma D.4, the condition $k \gtrsim \frac{1-\rho}{\eta}$ ensures the label noise summation captures the saturated floor:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{polyak}}(\tau) \eqsim \frac{\sigma^2\eta}{B(1-\rho)}. \tag{72}$$

To evaluate the deterministic bias $S_{\text{polyak}}(k)$ over the entire domain $k \gtrsim \frac{1-\rho}{\eta}$, we analyze the bounds derived in Proposition D.9 by splitting the domain into two consecutive sub-regimes at the momentum relaxation timescale $\frac{1}{1-\rho}$.

For $\frac{1-\rho}{\eta} \lesssim k \lesssim \frac{1}{1-\rho}$, Proposition D.9 gives:

$$S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{(1-\rho)^2}{\eta}\right)^s.$$

Since $k(1-\rho) \lesssim 1$, the exponential transient satisfies:

$$\rho^k = (1-(1-\rho))^k \eqsim e^{-k(1-\rho)} \eqsim 1.$$

In the mixed regime $\eta \gtrsim (1-\rho)^2$, the second term satisfies $\left(\frac{(1-\rho)^2}{\eta}\right)^s \lesssim 1$. Consequently, the deterministic bias is dominated by the non-decayed transient:

$$S_{\text{polyak}}(k) \eqsim 1. \tag{73}$$

Concurrently, we evaluate the target analytical formula $\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s$ in this same sub-regime. Utilizing the condition $k \gtrsim \frac{1-\rho}{\eta}$, we obtain:

$$\left(\frac{1-\rho}{\eta k}\right)^s \lesssim 1.$$

Combining this with $\rho^k \eqsim 1$, the target formula satisfies:

$$\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s \eqsim 1 + \mathcal{O}(1) \eqsim 1. \tag{74}$$

Equating (73) and (74) establishes the equivalence $S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s$ for $\frac{1-\rho}{\eta} \lesssim k \lesssim \frac{1}{1-\rho}$, where the polynomial decay is rigorously absorbed by the $\mathcal{O}(1)$ exponential term.

For $k \gtrsim \frac{1}{1-\rho}$, Proposition D.9 explicitly gives:

$$S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

Synthesizing both sub-regimes, the following unified equivalence holds uniformly for all $k \gtrsim \frac{1-\rho}{\eta}$:

$$S_{\text{polyak}}(k) \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

Substituting this and (72) into (71) yields:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s + \frac{\sigma^2 \eta}{B(1-\rho)}.$$

This completes the proof. □

E Nesterov scaling law

In this section, we prove the Nesterov scaling law. Following the structure of Appendix D, we first decompose the error into deterministic signal and stochastic noise contributions, then classify the spectrum into overdamped and underdamped modes. We finally evaluate the loss in the fully overdamped regime and in the mixed overdamped–underdamped regime.

The main distinction from Polyak is that the squared modulus of an underdamped Nesterov root is $\rho(1-\eta\lambda_j)$ rather than ρ, which changes both the oscillatory transient and the label-noise floor.

E.1 Signal–noise decomposition

Recall the parameter error

$$\mathbf{e}_k := \boldsymbol{\theta}_k - \boldsymbol{\theta}^*.$$

For Nesterov, the gradient is evaluated at the look-ahead point $\boldsymbol{\theta}_k^{\mathrm{la}} = \boldsymbol{\theta}_k + \rho(\boldsymbol{\theta}_k - \boldsymbol{\theta}_{k-1})$. Let

$$\mathbf{e}_k^{\mathrm{la}} := \boldsymbol{\theta}_k^{\mathrm{la}} - \boldsymbol{\theta}^*$$

denote the look-ahead error. Substituting the gradient decomposition into the Nesterov update (4), the error recursion in vector form is:

$$\mathbf{e}_{k+1} = (\mathbf{I} - \eta\mathbf{H})\mathbf{e}_k^{\mathrm{la}} - \eta\boldsymbol{\xi}_k(\boldsymbol{\theta}_k^{\mathrm{la}}).$$

In the diagonal eigenbasis of $\mathbf{H}$, since $\mathbf{e}_k^{\mathrm{la}} = (1+\rho)\mathbf{e}_k - \rho\mathbf{e}_{k-1}$, the j-th coordinate of the Nesterov error recursion satisfies:

$$e_{k+1}(j) = (1+\rho)(1-\eta\lambda_j)e_k(j) - \rho(1-\eta\lambda_j)e_{k-1}(j) - \eta\xi_k^{\mathrm{la}}(j),$$

where $\xi_k^{\mathrm{la}}(j)$ is the j-th coordinate of the stochastic gradient noise $\boldsymbol{\xi}_k(\boldsymbol{\theta}_k^{\mathrm{la}})$ evaluated at the look-ahead point.

Its homogeneous characteristic polynomial is

$$y^2 - (1+\rho)(1-\eta\lambda_j)y + \rho(1-\eta\lambda_j) = 0.$$

Let $y_{j,+}$ and $y_{j,-}$ be the two roots of

$$y^2 - (1+\rho)(1-\eta\lambda_j)y + \rho(1-\eta\lambda_j) = 0.$$

Define

$$G_k^{\mathrm{nesterov}}(j) := \frac{y_{j,+}^{k+1} - y_{j,-}^{k+1}}{y_{j,+} - y_{j,-}}, \qquad k \geqslant 0,$$

with the usual limiting definition at a repeated root. Then

$$e_k(j) = e_k^{\mathrm{det}}(j) - \eta\sum_{m=0}^{k-1} G_{k-1-m}^{\mathrm{nesterov}}(j)\xi_m^{\mathrm{la}}(j), \tag{75}$$

where $e_k^{\mathrm{det}}(j)$ is the deterministic Nesterov error trajectory.

Proposition E.1 (Impulse-response representation)**.** *The Green function is the unique solution of*

$$G_{k+1}^{\mathrm{nesterov}}(j) = (1+\rho)(1-\eta\lambda_j)G_k^{\mathrm{nesterov}}(j) - \rho(1-\eta\lambda_j)G_{k-1}^{\mathrm{nesterov}}(j), \qquad G_0^{\mathrm{nesterov}}(j) = 1, \qquad G_{-1}^{\mathrm{nesterov}}(j) = 0.$$

Moreover, if $e_0^{\mathrm{det}}(j) = -\theta_j^*$ *and* $e_1^{\mathrm{det}}(j) = -(1-\eta\lambda_j)\theta_j^*$*, then*

$$e_k^{\mathrm{det}}(j) = -\frac{(1-\eta\lambda_j - y_{j,-})y_{j,+}^k - (1-\eta\lambda_j - y_{j,+})y_{j,-}^k}{y_{j,+} - y_{j,-}}\,\theta_j^*.$$

Proof. Fix a coordinate j. The inhomogeneous scalar recursion is

$$e_{k+1}(j) = (1+\rho)(1-\eta\lambda_j)e_k(j) - \rho(1-\eta\lambda_j)e_{k-1}(j) - \eta\xi_k^{\mathrm{la}}(j).$$

The impulse response for this second-order recursion is the sequence $G_k^{\mathrm{nesterov}}(j)$ satisfying

$$G_{k+1}^{\mathrm{nesterov}}(j) = (1+\rho)(1-\eta\lambda_j)G_k^{\mathrm{nesterov}}(j) - \rho(1-\eta\lambda_j)G_{k-1}^{\mathrm{nesterov}}(j), \qquad G_0^{\mathrm{nesterov}}(j) = 1, \qquad G_{-1}^{\mathrm{nesterov}}(j) = 0.$$

Indeed, if an impulse $-\eta\xi_m^{\mathrm{la}}(j)$ is inserted at time m, then its contribution to $e_{m+1}(j)$ is $-\eta\xi_m^{\mathrm{la}}(j)$, its contribution to $e_{m+2}(j)$ is $-\eta(1+\rho)(1-\eta\lambda_j)\xi_m^{\mathrm{la}}(j)$, and subsequent contributions obey the same homogeneous recursion. Hence the contribution at time k is $-\eta G_{k-1-m}^{\mathrm{nesterov}}(j)\xi_m^{\mathrm{la}}(j)$ for $0 \leqslant m \leqslant k-1$. Summing over all previous impulses gives

$$e_k(j) = e_k^{\mathrm{det}}(j) - \eta\sum_{m=0}^{k-1} G_{k-1-m}^{\mathrm{nesterov}}(j)\xi_m^{\mathrm{la}}(j),$$

where $e_k^{\mathrm{det}}(j)$ solves the homogeneous recursion with the same initialization.

It remains to compute the closed form of $G_k^{\mathrm{nesterov}}(j)$. If $y_{j,+} \neq y_{j,-}$, then the general solution of the homogeneous recurrence is a linear combination of $y_{j,+}^k$ and $y_{j,-}^k$. The initial conditions $G_{-1}^{\mathrm{nesterov}}(j) = 0$ and $G_0^{\mathrm{nesterov}}(j) = 1$ give

$$G_k^{\mathrm{nesterov}}(j) = \frac{y_{j,+}^{k+1} - y_{j,-}^{k+1}}{y_{j,+} - y_{j,-}}, \qquad k \geqslant 0.$$

If $y_{j,+} = y_{j,-}$, this formula is interpreted by continuity, giving $G_k^{\mathrm{nesterov}}(j) = (k+1)y_{j,+}^k$. Thus the displayed formula is exactly the unique Green function of the Nesterov coordinate recursion.

Now consider the deterministic trajectory. It satisfies

$$e_{k+1}^{\mathrm{det}}(j) = (1+\rho)(1-\eta\lambda_j)e_k^{\mathrm{det}}(j) - \rho(1-\eta\lambda_j)e_{k-1}^{\mathrm{det}}(j), \qquad e_0^{\mathrm{det}}(j) = -\theta_j^*, \qquad e_1^{\mathrm{det}}(j) = -(1-\eta\lambda_j)\theta_j^*.$$

For $y_{j,+} \neq y_{j,-}$, write the deterministic solution as a linear combination of $y_{j,+}^k$ and $y_{j,-}^k$. Solving the two-by-two linear system determined by the two displayed initial conditions gives

$$e_k^{\mathrm{det}}(j) = -\frac{(1-\eta\lambda_j - y_{j,-})y_{j,+}^k - (1-\eta\lambda_j - y_{j,+})y_{j,-}^k}{y_{j,+} - y_{j,-}}\,\theta_j^*.$$

The repeated-root case follows by taking the same continuous limit. This proves the claim. □

E.2 Overdamped and underdamped Modes decomposition

Since both the deterministic error signal $e_k^{\mathrm{det}}(j)$ and the discrete Green function $G_k^{\mathrm{nesterov}}(j)$ obey the same homogeneous difference equation, their behavior is dictated by the same characteristic roots. Depending on the sign of the discriminant, these roots transition from real roots to a complex-conjugate pair. The following proposition establishes the exact Nesterov damping threshold, defines the crossover index $J_{\max}$, and partitions the spectrum into the two global regimes.

Proposition E.2 (Exact spectral decomposition and crossover boundary)**.** *For each coordinate j, the characteristic roots of the Nesterov recursion are*

$$y_{j,\pm} = \frac{(1+\rho)(1-\eta\lambda_j) \pm \sqrt{\Delta_j^{\mathrm{nesterov}}}}{2},$$

where

$$\Delta_j^{\mathrm{nesterov}} := (1+\rho)^2(1-\eta\lambda_j)^2 - 4\rho(1-\eta\lambda_j).$$

In the stable small-step range $0 < \eta\lambda_j < 1$, the critical transition $\Delta_j^{\text{nesterov}} = 0$ corresponds exactly to

$$\eta\lambda_j = \frac{(1-\rho)^2}{(1+\rho)^2},$$

which is asymptotically equivalent to $\eta\lambda_j \eqsim (1-\rho)^2$. Thus the individual eigen-directions are classified as follows:

1. ***Overdamped Modes*** $\big(\eta\lambda_j < (1-\rho)^2/(1+\rho)^2\big)$*: the roots are real.*
2. ***Underdamped Modes*** $\big(\eta\lambda_j > (1-\rho)^2/(1+\rho)^2\big)$*: the roots form a complex-conjugate pair.*

In the Mixed Overdamped–Underdamped Regime, the boundary index $J_{\max}$ separating the underdamped modes from the overdamped modes is defined by the crossover relation $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$:

$$J_{\max} \eqsim \left(\frac{\eta}{(1-\rho)^2}\right)^{1/\beta}. \tag{76}$$

Consequently, based on the maximum eigenvalue $\lambda_{\max}$, Nesterov operates in one of two global regimes:

- ***Fully Overdamped Regime*** $\big(\eta\lambda_{\max} \lesssim (1-\rho)^2\big)$*: all spectral modes are overdamped.*
- ***Mixed Overdamped–Underdamped Regime*** $\big(\eta\lambda_{\max} \gtrsim (1-\rho)^2\big)$*: the high-curvature modes $j \leqslant J_{\max}$ are underdamped, whereas the flat modes $j > J_{\max}$ are overdamped.*

Proof. In the stable small-step range, $1 - \eta\lambda_j \in (0,1)$. The condition $\Delta_j^{\text{nesterov}} < 0$ is

$$(1+\rho)^2(1-\eta\lambda_j)^2 - 4\rho(1-\eta\lambda_j) < 0.$$

Since $1-\eta\lambda_j > 0$, this is equivalent to $1-\eta\lambda_j < \frac{4\rho}{(1+\rho)^2}$, and hence $\eta\lambda_j > 1 - \frac{4\rho}{(1+\rho)^2} = \frac{(1-\rho)^2}{(1+\rho)^2}$. Therefore the equality case gives the exact damping boundary. Since $(1+\rho)^2 \eqsim 1$ for $0 \leqslant \rho < 1$, this boundary has the scaling form $\eta\lambda_j \eqsim (1-\rho)^2$.

To locate the crossover index in the Mixed Overdamped–Underdamped Regime, use $\lambda_j \eqsim j^{-\beta}$ and impose $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$. Then $\eta J_{\max}^{-\beta} \eqsim (1-\rho)^2$, which gives

$$J_{\max} \eqsim \left(\frac{\eta}{(1-\rho)^2}\right)^{1/\beta}.$$

Because the eigenvalues are non-increasing, the modes $j \leqslant J_{\max}$ lie above the damping threshold and are underdamped, while the modes $j > J_{\max}$ lie below the threshold and are overdamped. Examining whether this crossover occurs below the leading eigenvalue yields the two global regimes stated above. □

E.3 Risk recursion

Define the deterministic bias and the Nesterov convolution kernel by

$$S_{\text{nesterov}}(k) := \frac{1}{2}\sum_{j\geqslant 1} \lambda_j \big(e_k^{\text{det}}(j)\big)^2, \qquad \mathcal{K}_{\text{nesterov}}(k) := \sum_{j\geqslant 1} \lambda_j^2 \big(G_k^{\text{nesterov}}(j)\big)^2.$$

Because the stochastic gradient is evaluated at the look-ahead point, define

$$\mathbf{e}_k^{\text{la}} := \mathbf{e}_k + \rho(\mathbf{e}_k - \mathbf{e}_{k-1}), \qquad \mathcal{L}_k := \frac{1}{2}\|\mathbf{e}_k^{\text{la}}\|_{\mathbf{H}}^2.$$

Lemma E.3 (Nesterov look-ahead loss control)**.** *Assume the stable small-step condition $\eta\lambda_1 \leqslant c_{\mathrm{st}}$, where $c_{\mathrm{st}} > 0$ is sufficiently small. Then the look-ahead loss admits the coordinate expansion*

$$\mathcal{L}_m = \frac{1}{2}\sum_{j\geqslant 1}\lambda_j\left((1+\rho)e_m(j) - \rho e_{m-1}(j)\right)^2.$$

Moreover, the deterministic look-ahead loss is comparable to the next deterministic Nesterov loss:

$$\frac{1}{2}\sum_{j\geqslant 1}\lambda_j\left((1+\rho)e_m^{\mathrm{det}}(j) - \rho e_{m-1}^{\mathrm{det}}(j)\right)^2 \eqsim S_{\mathrm{nesterov}}(m+1).$$

For the stochastic iterates, the corresponding expectation-level control is

$$c\,\mathbb{E}\mathcal{E}_{m+1} - C\frac{\eta^2\sigma^2}{B} \leqslant \mathbb{E}\mathcal{L}_m \leqslant C\,\mathbb{E}\mathcal{E}_{m+1},$$

where the constants are independent of m, η, ρ, B.

Proof. The coordinate expansion follows directly from $\mathbf{e}_m^{\mathrm{la}} = (1+\rho)\mathbf{e}_m - \rho\mathbf{e}_{m-1}$ and the definition of the H-norm.

For the deterministic comparison, the deterministic Nesterov update gives, for each coordinate j,

$$e_{m+1}^{\mathrm{det}}(j) = (1-\eta\lambda_j)\left((1+\rho)e_m^{\mathrm{det}}(j) - \rho e_{m-1}^{\mathrm{det}}(j)\right).$$

Since $0 < 1-\eta\lambda_j \leqslant 1$ and $1-\eta\lambda_j \geqslant 1-c_{\mathrm{st}}$, we have

$$\left((1+\rho)e_m^{\mathrm{det}}(j) - \rho e_{m-1}^{\mathrm{det}}(j)\right)^2 \eqsim \left(e_{m+1}^{\mathrm{det}}(j)\right)^2.$$

Multiplying by λ_j and summing over $j \geqslant 1$ yields the deterministic claim.

For the stochastic comparison, the actual Nesterov update in coordinate j is

$$e_{m+1}(j) = (1-\eta\lambda_j)\left((1+\rho)e_m(j) - \rho e_{m-1}(j)\right) - \eta\xi_m^{\mathrm{la}}(j).$$

Conditioning on the past, the noise is mean zero at the look-ahead point. Therefore the mixed term vanishes and

$$\begin{aligned}\mathbb{E}\mathcal{E}_{m+1} \eqsim &\sum_{j\geqslant 1}\lambda_j(1-\eta\lambda_j)^2\mathbb{E}\left((1+\rho)e_m(j) - \rho e_{m-1}(j)\right)^2 \\ &+ \eta^2\sum_{j\geqslant 1}\lambda_j\mathbb{E}\left(\xi_m^{\mathrm{la}}(j)\right)^2.\end{aligned}$$

The first term is comparable to $\mathbb{E}\mathcal{L}_m$, because $1-\eta\lambda_j$ is bounded above and below by positive constants. For the second term, Proposition 3.4 gives

$$\sum_{j\geqslant 1}\lambda_j\mathbb{E}\left(\xi_m^{\mathrm{la}}(j)\right)^2 \lesssim \frac{1}{B}\left(\sigma^2 + \mathbb{E}\mathcal{L}_m\right)\sum_{j\geqslant 1}\lambda_j^2.$$

Since $\sum_{j\geqslant 1}\lambda_j^2 < \infty$ and c_{st} is chosen small, the term proportional to $\eta^2\mathbb{E}\mathcal{L}_m/B$ is absorbed into the comparable first term. Thus

$$\mathbb{E}\mathcal{E}_{m+1} \eqsim \mathbb{E}\mathcal{L}_m + O\left(\frac{\eta^2\sigma^2}{B}\right),$$

which gives

$$c\,\mathbb{E}\mathcal{E}_{m+1} - C\frac{\eta^2\sigma^2}{B} \leqslant \mathbb{E}\mathcal{L}_m \leqslant C\,\mathbb{E}\mathcal{E}_{m+1}.$$

This proves the lemma. □

Proposition E.4 (Nesterov risk recursion)**.** *Under the covariance relation in Proposition 3.4, the expected excess risk satisfies the two-sided recursion*

$$\mathbb{E}\mathcal{E}_k \eqsim S_{\text{nesterov}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau) + \frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau)\mathbb{E}\mathcal{L}_{k-1-\tau}. \tag{77}$$

Moreover, Lemma E.3 gives the required control of the look-ahead loss by the next-step loss. In particular, under the stable small-step condition,

$$c\,\mathbb{E}\mathcal{E}_{m+1} - C\frac{\eta^2\sigma^2}{B} \leqslant \mathbb{E}\mathcal{L}_m \leqslant C\,\mathbb{E}\mathcal{E}_{m+1}.$$

Proof. Substitute the Nesterov impulse-response representation (75) into the coordinate-wise excess risk. For every coordinate j,

$$e_k(j) = e_k^{\text{det}}(j) - \eta\sum_{m=0}^{k-1} G_{k-1-m}^{\text{nesterov}}(j)\xi_m^{\text{la}}(j).$$

The noise $\xi_m^{\text{la}}(j)$ is conditionally mean zero given the past and the look-ahead point $\mathbf{e}_m^{\text{la}}$, and the fresh mini-batches are independent across time. Therefore all mixed deterministic–stochastic and cross-time stochastic terms vanish after taking expectation. Hence

$$\mathbb{E}[e_k(j)^2] = \big(e_k^{\text{det}}(j)\big)^2 + \eta^2\sum_{m=0}^{k-1}\big(G_{k-1-m}^{\text{nesterov}}(j)\big)^2\mathbb{E}\big(\xi_m^{\text{la}}(j)\big)^2,$$

up to the same universal constants in the covariance comparison of Proposition 3.4. Multiplying by λ_j and summing over $j \geqslant 1$ gives

$$\mathbb{E}\mathcal{E}_k \eqsim \sum_{j\geqslant 1}\lambda_j\big(e_k^{\text{det}}(j)\big)^2 + \eta^2\sum_{m=0}^{k-1}\sum_{j\geqslant 1}\lambda_j\big(G_{k-1-m}^{\text{nesterov}}(j)\big)^2\mathbb{E}\big(\xi_m^{\text{la}}(j)\big)^2.$$

Since Nesterov evaluates the stochastic gradient at the look-ahead point $\mathbf{e}_m^{\text{la}}$, Proposition 3.4 gives the coordinate-wise variance relation

$$\mathbb{E}\big(\xi_m^{\text{la}}(j)\big)^2 \eqsim \frac{1}{B}\lambda_j\big(\sigma^2 + \mathbb{E}\mathcal{L}_m\big).$$

Substituting this into the previous display yields

$$\mathbb{E}\mathcal{E}_k \eqsim \sum_{j\geqslant 1}\lambda_j\big(e_k^{\text{det}}(j)\big)^2 + \frac{\eta^2}{B}\sum_{m=0}^{k-1}\big(\sigma^2 + \mathbb{E}\mathcal{L}_m\big)\sum_{j\geqslant 1}\lambda_j^2\big(G_{k-1-m}^{\text{nesterov}}(j)\big)^2.$$

Using the definitions of S_{nesterov} and $\mathcal{K}_{\text{nesterov}}$, and then changing variables $\tau = k-1-m$, we obtain

$$\mathbb{E}\mathcal{E}_k \eqsim S_{\text{nesterov}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau) + \frac{\eta^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau)\mathbb{E}\mathcal{L}_{k-1-\tau}.$$

This proves the claimed two-sided recursion.

It remains to control the look-ahead risk. This is exactly Lemma E.3, which expands $\mathcal{L}_m$ in coordinates and then uses the Nesterov update from step m to step $m+1$ to compare the look-ahead loss with the next-step loss. □

E.4 Label noise calculation

The convolution estimates in Appendix D.3 will be used throughout this section. It remains to identify the Nesterov label-noise floor and to replace the look-ahead loss in the multiplicative-noise recursion.

Lemma E.5 (Nesterov label-noise contribution)**.** *Assume the stable small-step condition $\eta\lambda_1 \leqslant c_{\mathrm{st}}$, where $c_{\mathrm{st}} > 0$ is a sufficiently small universal constant. There exist constants $C, c_-, c_+ > 0$ such that if*

$$k \geqslant C \max\left\{\frac{1}{1-\rho}, \frac{1-\rho}{\eta}\right\},$$

then

$$c_- \frac{\sigma^2}{B} \min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\} \leqslant \frac{\eta^2\sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\mathrm{nesterov}}(\tau) \leqslant c_+ \frac{\sigma^2}{B} \min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\}.$$

Hence the Nesterov label-noise floor is

$$\frac{\eta^2\sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\mathrm{nesterov}}(\tau) \eqsim \frac{\sigma^2}{B} \min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\}.$$

Proof. We first compute the saturated Green energy in one eigendirection. The Nesterov Green function satisfies

$$G_{k+1}^{\mathrm{nesterov}}(j) = (1+\rho)(1-\eta\lambda_j)G_k^{\mathrm{nesterov}}(j) - \rho(1-\eta\lambda_j)G_{k-1}^{\mathrm{nesterov}}(j), \qquad G_0^{\mathrm{nesterov}}(j) = 1, \qquad G_{-1}^{\mathrm{nesterov}}(j) = 0.$$

Multiplying this recursion by $G_k^{\mathrm{nesterov}}(j)$, summing over $k \geqslant 0$, and then squaring the recursion and summing over $k \geqslant 0$ gives the standard second-order energy identity

$$\sum_{k=0}^{\infty} \big(G_k^{\mathrm{nesterov}}(j)\big)^2 = \frac{1+\rho(1-\eta\lambda_j)}{\big(1-\rho(1-\eta\lambda_j)\big)\Big(\big(1+\rho(1-\eta\lambda_j)\big)^2 - (1+\rho)^2(1-\eta\lambda_j)^2\Big)}.$$

The second factor in the denominator simplifies as

$$\big(1+\rho(1-\eta\lambda_j)\big)^2 - (1+\rho)^2(1-\eta\lambda_j)^2 = \eta\lambda_j\left(1+(1+2\rho)(1-\eta\lambda_j)\right).$$

Under $\eta\lambda_1 \leqslant c_{\mathrm{st}}$, the remaining factors are bounded above and below by constants except for $1-\rho+\eta\lambda_j$ and $\eta\lambda_j$. Therefore

$$\sum_{k=0}^{\infty} \big(G_k^{\mathrm{nesterov}}(j)\big)^2 \eqsim \frac{1}{\eta\lambda_j(1-\rho+\eta\lambda_j)}.$$

Consequently the saturated label-noise contribution is

$$\begin{aligned}\frac{\eta^2\sigma^2}{B} \sum_{j\geqslant 1} \lambda_j^2 \sum_{k=0}^{\infty} \big(G_k^{\mathrm{nesterov}}(j)\big)^2 &\eqsim \frac{\eta\sigma^2}{B} \sum_{j\geqslant 1} \frac{\lambda_j}{1-\rho+\eta\lambda_j} \\ &= \frac{\sigma^2}{B} \sum_{j\geqslant 1} \frac{\lambda_j}{\lambda_j + \frac{1-\rho}{\eta}}.\end{aligned}$$

This is the point where the Nesterov computation differs from the Heavy Ball computation: the effective root product is $\rho(1-\eta\lambda_j)$, so the saturated spectral weight contains $\lambda_j/(\lambda_j+(1-\rho)/\eta)$, rather than simply λ_j.

We next estimate the spectral sum. If $\eta/(1-\rho) \leqslant 1$, then $(1-\rho)/\eta \gtrsim 1$, and hence

$$\sum_{j\geqslant 1} \frac{\lambda_j}{\lambda_j + \frac{1-\rho}{\eta}} \eqsim \frac{\eta}{1-\rho} \sum_{j\geqslant 1} \lambda_j \eqsim \frac{\eta}{1-\rho}.$$

If $\eta/(1-\rho) \geqslant 1$, then the spectral cutoff is $\lambda_j \eqsim \frac{1-\rho}{\eta}$. Since $\lambda_j \eqsim j^{-\beta}$, the number of indices above this cutoff is $\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}$. The modes above the cutoff each contribute a constant amount to the summand, while the tail below the cutoff contributes the same order:

$$\sum_{\lambda_j < \frac{1-\rho}{\eta}} \frac{\lambda_j}{\frac{1-\rho}{\eta}} \eqsim \left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}.$$

Thus, in all cases,

$$\sum_{j\geqslant 1} \frac{\lambda_j}{\lambda_j + \frac{1-\rho}{\eta}} \eqsim \min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\}.$$

It remains to pass from the saturated energy to the finite-time sum. The root formula for $G_k^{\mathrm{nesterov}}(j)$ gives a uniform tail estimate: after increasing C if necessary, the time condition

$$k \geqslant C \max\left\{\frac{1}{1-\rho}, \frac{1-\rho}{\eta}\right\}$$

ensures that the finite partial sum captures a fixed positive fraction of the saturated Green energy on the spectral region responsible for the lower bounds above. When $\eta/(1-\rho) \leqslant 1$, this region may be taken to be a fixed finite block of leading eigendirections, whose relaxation time is of order $(1-\rho)/\eta$. When $\eta/(1-\rho) \geqslant 1$, it may be taken to be the block $\lambda_j \gtrsim (1-\rho)/\eta$, whose relaxation time is at most of order $(1-\rho)^{-1}$. Therefore

$$\sum_{\tau=0}^{k-1} \mathcal{K}_{\mathrm{nesterov}}(\tau) \eqsim \sum_{j\geqslant 1} \lambda_j^2 \sum_{\tau=0}^{\infty} \left(G_\tau^{\mathrm{nesterov}}(j)\right)^2$$

up to constants for the purpose of the two-sided label-noise estimate. Combining this finite-time comparison with the saturated computation above proves the lemma. □

E.5 Uniform boundedness of expected risk

As long as the additive label noise is strictly positive ($\sigma > 0$), we can prove that the expected risk is uniformly bounded, which allows the multiplicative gradient noise to be completely absorbed by the additive label noise floor.

Lemma E.6 (Uniform boundedness of expected risk)**.** *Suppose $\sigma > 0$. Under the stability condition $\eta \leqslant cB^{\beta}(1-\rho)$ for a sufficiently small universal constant $c > 0$, the expected excess risk of Nesterov is uniformly bounded for all $k \geqslant 0$:*

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M,$$

where $M > 0$ is a constant independent of k, η, B, and ρ.

Proof. We proceed by induction on k. By Proposition E.4, there exist universal constants $C_1, C_2, C_3 > 0$ such that for any $k \geqslant 1$:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant C_1 S_{\mathrm{nesterov}}(k) + C_2 \frac{\eta^2\sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\mathrm{nesterov}}(\tau) + C_3 \frac{\eta^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\mathrm{nesterov}}(\tau) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-\tau})]. \quad (78)$$

First, Propositions E.8 and E.10 establish that the deterministic bias monotonically decays or is uniformly bounded by its initialization across all regimes. Thus:

$$S_{\text{nesterov}}(k) \leqslant C_h^2 \mathcal{E}(\boldsymbol{\theta}_0) =: M_1.$$

Second, applying Lemma E.5, the infinite sum of the convolution kernel satisfies:

$$\frac{\eta^2}{B} \sum_{\tau=0}^{\infty} \mathcal{K}_{\text{nesterov}}(\tau) \leqslant \frac{C_K}{B} \min\left\{ \left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho} \right\}.$$

Therefore, the label noise contribution is universally bounded by:

$$C_2 \frac{\eta^2 \sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau) \leqslant C_2 \sigma^2 \frac{C_K}{B} \left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}.$$

Under the stability condition $\eta \leqslant cB^{\beta}(1-\rho)$, we have $\left(\frac{\eta}{1-\rho}\right)^{1/\beta} \leqslant c^{1/\beta} B$. Thus, the label noise is bounded by

$$C_2 \sigma^2 C_K c^{1/\beta} =: M_2.$$

Assuming inductively $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_m)] \leqslant M$ for $m < k$, the multiplicative noise term yields:

$$C_3 \frac{\eta^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-\tau})] \leqslant C_3 \left(\frac{\eta^2}{B} \sum_{\tau=0}^{\infty} \mathcal{K}_{\text{nesterov}}(\tau) \right) M \leqslant C_3 C_K \frac{1}{B} \left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}} M \leqslant C_3 C_K c^{1/\beta} M.$$

Choosing c sufficiently small such that $c^{1/\beta} \leqslant \frac{1}{2C_3 C_K}$ limits this term to $M/2$. Substituting into (78) yields:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M_1 + M_2 + \frac{M}{2}.$$

Setting $M = 2(M_1 + M_2)$ guarantees $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \leqslant M$. The base case $k = 0$ holds trivially since $\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_0)] = S_{\text{nesterov}}(0) \leqslant M_1 \leqslant M$. This completes the induction. □

Corollary E.7 (Absorption of multiplicative noise)**.** *For $\sigma > 0$, under the stability condition $\eta \lesssim \min\{1, B^{\beta}(1-\rho)\}$, the multiplicative gradient noise is strictly bounded by the additive label noise up to a constant factor:*

$$\frac{\eta^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau) \mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_{k-\tau})] \leqslant \frac{M}{\sigma^2} \left(\frac{\eta^2 \sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau) \right).$$

Consequently, the multiplicative noise can be safely absorbed, and the expected excess risk is asymptotically equivalent to the sum of the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{nesterov}}(k) + \frac{\eta^2 \sigma^2}{B} \sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau).$$

E.6 Risk dynamics in the fully overdamped regime

In the fully overdamped regime, every characteristic root is real and the deterministic bias has no oscillatory component.

Proposition E.8 (Deterministic bias, fully overdamped regime)**.** *Under Assumptions 3.1 and 3.3, in the fully overdamped regime $\eta\lambda_1 \leqslant c_0(1-\rho)^2$ with a sufficiently small universal constant $c_0 > 0$, we have:*

$$S_{\text{nesterov}}(k) \eqsim \begin{cases} 1, & k \lesssim \frac{1-\rho}{\eta}, \\ \left(\frac{1-\rho}{\eta k}\right)^s, & k \gtrsim \frac{1-\rho}{\eta}. \end{cases}$$

Proof. For each coordinate, write the deterministic multiplier as

$$e_k^{\mathrm{det}}(j) = -h_k^{\mathrm{nesterov}}(j)\theta_j^*.$$

The deterministic Nesterov coordinate recursion gives

$$h_{k+1}^{\mathrm{nesterov}}(j) = (1+\rho)(1-\eta\lambda_j)h_k^{\mathrm{nesterov}}(j) - \rho(1-\eta\lambda_j)h_{k-1}^{\mathrm{nesterov}}(j), \qquad h_0^{\mathrm{nesterov}}(j) = 1, \qquad h_1^{\mathrm{nesterov}}(j) = 1-\eta\lambda_j.$$

Its characteristic roots are

$$y_{j,+}, y_{j,-} = \frac{(1+\rho)(1-\eta\lambda_j) \pm \sqrt{(1+\rho)^2(1-\eta\lambda_j)^2 - 4\rho(1-\eta\lambda_j)}}{2}.$$

In the fully overdamped regime, $\eta\lambda_j \leqslant c_0(1-\rho)^2$ for every j. To locate the two roots, evaluate the characteristic polynomial at $1-u$. A direct expansion gives

$$\begin{aligned}(1-u)^2 - (1+\rho)(1-\eta\lambda_j)(1-u) + \rho(1-\eta\lambda_j) \\ = \eta\lambda_j - \big(1-\rho+(1+\rho)\eta\lambda_j\big)u + u^2.\end{aligned}$$

Taking $c_0 > 0$ sufficiently small, the two evaluations

$$\eta\lambda_j - \big(1-\rho+(1+\rho)\eta\lambda_j\big)\frac{\eta\lambda_j}{2(1-\rho)} + \left(\frac{\eta\lambda_j}{2(1-\rho)}\right)^2 > 0$$

and

$$\eta\lambda_j - \big(1-\rho+(1+\rho)\eta\lambda_j\big)\frac{2\eta\lambda_j}{1-\rho} + \left(\frac{2\eta\lambda_j}{1-\rho}\right)^2 < 0$$

hold uniformly for all $0 < \eta\lambda_j \leqslant c_0(1-\rho)^2$. Hence the slow root satisfies

$$1 - \frac{2\eta\lambda_j}{1-\rho} \leqslant y_{j,+} \leqslant 1 - \frac{\eta\lambda_j}{2(1-\rho)}.$$

Moreover,

$$y_{j,+}y_{j,-} = \rho(1-\eta\lambda_j).$$

Using the lower bound on $y_{j,+}$, and decreasing c_0 if necessary, the fast root obeys

$$0 \leqslant y_{j,-} \leqslant 1 - \frac{1-\rho}{2}.$$

The solution of the two-point initial value problem is

$$h_k^{\mathrm{nesterov}}(j) = \frac{1-\eta\lambda_j - y_{j,-}}{y_{j,+}-y_{j,-}}y_{j,+}^k + \frac{y_{j,+} - (1-\eta\lambda_j)}{y_{j,+}-y_{j,-}}y_{j,-}^k.$$

The root bounds above imply

$$\frac{1}{2} \leqslant \frac{1-\eta\lambda_j - y_{j,-}}{y_{j,+}-y_{j,-}} \leqslant 2, \qquad \left|\frac{y_{j,+} - (1-\eta\lambda_j)}{y_{j,+}-y_{j,-}}\right| \leqslant Cc_0.$$

Indeed, $y_{j,+} - y_{j,-} \eqsim 1-\rho$, while $1-\eta\lambda_j - y_{j,-} \eqsim 1-\rho$, and

$$|y_{j,+} - (1-\eta\lambda_j)| \lesssim \frac{\eta\lambda_j}{1-\rho} \leqslant c_0(1-\rho).$$

Thus the slow component has a uniformly positive coefficient and the fast component has a coefficient of order c_0.

To evaluate the deterministic bias $S_{\text{nesterov}}(k) = \frac{1}{2}\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2\left(h_k^{\text{nesterov}}(j)\right)^2$, we divide the time k into three successive stages.

Stage 1: $k \lesssim \frac{1}{1-\rho}$.

In this extremely short-time regime, for the leading modes ($j = \mathcal{O}(1)$), since $\eta\lambda_j \leqslant c_0(1-\rho)^2$, the slow root satisfies $y_{j,+} \geqslant 1 - 2c_0(1-\rho)$. For $k \lesssim \frac{1}{1-\rho}$, we have

$$y_{j,+}^k \geqslant (1 - 2c_0(1-\rho))^{\frac{C}{1-\rho}} \eqsim 1.$$

Since the dynamics are unconditionally stable ($h_k^{\text{nesterov}}(j) \leqslant 1$), it follows that $h_k^{\text{nesterov}}(j) \eqsim 1$ for the leading modes. Consequently, the bias is dominated by these non-decayed modes:

$$S_{\text{nesterov}}(k) \eqsim \sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2 \eqsim 1.$$

Stage 2: $\frac{1}{1-\rho} \lesssim k \lesssim \frac{1-\rho}{\eta}$.

For $k \gtrsim \frac{1}{1-\rho}$, the fast root decays exponentially since $y_{j,-}^k \leqslant \left(1 - \frac{1-\rho}{2}\right)^k \ll 1$. The multiplier is overwhelmingly dominated by the slow root:

$$h_k^{\text{nesterov}}(j) \simeq \frac{1 - \eta\lambda_j - y_{j,-}}{y_{j,+} - y_{j,-}} y_{j,+}^k \simeq y_{j,+}^k.$$

For the leading modes with $\lambda_j = \Theta(1)$, within the time window $k \lesssim \frac{1-\rho}{\eta}$, the exponent is bounded by:

$$y_{j,+}^k \eqsim \left(1 - c\frac{\eta\lambda_j}{1-\rho}\right)^k \eqsim \exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right) \eqsim 1,$$

where we used the condition $\frac{\eta\lambda_j}{1-\rho}k \lesssim \lambda_j \leqslant \lambda_1 = \mathcal{O}(1)$. Thus, the leading modes have not yet experienced significant exponential decay, preserving $h_k^{\text{nesterov}}(j) \eqsim 1$. As a result, the deterministic bias remains unchanged:

$$S_{\text{nesterov}}(k) \eqsim 1.$$

Stage 3: $k \gtrsim \frac{1-\rho}{\eta}$.

In this long-time regime, the exponential decay of the slow root becomes significant across the spectrum. For the slow root, the preceding bounds give

$$\exp\left(-C\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant y_{j,+}^k \leqslant \exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right).$$

For the fast root, since $\eta\lambda_j/(1-\rho) \leqslant c_0(1-\rho)$, the ratio of the fast component to the slow component is bounded by

$$Cc_0\exp\left(-c(1-\rho)k + C\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant Cc_0\exp\left(-c(1-\rho)k\right),$$

after decreasing c_0. Therefore, for all $k \geqslant C(1-\rho)^{-1}$, the fast component is at most a fixed fraction of the slow component. Consequently there exist constants $c, C > 0$ such that

$$c\exp\left(-C\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant \left|h_k^{\text{nesterov}}(j)\right| \leqslant C\exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right).$$

Substituting this coordinate estimate into the deterministic bias gives

$$c\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2\exp\left(-C\frac{\eta\lambda_j}{1-\rho}k\right) \leqslant S_{\text{nesterov}}(k) \leqslant C\sum_{j\geqslant 1}\lambda_j(\theta_j^*)^2\exp\left(-c\frac{\eta\lambda_j}{1-\rho}k\right).$$

By Assumption 3.3,

$$\lambda_j(\theta_j^*)^2 \eqsim j^{-1-\beta s}, \qquad \lambda_j \eqsim j^{-\beta}.$$

It remains to estimate

$$\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho} j^{-\beta}\right).$$

The condition $k \gtrsim \frac{1-\rho}{\eta}$ gives $\frac{\eta k}{1-\rho} \geqslant C$.

For the lower bound, take the tail $j \geqslant \left\lceil \left(\frac{\eta k}{1-\rho}\right)^{1/\beta} \right\rceil$. On this tail the exponential factor is bounded below by a positive constant, hence

$$\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-C\frac{\eta k}{1-\rho} j^{-\beta}\right) \geqslant c\left(\frac{\eta k}{1-\rho}\right)^{-s}.$$

For the upper bound, an integral comparison gives

$$\begin{aligned}\sum_{j\geqslant 1} j^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho} j^{-\beta}\right) &\leqslant C\int_1^\infty x^{-1-\beta s} \exp\left(-c\frac{\eta k}{1-\rho} x^{-\beta}\right) dx \\ &\leqslant C\left(\frac{\eta k}{1-\rho}\right)^{-s}.\end{aligned}$$

Combining the spectral upper and lower bounds yields

$$S_{\text{nesterov}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s.$$

Synthesizing the three stages completes the proof. □

Theorem E.9 (Scaling law, fully overdamped regime)**.** *Under Assumptions 3.1 and 3.3, the stability condition $\eta \lesssim B^\beta(1-\rho)$, and for the fully overdamped regime $\eta \lesssim (1-\rho)^2$ with $k \gtrsim \frac{1-\rho}{\eta}$, the expected excess risk satisfies:*

$$\mathbb{E}\mathcal{E}_k \eqsim \left(\frac{1-\rho}{\eta k}\right)^s + \frac{\sigma^2\eta}{B(1-\rho)}.$$

Proof. By Corollary E.7, the expected risk is completely determined by the deterministic bias and the label noise:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{nesterov}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau).$$

For $k \gtrsim \frac{1-\rho}{\eta}$, we substitute the deterministic bias from Proposition E.8:

$$S_{\text{nesterov}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s.$$

We substitute the label noise floor from Lemma E.5. In the fully overdamped regime $\eta \lesssim (1-\rho)^2$, we have $\frac{\eta}{1-\rho} \lesssim 1-\rho \ll 1$. Since $\beta > 1$, the minimum is attained by the linear term:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1} \mathcal{K}_{\text{nesterov}}(\tau) \eqsim \frac{\sigma^2}{B}\min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\} = \frac{\sigma^2\eta}{B(1-\rho)}.$$

Summing these two terms directly yields the exact scaling law. □

E.7 Risk dynamics in the mixed regime

When the leading spectral modes are underdamped, the loss contains an exponentially damped edge contribution in addition to the polynomial overdamped tail.

Define the Nesterov edge damping factor:

$$p_{\text{nesterov}} := \rho\left(1 - \frac{(1-\rho)^2}{(1+\rho)^2}\right) = \frac{4\rho^2}{(1+\rho)^2}.$$

It strictly satisfies $p_{\text{nesterov}} = 1 - (1-\rho) + \mathcal{O}((1-\rho)^2) < \rho$, ensuring that the underdamped edge contribution decays exponentially.

Proposition E.10 (Deterministic bias, mixed regime)**.** *Under Assumptions 3.1 and 3.3, assume the stability condition $\eta \lesssim \min\{1, B(1-\rho)\}$. For the mixed regime $\eta \gtrsim (1-\rho)^2$, the deterministic bias decomposes into $S_{\text{nesterov}}(k) = S_{\text{over}}(k) + S_{\text{under}}(k)$, where:*

$$S_{\text{over}}(k) \eqsim \begin{cases} \left(\frac{(1-\rho)^2}{\eta}\right)^s, & k \lesssim \frac{1}{1-\rho}, \\ \left(\frac{1-\rho}{\eta k}\right)^s, & k \gtrsim \frac{1}{1-\rho}, \end{cases}$$

and

$$S_{\text{under}}(k) \eqsim \begin{cases} \rho^k, & k \lesssim \frac{1}{\eta}, \\ \frac{\rho^k}{(\eta k)^s}, & \frac{1}{\eta} \lesssim k \lesssim \frac{1}{(1-\rho)^2}, \\ \frac{1}{\eta k}\left(\frac{(1-\rho)^2}{\eta}\right)^{s-1} p_{\text{nesterov}}^k, & \frac{1}{(1-\rho)^2} \lesssim k \lesssim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}, \\ \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k, & k \gtrsim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}. \end{cases}$$

Proof. For each coordinate, write $e_k^{\text{det}}(j) = -h_k^{\text{nesterov}}(j)\theta_j^*$. The multiplier satisfies

$$h_{k+1}^{\text{nesterov}}(j) = (1+\rho)(1-\eta\lambda_j)h_k^{\text{nesterov}}(j) - \rho(1-\eta\lambda_j)h_{k-1}^{\text{nesterov}}(j), \qquad h_0^{\text{nesterov}}(j) = 1, \qquad h_1^{\text{nesterov}}(j) = 1-\eta\lambda_j.$$

The roots are

$$y_{j,+}, y_{j,-} = \frac{(1+\rho)(1-\eta\lambda_j) \pm \sqrt{(1+\rho)^2(1-\eta\lambda_j)^2 - 4\rho(1-\eta\lambda_j)}}{2}.$$

The discriminant is negative precisely when

$$(1-\sqrt{\rho})^2 < \eta\lambda_j < (1+\sqrt{\rho})^2.$$

Thus the comparison between η and $(1-\rho)^2$ determines whether a coordinate is overdamped or underdamped. In the mixed regime $\eta \gtrsim (1-\rho)^2$, the crossover index is defined by $\eta\lambda_{J_{\max}} \eqsim (1-\rho)^2$. We decompose the deterministic bias into the overdamped tail ($j > J_{\max}$) and the underdamped leading block ($j \leqslant J_{\max}$).

1. Overdamped Tail (S_{over}):

For overdamped coordinates in the stable small-step range, the root comparison used in Proposition E.8 gives constants $c_1, C_1 > 0$ such that

$$|h_k^{\text{nesterov}}(j)| \leqslant C_1 \exp\left(-c_1 \frac{\eta\lambda_j}{1-\rho} k\right).$$

Moreover, if $\eta\lambda_j \leqslant c_1(1-\rho)^2$ and $k \geqslant C_1(1-\rho)^{-1}$, the slow root dominates the fast root and

$$|h_k^{\text{nesterov}}(j)| \geqslant c_1 \exp\left(-C_1 \frac{\eta\lambda_j}{1-\rho} k\right).$$

For $k \lesssim \frac{1}{1-\rho}$, the exponent satisfies $\frac{\eta\lambda_j}{1-\rho}k \lesssim \frac{(1-\rho)^2}{1-\rho}\frac{1}{1-\rho} = 1$. The exponential decay is bounded away from zero, so $|h_k^{\text{nesterov}}(j)| \eqsim 1$. The sum is strictly dominated by its lower limit at the crossover:

$$S_{\text{over}}(k) = \sum_{j>J_{\max}} \lambda_j(\theta_j^*)^2 h_k^{\text{nesterov}}(j)^2 \eqsim \sum_{j>J_{\max}} j^{-1-\beta s} \eqsim (J_{\max})^{-\beta s} \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^s.$$

For $k \gtrsim \frac{1}{1-\rho}$, applying $\lambda_j \eqsim j^{-\beta}$, the integral comparison yields the strict upper bound:

$$S_{\text{over}}(k) \leqslant C \sum_{j>J_{\max}} j^{-1-\beta s} \exp\left(-2c_1 \frac{\eta k}{1-\rho} j^{-\beta}\right) \leqslant C\left(\frac{1-\rho}{\eta k}\right)^s.$$

By restricting the sum to $j \geqslant (C\frac{\eta k}{1-\rho})^{1/\beta}$, we obtain the matching lower bound $c\left(\frac{1-\rho}{\eta k}\right)^s$.

2. Underdamped Leading Block (S_{under}):
For $j \leqslant J_{\max}$, write the two roots as

$$\sqrt{\rho(1-\eta\lambda_j)}e^{i\omega_j}, \quad \sqrt{\rho(1-\eta\lambda_j)}e^{-i\omega_j}, \qquad \text{where} \quad \cos\omega_j = \frac{(1+\rho)\sqrt{1-\eta\lambda_j}}{2\sqrt{\rho}}.$$

Solving the initial conditions gives the exact identity

$$h_k^{\text{nesterov}}(j) = \big(\rho(1-\eta\lambda_j)\big)^{k/2}\left[\cos(k\omega_j) + \frac{(1-\rho)\sqrt{1-\eta\lambda_j}}{2\sqrt{\rho}\sin\omega_j}\sin(k\omega_j)\right].$$

Set $A_j := \frac{(1-\rho)\sqrt{1-\eta\lambda_j}}{2\sqrt{\rho}\sin\omega_j}$ and define $\tan\psi_j = A_j$. The squared multiplier is bounded by the envelope $\big(\rho(1-\eta\lambda_j)\big)^k(1+A_j^2)\cos^2(k\omega_j - \psi_j)$.

To evaluate the underdamped contribution, we define the positive spectral weights $w_j(k) = \lambda_j(\theta_j^*)^2\big(\rho(1-\eta\lambda_j)\big)^k(1+A_j^2)$ and decompose the cosine-squared terms:

$$S_{\text{under}}(k) = \frac{1}{2}\sum_{j\leqslant J_{\max}} w_j(k) + \frac{1}{2}\sum_{j\leqslant J_{\max}} w_j(k)\cos(2k\omega_j - 2\psi_j).$$

The first sum represents the stable static mass $M_w(k) = \frac{1}{2}\sum_{j\leqslant J_{\max}} w_j(k)$. To analyze the oscillatory second sum, we view the indices as a continuous variable $x \in [1, J_{\max}]$. For the continuous oscillatory integral $I(k)$, we apply the substitution $u = \omega(x)$ mapping $[1, J_{\max}]$ to $[u_{\min}, u_{\max}]$:

$$I(k) = \int_{u_{\min}}^{u_{\max}} g(u,k)\cos(2ku - \phi(u))\,du, \qquad \text{where } g(u,k) = w(x(u),k)\left|x'(u)\right|, \phi(u) = 2\psi(x(u)).$$

By integration by parts with the differential $d\left(\sin(2ku-\phi(u))\right) = (2k-\phi'(u))\cos(2ku-\phi(u))\,du$:

$$I(k) = \left[v_k(u)\sin(2ku-\phi(u))\right]_{u_{\min}}^{u_{\max}} - \int_{u_{\min}}^{u_{\max}} v_k'(u)\sin(2ku-\phi(u))\,du,$$

where $v_k(u) = \frac{g(u,k)}{2k-\phi'(u)}$. This establishes the temporal decay $|I(k)| \leqslant \frac{C_1}{k}\max_u|g(u,k)| = \mathcal{O}(1/k)$. Thus, for sufficiently large k, the oscillatory integral is strictly subsumed by the static mass: $|I(k)| \leqslant \frac{1}{4}M_w(k)$. This guarantees the discrete sum is strictly bounded away from zero by the static part.

We now rigorously evaluate $M_w(k)$. Asymptotically, $1 + A_j^2 \asymp 1$ everywhere except for a subdominant logarithmic singularity strictly at the continuous edge, which does not alter the polynomial/exponential macroscopic order. Thus:

$$M_w(k) \asymp \sum_{j=1}^{J_{\max}} j^{-1-\beta s} \big(\rho(1-\eta\lambda_j)\big)^k \asymp \sum_{j=1}^{J_{\max}} j^{-1-\beta s} p_{\text{nesterov}}^k e^{-ckx_j},$$

where $p_{\text{nesterov}} = \rho(1-u_c)$, $u_c \asymp (1-\rho)^2$, and the distance to the edge is $x_j = \eta\lambda_j - u_c \approx \eta\beta J_{\max}^{-\beta-1}(J_{\max} - j)$. We proceed in four temporal stages:

Stage 1: $k \lesssim \frac{1}{\eta}$.

The exponential decay $e^{-ckx_j} \approx 1$. The sum is overwhelmingly dominated by the leading modes ($j \sim 1$):

$$S_{\text{under}}(k) \asymp \rho^k \sum_{j=1}^{J_{\max}} j^{-1-\beta s} \asymp \rho^k.$$

Stage 2: $\frac{1}{\eta} \lesssim k \lesssim \frac{1}{(1-\rho)^2}$.

The dominant bulk peak $j_k \asymp (\eta k)^{1/\beta} \ll J_{\max}$ is fully captured inside the interval. The integral approximation perfectly applies:

$$S_{\text{under}}(k) \asymp \int_1^\infty x^{-1-\beta s} \rho^k e^{-ck\eta x^{-\beta}} dx \asymp \rho^k (\eta k)^{-s}.$$

Stage 3 & 4: $k \gtrsim \frac{1}{(1-\rho)^2}$.

The bulk peak j_k passes the boundary $J_{\max}$, so the summation is strictly dominated by the discrete tail items $j \to J_{\max}$. Let $m = J_{\max} - j \geqslant 0$. The distance to the edge is $x_j \approx \eta\beta J_{\max}^{-\beta-1} m \asymp (1-\rho)^{2+2/\beta}\eta^{-1/\beta} m$. Using the discrete geometric series sum for the dominant tail terms:

$$\begin{aligned} S_{\text{under}}(k) &\asymp \sum_{m=0}^{J_{\max}} J_{\max}^{-1-\beta s} p_{\text{nesterov}}^k \exp\left(-ck(1-\rho)^{2+2/\beta}\eta^{-1/\beta} m\right) \\ &\asymp J_{\max}^{-1-\beta s} p_{\text{nesterov}}^k \sum_{m=0}^{\infty} \left(\exp\left(-ck(1-\rho)^{2+2/\beta}\eta^{-1/\beta}\right)\right)^m \\ &\asymp \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k \frac{1}{1-\exp\left(-ck(1-\rho)^{2+2/\beta}\eta^{-1/\beta}\right)}. \end{aligned}$$

Stage 3: $\frac{1}{(1-\rho)^2} \lesssim k \lesssim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}$.

In this stage, the exponent is small, $k(1-\rho)^{2+2/\beta}\eta^{-1/\beta} \lesssim 1$, meaning the decay window covers multiple discrete points near the edge. We apply the Taylor expansion $1 - e^{-x} \asymp x$:

$$\begin{aligned} S_{\text{under}}(k) &\asymp \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k \frac{1}{k(1-\rho)^{2+2/\beta}\eta^{-1/\beta}} \\ &= \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k \frac{1}{k(1-\rho)^2} \left(\frac{\eta}{(1-\rho)^2}\right)^{\frac{1}{\beta}} \\ &= \frac{1}{\eta k} \left(\frac{(1-\rho)^2}{\eta}\right)^{s-1} p_{\text{nesterov}}^k. \end{aligned}$$

Stage 4: $k \gtrsim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}$.

In this extreme long-time regime, the exponent is large, $k(1-\rho)^{2+2/\beta}\eta^{-1/\beta} \gtrsim 1$. The exponential

decay is so sharp that the single point $j = J_{\max}$ strictly dominates the entire sum. The denominator satisfies $1 - \exp\left(-ck(1-\rho)^{2+2/\beta}\eta^{-1/\beta}\right) \eqsim 1$:

$$S_{\text{under}}(k) \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k \cdot 1 = \left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k.$$

This concludes the proof. □

Theorem E.11 (Two-sided scaling law, mixed regime)**.** *Under Assumption 3.3 and Proposition 3.4, the conditions of Proposition E.10, and the stability condition $\eta \leqslant cB^{\beta}(1-\rho)$ with $c > 0$ sufficiently small, for the mixed regime $\eta \gtrsim (1-\rho)^2$ with $k \gtrsim \frac{1-\rho}{\eta}$, the expected excess risk strictly satisfies:*

$$\mathbb{E}\mathcal{E}_k \quad \eqsim \quad \min\left\{1, (\eta k)^{-s}\right\}\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s + \frac{\sigma^2}{B}\min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\}. \tag{79}$$

Proof. By Corollary E.7, the multiplicative gradient noise is absorbed by the additive label noise floor:

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_k)] \eqsim S_{\text{nesterov}}(k) + \frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau).$$

By Lemma E.5, the domain condition $k \gtrsim \frac{1-\rho}{\eta}$ guarantees the label noise summation is saturated:

$$\frac{\eta^2\sigma^2}{B}\sum_{\tau=0}^{k-1}\mathcal{K}_{\text{nesterov}}(\tau) \eqsim \frac{\sigma^2}{B}\min\left\{\left(\frac{\eta}{1-\rho}\right)^{\frac{1}{\beta}}, \frac{\eta}{1-\rho}\right\}.$$

It remains to establish that the true deterministic bias $S_{\text{nesterov}}(k) = S_{\text{under}}(k) + S_{\text{over}}(k)$ from Proposition E.10 matches the target analytical formula:

$$\widehat{S}_{\text{nesterov}}(k) := \min\left\{1, (\eta k)^{-s}\right\}\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

We partition the domain $k \gtrsim \frac{1-\rho}{\eta}$ into four sequential stages. Since the relative order of $\frac{1}{\eta}$ and $\frac{1}{1-\rho}$ is unfixed, we explicitly discuss their relation in the intermediate regime.

Stage 1: $\frac{1-\rho}{\eta} \lesssim k \lesssim \min\left\{\frac{1}{\eta}, \frac{1}{1-\rho}\right\}$**.**

Here $\eta k \lesssim 1 \implies \min\{1, (\eta k)^{-s}\} = 1$, and $k(1-\rho) \lesssim 1 \implies \rho^k = (1-(1-\rho))^k \eqsim 1$. Thus:

$$\widehat{S}_{\text{nesterov}}(k) \eqsim 1 + \left(\frac{1-\rho}{\eta k}\right)^s.$$

Since $k \gtrsim \frac{1-\rho}{\eta}$, the polynomial term satisfies $\left(\frac{1-\rho}{\eta k}\right)^s \lesssim 1$, yielding $\widehat{S}_{\text{nesterov}}(k) \eqsim 1$.

Concurrently, Proposition E.10 asserts $S_{\text{under}}(k) \eqsim \rho^k \eqsim 1$ and $S_{\text{over}}(k) \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^s$. Because $\eta \gtrsim (1-\rho)^2$, we have $S_{\text{over}}(k) \lesssim 1$. Hence, the true bias $S_{\text{nesterov}}(k) \eqsim 1 + \mathcal{O}(1) \eqsim 1 \eqsim \widehat{S}_{\text{nesterov}}(k)$.

Stage 2: The intermediate regime.

We verify the equivalence across the intermediate window bounded by $\frac{1}{\eta}$ and $\frac{1}{1-\rho}$, examining two exhaustive cases:

- *Case A:* $\eta \gtrsim 1-\rho \implies \frac{1}{\eta} \lesssim \frac{1}{1-\rho}$. The interval is $\frac{1}{\eta} \lesssim k \lesssim \frac{1}{1-\rho}$.

Here $\eta k \gtrsim 1 \implies \min\{1, (\eta k)^{-s}\} = (\eta k)^{-s}$, and $k(1-\rho) \lesssim 1 \implies \rho^k \eqsim 1$. The target evaluates to:

$$\widehat{S}_{\text{nesterov}}(k) \eqsim (\eta k)^{-s} + \left(\frac{1-\rho}{\eta k}\right)^s \eqsim (\eta k)^{-s},$$

where the second term is absorbed because $1 - \rho \lesssim \eta \lesssim 1$.

By Proposition E.10, $S_{\text{under}}(k) \eqsim \frac{\rho^k}{(\eta k)^s} \eqsim (\eta k)^{-s}$ and $S_{\text{over}}(k) \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^s$. Since $k \lesssim \frac{1}{1-\rho}$, we have $(\eta k)^{-s} \gtrsim \left(\frac{\eta}{1-\rho}\right)^{-s} = \left(\frac{1-\rho}{\eta}\right)^s \geqslant \left(\frac{(1-\rho)^2}{\eta}\right)^s$. Thus, $S_{\text{nesterov}}(k) \eqsim (\eta k)^{-s} \eqsim \widehat{S}_{\text{nesterov}}(k)$.

- *Case B:* $\eta \lesssim 1 - \rho \implies \frac{1}{1-\rho} \lesssim \frac{1}{\eta}$. The interval is $\frac{1}{1-\rho} \lesssim k \lesssim \frac{1}{\eta}$.

  Here $\eta k \lesssim 1 \implies \min\{1, (\eta k)^{-s}\} = 1$. The target directly yields:

  $$\widehat{S}_{\text{nesterov}}(k) = \rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

  By Proposition E.10, $S_{\text{under}}(k) \eqsim \rho^k$ and $S_{\text{over}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s$. Summing them exactly matches $\widehat{S}_{\text{nesterov}}(k)$.

Stage 3: $\max\left\{\frac{1}{\eta}, \frac{1}{1-\rho}\right\} \lesssim k \lesssim \frac{1}{(1-\rho)^2}$.
Here $\eta k \gtrsim 1 \implies \min\{1, (\eta k)^{-s}\} = (\eta k)^{-s}$. The target yields:

$$\widehat{S}_{\text{nesterov}}(k) = (\eta k)^{-s}\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

By Proposition E.10, the components are exactly $S_{\text{under}}(k) \eqsim \frac{\rho^k}{(\eta k)^s}$ and $S_{\text{over}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s$, which precisely reconstructs $\widehat{S}_{\text{nesterov}}(k)$.

Stage 4: $k \gtrsim \frac{1}{(1-\rho)^2}$.
The mixed regime condition $\eta \gtrsim (1-\rho)^2$ dictates $\eta k \gtrsim \frac{\eta}{(1-\rho)^2} \gtrsim 1 \implies \min\{1, (\eta k)^{-s}\} = (\eta k)^{-s}$. Thus:

$$\widehat{S}_{\text{nesterov}}(k) = (\eta k)^{-s}\rho^k + \left(\frac{1-\rho}{\eta k}\right)^s.$$

We rigorously demonstrate that the exponential transient is absorbed. The ratio satisfies:

$$\frac{(\eta k)^{-s}\rho^k}{\left(\frac{1-\rho}{\eta k}\right)^s} = (1-\rho)^{-s}\rho^k \leqslant (1-\rho)^{-s}e^{-k(1-\rho)}.$$

For $k \gtrsim \frac{1}{(1-\rho)^2}$, the exponent $k(1-\rho) \gtrsim \frac{1}{1-\rho} \gg 1$. Factoring the decay gives:

$$(1-\rho)^{-s}e^{-k(1-\rho)} \leqslant (1-\rho)^{-s}e^{-\frac{1}{2(1-\rho)}}e^{-\frac{k(1-\rho)}{2}}.$$

Since $x^{-s}e^{-x/2} \to 0$ as $x \to \infty$, the term $(1-\rho)^{-s}e^{-\frac{1}{2(1-\rho)}} \ll 1$. Thus, the transient is fully absorbed:

$$\widehat{S}_{\text{nesterov}}(k) \eqsim \left(\frac{1-\rho}{\eta k}\right)^s.$$

Concurrently, we bound the true underdamped bias $S_{\text{under}}(k)$ from Proposition E.10, where $p_{\text{nesterov}} < \rho \leqslant e^{-(1-\rho)}$. We split into its two sub-branches:

- *Sub-branch* $\frac{1}{(1-\rho)^2} \lesssim k \lesssim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}$*:*

$$\begin{aligned}\frac{S_{\text{under}}(k)}{S_{\text{over}}(k)} &\asymp \frac{\frac{1}{\eta k}\left(\frac{(1-\rho)^2}{\eta}\right)^{s-1} p_{\text{nesterov}}^k}{\left(\frac{1-\rho}{\eta k}\right)^s} \\ &= k^{s-1}(1-\rho)^{s-2} p_{\text{nesterov}}^k \\ &\leqslant \left[k^{s-1} e^{-\frac{k(1-\rho)}{2}}\right](1-\rho)^{s-2} e^{-\frac{k(1-\rho)}{2}}.\end{aligned}$$

  Since $k(1-\rho) \gtrsim \frac{1}{1-\rho}$, we have $e^{-\frac{k(1-\rho)}{2}} \leqslant e^{-\frac{1}{2(1-\rho)}}$. Thus, $(1-\rho)^{s-2} e^{-\frac{1}{2(1-\rho)}} \to 0$, absorbing $S_{\text{under}}(k)$.

- *Sub-branch* $k \gtrsim \eta^{1/\beta}(1-\rho)^{-2-\frac{2}{\beta}}$*:* Using $\eta \gtrsim (1-\rho)^2 \implies \eta^{-\frac{1}{\beta}} \lesssim (1-\rho)^{-\frac{2}{\beta}}$, we have:

$$\begin{aligned}\frac{S_{\text{under}}(k)}{S_{\text{over}}(k)} &\asymp \frac{\left(\frac{(1-\rho)^2}{\eta}\right)^{s+\frac{1}{\beta}} p_{\text{nesterov}}^k}{\left(\frac{1-\rho}{\eta k}\right)^s} \\ &= (1-\rho)^{s+\frac{2}{\beta}} \eta^{-\frac{1}{\beta}} k^s p_{\text{nesterov}}^k \\ &\lesssim \left[\big(k(1-\rho)\big)^s e^{-\frac{k(1-\rho)}{2}}\right] e^{-\frac{k(1-\rho)}{2}} \to 0.\end{aligned}$$

In both cases, $S_{\text{under}}(k) \ll S_{\text{over}}(k) \asymp \left(\frac{1-\rho}{\eta k}\right)^s$. The true bias simplifies to $S_{\text{nesterov}}(k) \asymp \left(\frac{1-\rho}{\eta k}\right)^s$, perfectly matching $\widehat{S}_{\text{nesterov}}(k)$.

Synthesizing all stages, $S_{\text{nesterov}}(k) \asymp \widehat{S}_{\text{nesterov}}(k)$ holds uniformly for $k \gtrsim \frac{1-\rho}{\eta}$. Substituting this and the saturated label noise into the total risk equivalence completes the proof. □

F Optimal data scaling across batch-size regimes

This appendix proves the batch-size-dependent scaling laws stated in Theorem 6.2. Throughout this appendix, we work in the noisy setting $\sigma > 0$ considered in Section 6 and impose the data-budget constraint

$$D = BK, \qquad B \eqsim D^b, \qquad K \eqsim D^{1-b}, \qquad 0 \leqslant b < 1.$$

Thus, the batch exponent b is prescribed, and only the remaining hyperparameters are optimized. We use the transition exponents defined in Section 6:

$$b_1 = \frac{s}{s+1}, \qquad b_2 = \frac{2s+1/\beta}{2s+1+1/\beta} = \frac{2s\beta+1}{2s\beta+\beta+1}, \qquad b_3 = \frac{2s+1}{2s+2}.$$

For the momentum methods, we parameterize the momentum gap and learning rate by $1-\rho \eqsim D^{-m}$ and $\eta \eqsim D^{-c}$, where $m, c \geqslant 0$.

F.1 SGD

For SGD, write $\eta \eqsim D^{-c}$, where $c \geqslant 0$. By the SGD loss scaling law in Section 5,

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim (\eta K)^{-s} + \frac{\sigma^2 \eta}{B} \eqsim D^{-s(1-b-c)} + D^{-(b+c)}.$$

Therefore, for a fixed batch exponent b, the achievable decay exponent is $\min\{s(1-b-c), b+c\}$. If $0 \leqslant b < b_1$, the two terms can be balanced by taking $c = b_1 - b \geqslant 0$. Since $b_1 = s/(s+1)$, this gives

$$s(1-b-c) = b + c = b_1 = \frac{s}{s+1}.$$

Hence $r_{\mathrm{sgd}}(b) = s/(s+1)$ for $0 \leqslant b < b_1$.

If $b \geqslant b_1$, the unconstrained balancing choice would require $c = b_1 - b \leqslant 0$. Because the learning rate is not allowed to grow polynomially with D, we must have $c \geqslant 0$. Moreover, $b + c \geqslant b \geqslant b_1$, so the signal term is the slower-decaying term. It is optimized by taking $c = 0$, which yields $r_{\mathrm{sgd}}(b) = s(1-b)$ for $b_1 \leqslant b < 1$.

Consequently,

$$r_{\mathrm{sgd}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_1, \\ s(1-b), & b_1 \leqslant b < 1. \end{cases}$$

Thus, b_1 is precisely the largest batch exponent for which SGD preserves the data-optimal rate $s/(s+1)$.

F.2 Polyak

We next prove the Polyak part of Theorem 6.2. We retain $K \eqsim D^{1-b}$, $1-\rho \eqsim D^{-m}$, and $\eta \eqsim D^{-c}$, where $m, c \geqslant 0$. We require $b + m < 1$, so that $\rho^K \eqsim \exp(-D^{1-b-m})$ is smaller than any polynomial power of D. The stability condition $\eta \lesssim B(1-\rho)$ becomes $b + c - m \geqslant 0$. Moreover,

$$\eta_\rho K = \frac{\eta K}{1-\rho} \eqsim D^{1-b-c+m}, \qquad \frac{\eta_\rho}{B} = \frac{\eta}{B(1-\rho)} \eqsim D^{-(b+c-m)}.$$

The damping threshold $\eta \eqsim (1-\rho)^2$ corresponds to $c = 2m$. We consider the fully overdamped and mixed regimes separately.

Fully overdamped regime. If $c \geqslant 2m$, Theorem 5.1 gives

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim (\eta_\rho K)^{-s} + \frac{\sigma^2 \eta_\rho}{B} \eqsim D^{-s(1-b-c+m)} + D^{-(b+c-m)}.$$

Therefore, the polynomial decay exponent in the fully overdamped regime is $\min\{s(1-b-c+m), b+c-m\}$.

Mixed overdamped–underdamped regime. If $c \leqslant 2m$, Theorem 5.1 gives

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim \rho^K + (\eta_\rho K)^{-s} + \frac{\sigma^2 \eta_\rho}{B}.$$

Since $b + m < 1$, the transient ρ^K is smaller than any polynomial power of D. Hence

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim D^{-s(1-b-c+m)} + D^{-(b+c-m)},$$

and the polynomial decay exponent is again $\min\{s(1-b-c+m), b+c-m\}$. Thus, although the two regimes have different transient dynamics, they lead to the same polynomial optimization problem. The signal and additive-noise powers are balanced when

$$s(1-b-c+m) = b+c-m \quad \Longleftrightarrow \quad b+c-m = \frac{s}{s+1} = b_1.$$

If $0 \leqslant b < b_1$, this balance is attained by taking $m = 0$ and $c = b_1 - b$. Since $c \geqslant 2m$, this choice lies in the fully overdamped regime and yields $r_{\text{polyak}}(b) = s/(s+1)$.

If $b_1 \leqslant b < b_3$, the same balance is attained by taking $c = 0$ and $m = b - b_1$. For $b > b_1$, this choice satisfies $c \leqslant 2m$ and therefore lies in the mixed regime. Moreover, $b + c - m = b_1$, while the transient condition becomes $b + m = 2b - b_1 < 1$. Since $(1 + b_1)/2 = (2s+1)/(2s+2) = b_3$, this condition holds precisely when $b < b_3$. Hence $r_{\text{polyak}}(b) = s/(s+1)$ for $0 \leqslant b < b_3$.

Now suppose that $b \geqslant b_3$. Since $c \geqslant 0$ and $m < 1 - b$, every admissible choice satisfies $b + c - m > 2b - 1 \geqslant b_1$, so the signal term is the slower-decaying term and $s(1-b-c+m) < 2s(1-b)$. This upper bound is approached by taking $c = 0$ and $m \uparrow 1 - b$, which lies in the mixed regime and gives

$$s(1-b-c+m) \uparrow 2s(1-b), \qquad b+c-m \downarrow 2b-1.$$

Moreover, $2b - 1 \geqslant 2s(1-b)$ holds exactly when $b \geqslant b_3$, so the signal term is indeed dominant. Consequently, $r_{\text{polyak}}(b) = 2s(1-b)$ for $b_3 \leqslant b < 1$, where the endpoint value is understood in the supremal sense represented by the $o(1)$ term in Definition 6.1. Combining the two regimes gives

$$r_{\text{polyak}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_3, \\ 2s(1-b), & b_3 \leqslant b < 1. \end{cases}$$

Thus, Polyak preserves the optimal data-scaling exponent $s/(s+1)$ up to the critical batch exponent b_3. Compared with SGD, it extends the critical batch size from D^{b_1} to D^{b_3}, while leaving the optimal small-batch exponent unchanged.

F.3 Nesterov

We finally prove the Nesterov part of Theorem 6.2. We retain $B \eqsim D^b$, $K \eqsim D^{1-b}$, $1 - \rho \eqsim D^{-m}$, and $\eta \eqsim D^{-c}$, where $m, c \geqslant 0$. We require $b + m < 1$, so all terms containing the Nesterov transient p^K_{nesterov}, including those multiplied by polynomial factors, are smaller than any polynomial power of D. The stability condition $\eta \lesssim B^\beta(1-\rho)$ becomes $\beta b + c - m \geqslant 0$, while the damping threshold $\eta \eqsim (1-\rho)^2$ corresponds to $c = 2m$. We optimize the fully overdamped and mixed regimes separately and then compare their optimal decay exponents.

Fully overdamped regime. If $c \geqslant 2m$, Theorem 5.2 gives

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim (\eta_\rho K)^{-s} + \frac{\sigma^2 \eta_\rho}{B} \eqsim D^{-s(1-b-c+m)} + D^{-(b+c-m)}.$$

Thus, the decay exponent is $\min\{s(1-b-c+m), b+c-m\}$. Since $c \geqslant 2m$ implies $c-m \geqslant 0$, we have $b+c-m \geqslant b$. If $0 \leqslant b < b_1$, the signal and additive-noise powers can be balanced by taking $m=0$ and $c=b_1-b$, which gives $b+c-m=b_1$ and decay exponent $s/(s+1)$. If $b \geqslant b_1$, then $b+c-m \geqslant b \geqslant b_1$, so the signal term is limiting. Its exponent is maximized by taking $m=c=0$, which gives $s(1-b)$. Therefore, the optimal exponent in the fully overdamped regime is

$$r^{\text{full}}_{\text{nesterov}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_1, \\ s(1-b), & b_1 \leqslant b < 1. \end{cases}$$

In particular, the fully overdamped Nesterov dynamics have the same batch-size-dependent scaling exponent as SGD.

Mixed overdamped–underdamped regime. If $c \leqslant 2m$, Theorem 5.2 gives

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim \left(\frac{(1-\rho)^2}{\eta}\right)^{s+1/\beta} p^K_{\text{nesterov}} + (\eta_\rho K)^{-s} + \frac{\sigma^2}{B} \min\left\{\eta_\rho, \eta_\rho^{1/\beta}\right\}.$$

Since $b+m<1$, the first term is smaller than any polynomial power of D. Hence

$$\mathbb{E}[\mathcal{E}(\boldsymbol{\theta}_K)] \eqsim D^{-s(1-b-c+m)} + \frac{\sigma^2}{B} \min\left\{\frac{\eta}{1-\rho}, \left(\frac{\eta}{1-\rho}\right)^{1/\beta}\right\}.$$

The signal term has decay power $s(1-b-c+m)$. Since $\eta/(1-\rho) \eqsim D^{m-c}$, the additive-noise decay power is

$$\begin{cases} b+c-m, & c \geqslant m, \\ b + \dfrac{c-m}{\beta}, & c \leqslant m. \end{cases}$$

We optimize these two branches separately.

The branch $c \geqslant m$. In this branch, the decay exponent is

$$\min\{s(1-b-c+m), b+c-m\}.$$

Because $c-m \geqslant 0$, the same argument as in the fully overdamped regime applies. For $0 \leqslant b < b_1$, the balance $b+c-m=b_1$ is feasible in the mixed regime. For example, one may take $m=b_1-b$ and $c=2(b_1-b)$, for which $b+m=b_1<1$. For $b \geqslant b_1$, the optimal choice satisfies $c-m=0$ and gives exponent $s(1-b)$. Therefore,

$$r^{\text{mixed, c}\geqslant\text{m}}_{\text{nesterov}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_1, \\ s(1-b), & b_1 \leqslant b < 1. \end{cases}$$

The branch $c \leqslant m$. In this branch, the decay exponent is

$$\min\left\{s(1-b-c+m),\ b + \frac{c-m}{\beta}\right\}.$$

If $0 \leqslant b < b_1$, then at $m - c = 0$ we have $b < s(1-b)$. Increasing $m - c$ increases the signal power but decreases the additive-noise power, so the optimal choice is $m - c = 0$, which gives decay exponent b. If $b \geqslant b_1$, balancing the signal and additive-noise powers gives

$$s(1-b-c+m) = b + \frac{c-m}{\beta}, \qquad m - c = \frac{\beta\big((s+1)b - s\big)}{s\beta + 1}.$$

For a fixed value of $m - c$, the least restrictive choice for the transient condition is $c = 0$, so we take

$$c = 0, \qquad m = \frac{\beta\big((s+1)b - s\big)}{s\beta + 1}.$$

This choice satisfies the stability condition because

$$\beta b + c - m = \frac{\beta s\big(1 + (\beta - 1)b\big)}{s\beta + 1} > 0.$$

The signal and additive-noise powers are then equal to

$$s(1 - b + m) = b - \frac{m}{\beta} = \frac{s\big(1 + (\beta - 1)b\big)}{s\beta + 1}.$$

The transient condition holds precisely when

$$b + m < 1 \quad \Longleftrightarrow \quad b < \frac{2s\beta + 1}{2s\beta + \beta + 1} = \frac{2s + 1/\beta}{2s + 1 + 1/\beta} = b_2.$$

Therefore, for $b_1 \leqslant b < b_2$, the optimal exponent on the branch $c \leqslant m$ is

$$\frac{s\big(1 + (\beta - 1)b\big)}{s\beta + 1}.$$

If $b \geqslant b_2$, the preceding balance is no longer compatible with $b + m < 1$. Since $c \geqslant 0$ and $m < 1 - b$, every admissible choice satisfies

$$s(1 - b - c + m) < 2s(1 - b).$$

This upper bound is approached by taking $c = 0$ and $m \uparrow 1 - b$. Under this choice,

$$s(1 - b + m) \uparrow 2s(1 - b), \qquad b - \frac{m}{\beta} \longrightarrow \frac{(\beta + 1)b - 1}{\beta}.$$

At $b = b_2$, the two limiting powers agree, while for $b > b_2$,

$$\frac{(\beta + 1)b - 1}{\beta} > 2s(1 - b).$$

Thus, the signal term is limiting and the supremal decay exponent is $2s(1 - b)$. Consequently,

$$r_{\text{nesterov}}^{\text{mixed, c}\leqslant\text{m}}(b) = \begin{cases} b, & 0 \leqslant b < b_1, \\ \dfrac{s\big(1 + (\beta - 1)b\big)}{s\beta + 1}, & b_1 \leqslant b < b_2, \\ 2s(1 - b), & b_2 \leqslant b < 1. \end{cases}$$

Taking the larger exponent between the two additive-noise branches gives the optimal exponent in the mixed regime:

$$r_{\text{nesterov}}^{\text{mixed}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_1, \\ \dfrac{s\big(1 + (\beta - 1)b\big)}{s\beta + 1}, & b_1 \leqslant b < b_2, \\ 2s(1 - b), & b_2 \leqslant b < 1. \end{cases}$$

Indeed, the branch $c \geqslant m$ is optimal for $b < b_1$, whereas the branch $c \leqslant m$ is optimal for $b > b_1$. Finally, optimizing over the fully overdamped and mixed regimes gives

$$r_{\text{nesterov}}(b) = \max\left\{ r^{\text{full}}_{\text{nesterov}}(b), r^{\text{mixed}}_{\text{nesterov}}(b) \right\}.$$

For $0 \leqslant b < b_1$, the two regimes attain the same exponent $s/(s+1)$. For $b_1 < b < b_2$, the mixed-regime exponent $s(1 + (\beta - 1)b)/(s\beta + 1)$ is strictly larger than the fully overdamped exponent $s(1-b)$. For $b \geqslant b_2$, the mixed-regime exponent $2s(1-b)$ is also larger than the fully overdamped exponent $s(1-b)$. We therefore obtain

$$r_{\text{nesterov}}(b) = \begin{cases} \dfrac{s}{s+1}, & 0 \leqslant b < b_1, \\ \dfrac{s\big(1 + (\beta - 1)b\big)}{s\beta + 1}, & b_1 \leqslant b < b_2, \\ 2s(1-b), & b_2 \leqslant b < 1. \end{cases}$$

Thus, Nesterov agrees with SGD and Polyak in the small-batch regime, improves its data-scaling exponent once b exceeds b_1, and reaches its maximal exponent at $b = b_2$. For $b \geqslant b_2$, its exponent becomes $2s(1-b)$, which coincides with the Polyak exponent in the ultra-large-batch regime.

G Experimental details and additional results

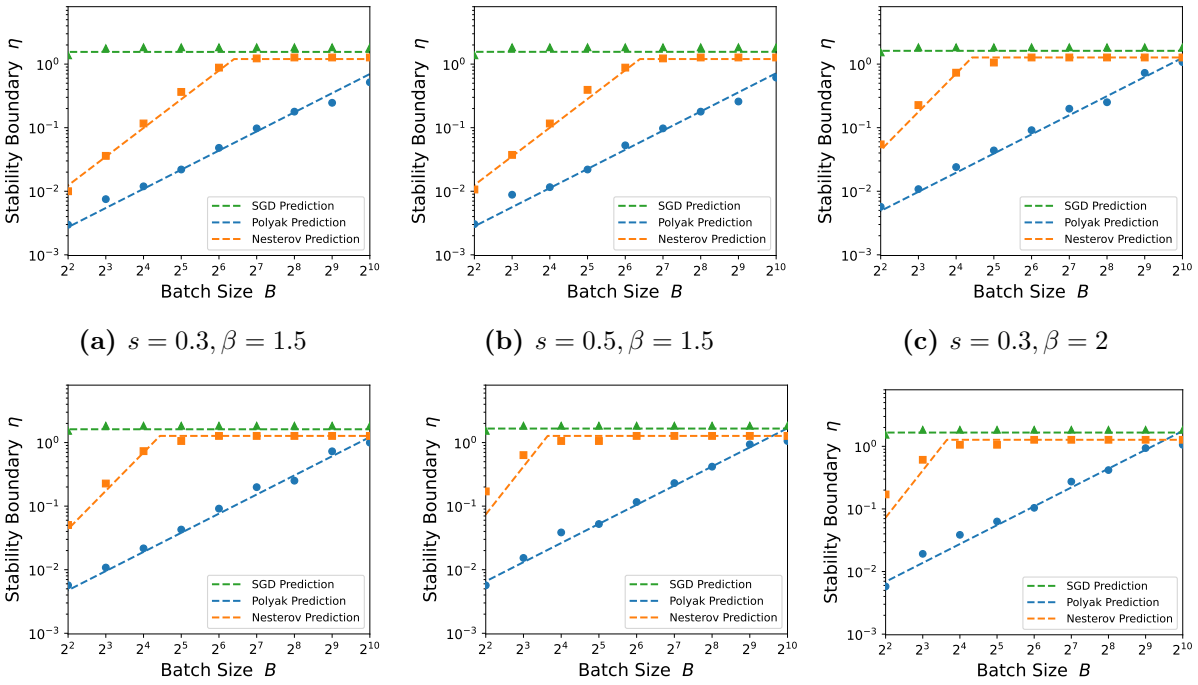
G.1 Empirical stability boundaries

We simulate the power-law kernel regression model from Section 3.1 in a finite-dimensional eigenbasis with $d = 4096$. At every iteration, a fresh Gaussian mini-batch is sampled and the corresponding stochastic-gradient update is applied. The three algorithms are initialized from the same point, with $\boldsymbol{\theta}_{-1} = \boldsymbol{\theta}_0 = 0$. For the momentum methods, we fix $1-\rho = 2^{-24}$, while SGD uses $\rho = 0$. We scan batch sizes $B \in \{2^0, 2^1, \ldots, 2^{12}\}$ and 50 logarithmically spaced learning rates between 2^{-10} and 8. Each run lasts $K = 2000$ iterations, and the exact population excess risk is evaluated after every update.

We classify a run as stable if the average population excess risk over the last 20 iterations is finite and smaller than 1. If the loss becomes non-finite or exceeds 10^{12} during training, the run is terminated and classified as unstable. For each algorithm, batch size, and random seed, the empirical stability boundary is the largest learning rate on the search grid that passes this criterion. We repeat the search over five independent random seeds. The markers in Figure 3(left) report the arithmetic mean of the five per-seed stability boundaries.

The dashed lines are fitted only after the empirical boundaries have been computed. For Polyak, we fix the slope to the theoretical form $B(1-\rho)$ and fit its multiplicative constant using all finite boundary points. For Nesterov, we divide the points at the onset of the visible large-batch plateau. The points before this split are fitted to $B^{\beta}(1-\rho)$, with the exponent fixed to its theoretical value, whereas the points after the split are fitted by a constant horizontal level. In each case, the multiplicative constant or horizontal level is the geometric mean of the pointwise ratios in log scale. The dotted vertical line indicates this split; it affects only the reference fits and not the empirical stability classification.

Figure 5 presents additional empirical stability-boundary results under different choices of the source exponent s and capacity exponent β.



(a) $s = 0.3, \beta = 1.5$ **(b)** $s = 0.5, \beta = 1.5$ **(c)** $s = 0.3, \beta = 2$

(d) $s = 0.5, \beta = 2$ **(e)** $s = 0.3, \beta = 2.5$ **(f)** $s = 0.5, \beta = 2.5$

Figure 5: Additional empirical stability boundaries under different choices of s and β.

G.2 Scaling law predictions

We simulate the power-law kernel regression model from Section 3.1 in a finite-dimensional eigenbasis with $d = 4096$. Both methods are initialized with $\boldsymbol{\theta}_{-1} = \boldsymbol{\theta}_0 = 0$, and a fresh realization of the curvature-aligned Gaussian gradient noise is generated at every iteration. For the representative trajectories in Figure 3(right), we use $\sigma = 0.5$, $B = 64$, $\eta = 0.1$, $\rho = 0.995$, run each method for 5×10^5 iterations, and report the exact population excess risk averaged over 100 independent runs.

We next describe how the theoretical curves are fitted. Recall the momentum-rescaled learning rate $\eta_\rho = \frac{\eta}{1-\rho}$ and the common signal term $S(k) = (\eta_\rho k)^{-s}$. For Polyak momentum, the transient and noise terms predicted by Theorem 5.1 are $T_{\text{polyak}}(k) = \rho^k, N_{\text{polyak}} = \frac{\sigma^2}{B}\eta_\rho$. We fit the empirical Polyak trajectory using

$$\widehat{L}_{\text{polyak}}(k) = C_T T_{\text{polyak}}(k) + C_S S(k) + C_N N_{\text{polyak}},$$

where the three coefficients are constrained to be nonnegative.

For Nesterov momentum, let $p = \frac{4\rho^2}{(1+\rho)^2}$. The transient and noise terms predicted by Theorem 5.2 are $T_{\text{nesterov}}(k) = \left(\frac{(1-\rho)^2}{\eta}\right)^{s+1/\beta} p^k, N_{\text{nesterov}} = \frac{\sigma^2}{B} \min\left\{\eta_\rho, \eta_\rho^{1/\beta}\right\}$. Accordingly, we fit the empirical Nesterov trajectory using

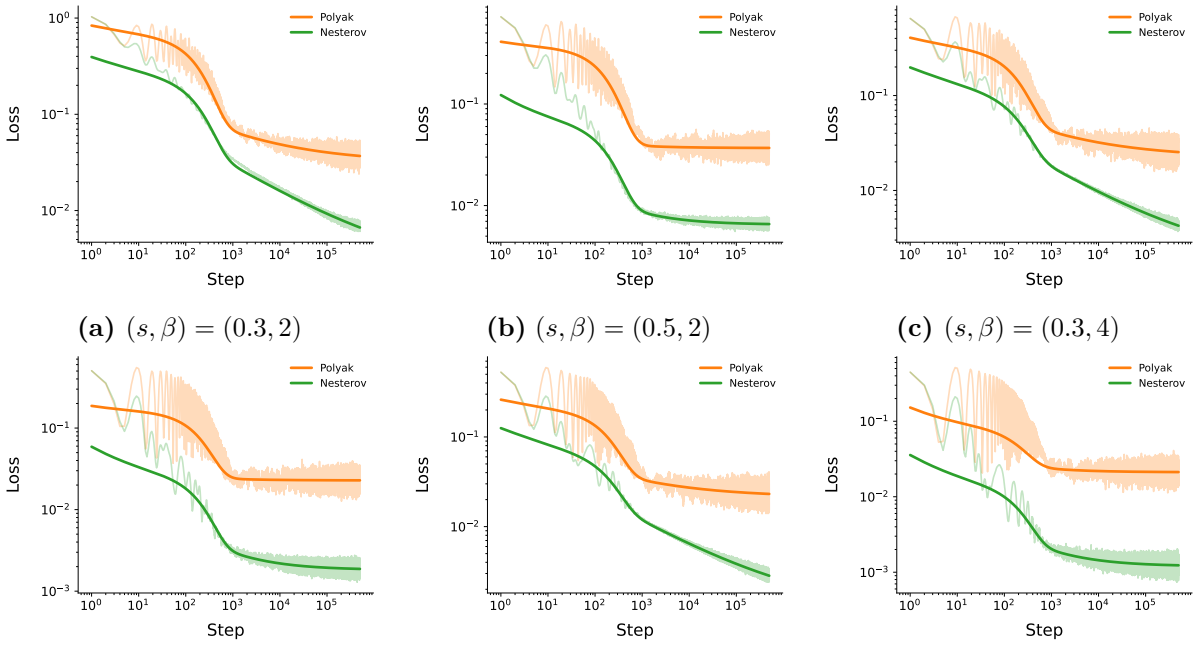
$$\widehat{L}_{\text{nesterov}}(k) = \tilde{C}_T T_{\text{nesterov}}(k) + \tilde{C}_S S(k) + \tilde{C}_N N_{\text{nesterov}},$$

again subject to nonnegative coefficients.

The coefficients are fitted independently for the two methods using all recorded iterations satisfying $\eta_\rho k \geqslant 1$. Let $\overline{L}(k)$ denote the empirical loss averaged over the 100 runs and let $\widehat{L}(k)$ denote the corresponding fitted expression above. For each method, the coefficients are obtained by solving the problem

$$\min_{C_T, C_S, C_N \geqslant 0} \frac{1}{|\mathcal{K}_{\text{fit}}|} \sum_{k \in \mathcal{K}_{\text{fit}}} \left(\frac{\widehat{L}(k) - \overline{L}(k)}{\overline{L}(k)} \right)^2, \qquad \mathcal{K}_{\text{fit}} = \{k \geqslant 1 : \eta_\rho k \geqslant 1\}.$$

The same simulation and fitting procedure is applied independently to each additional (s, β) setting reported in Figures 6. Across these settings, the fitted scaling laws consistently capture the transition from signal-dominated decay to noise-dominated long-time behavior and the lower noise contribution achieved by Nesterov.



(a) $(s, \beta) = (0.3, 2)$ **(b)** $(s, \beta) = (0.5, 2)$ **(c)** $(s, \beta) = (0.3, 4)$

(d) $(s, \beta) = (0.5, 4)$ **(e)** $(s, \beta) = (0.3, 6)$ **(f)** $(s, \beta) = (0.5, 6)$

Figure 6: Additional scaling-law trajectories for Polyak and Nesterov. All experiments use dimension $d = 4096$, run for 5×10^5 iterations, and report the mean trajectory over 100 independent runs.

G.3 Data scaling across three regimes

We simulate the power-law kernel regression model from Section 3.1 in a finite-dimensional eigenbasis with

$$d = 4096, \qquad s = 0.3, \qquad \beta = 4, \qquad \sigma = 0.1.$$

The data budgets are

$$D \in \{2000, 4000, 8000, 16000, 32000, 64000, 128000\}.$$

All algorithms are initialized with $\boldsymbol{\theta}_{-1} = \boldsymbol{\theta}_0 = 0$.

For a prescribed batch-size exponent a, we set

$$B = \mathrm{round}(C_B D^a), \qquad K = \left\lfloor \frac{D}{B} \right\rfloor,$$

where C_B is a fixed order-one constant. Once B is fixed, we search over order-one multiplicative constants in the theoretically prescribed scalings of η and $1 - \rho$. More precisely, if the theory prescribes $\eta \eqsim D^{-c}$ and $1 - \rho \eqsim D^{-b}$, we search over

$$\eta = C_\eta D^{-c}, \qquad 1 - \rho = C_\rho D^{-b},$$

with C_η and C_ρ chosen from fixed finite grids of order-one constants. For each method and data budget, we report the best mean excess risk among the stable configurations in this local search.

For each run, the reported loss is averaged over the final 1% of optimization steps, and each empirical point is obtained by further averaging over 10 independent runs. We consider the following three batch-size regimes.

- **The small batch regime.** We take

$$B \eqsim D^0, \qquad \eta_{\text{sgd}} \eqsim \eta_{\text{polyak}} \eqsim \eta_{\text{nesterov}} \eqsim D^{-\frac{s}{s+1}}, \qquad 1-\rho_{\text{polyak}} \eqsim 1-\rho_{\text{nesterov}} \eqsim 1.$$

These schedules yield

$$r_{\text{sgd}} = r_{\text{polyak}} = r_{\text{nesterov}} = \frac{s}{s+1}.$$

- **The large batch regime.** We choose

$$B \eqsim D^{a_{\text{large}}}, \qquad a_{\text{large}} = \frac{2s\beta+1}{2s\beta+\beta+1} - \varepsilon,$$

where $\varepsilon > 0$ is a small constant. SGD uses $\eta_{\text{sgd}} \eqsim 1$. For Polyak, we search around

$$\eta_{\text{polyak}} \eqsim 1, \qquad 1-\rho_{\text{polyak}} \eqsim D^{-\left(a_{\text{large}}-\frac{s}{s+1}\right)}.$$

For Nesterov, we search around

$$\eta_{\text{nesterov}} \eqsim 1, \qquad 1-\rho_{\text{nesterov}} \eqsim D^{-b_{\text{nesterov}}}, \qquad b_{\text{nesterov}} = a_{\text{large}} - \frac{s\beta-(\beta-1)a_{\text{large}}}{s\beta+1}.$$

The corresponding theoretical exponents are

$$r_{\text{sgd}} = s(1-a_{\text{large}}), \qquad r_{\text{polyak}} = \frac{s}{s+1}, \qquad r_{\text{nesterov}} = \frac{s\big(1+(\beta-1)a_{\text{large}}\big)}{s\beta+1},$$

and hence

$$r_{\text{sgd}} < r_{\text{polyak}} < r_{\text{nesterov}}.$$

- **The ultra-large batch regime.** We choose

$$B \eqsim D^{a_{\text{ultra}}}, \qquad b_3 < a_{\text{ultra}} < 1.$$

SGD uses $\eta_{\text{sgd}} \eqsim 1$. For both momentum methods, we search around

$$\eta_{\text{polyak}} \eqsim \eta_{\text{nesterov}} \eqsim 1, \qquad 1-\rho_{\text{polyak}} \eqsim 1-\rho_{\text{nesterov}} \eqsim D^{-(1-a_{\text{ultra}}-\delta)},$$

where $\delta > 0$ is a small constant ensuring $K(1-\rho) \to \infty$. These choices give

$$r_{\text{sgd}} = s(1-a_{\text{ultra}}), \qquad r_{\text{polyak}} = r_{\text{nesterov}} = 2s(1-a_{\text{ultra}}) - s\delta.$$

Thus,

$$r_{\text{sgd}} < r_{\text{polyak}} = r_{\text{nesterov}},$$

and the common momentum exponent approaches $2s(1-a_{\text{ultra}})$ as $\delta \downarrow 0$.

Figure 7, 8 and 9 shows additional experiments for data scaling with different (s, β).

G.4 Critical batch sizes

We simulate the power-law kernel regression model in a $d = 4096$-dimensional eigenbasis with $(s, \beta) = (0.3, 4), \sigma = 0.1$, all methods are initialized from $\boldsymbol{\theta}_{-1} = \boldsymbol{\theta}_0 = 0$. We fix the total data budget at $D = 2^{14}$. For each batch size, we search locally around the hyperparameter scalings predicted by Theorem 6.2. For each hyperparameter candidate, we run 50 independent runs. Let $T = \max\{1, \lceil 0.01K \rceil\}$. For each run r, its score is the excess risk averaged over the final T iterations,

$$\overline{\mathcal{E}}^{(r)} = \frac{1}{T} \sum_{k=K-T+1}^{K} \mathcal{E}(\boldsymbol{\theta}_k^{(r)}).$$

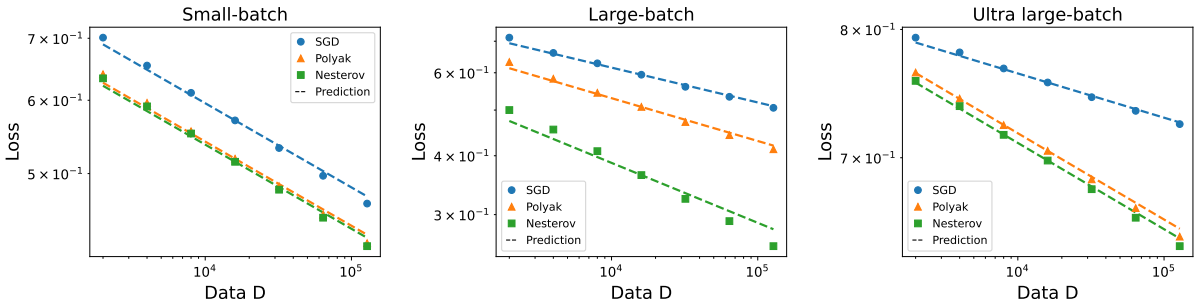


Figure 7: Data scaling with $(s, \beta) = (0.1, 4)$.

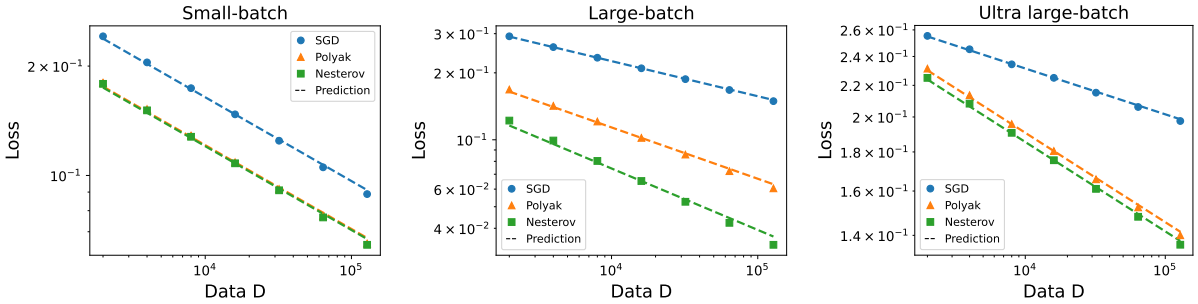


Figure 8: Data scaling with $(s, \beta) = (0.3, 2)$.

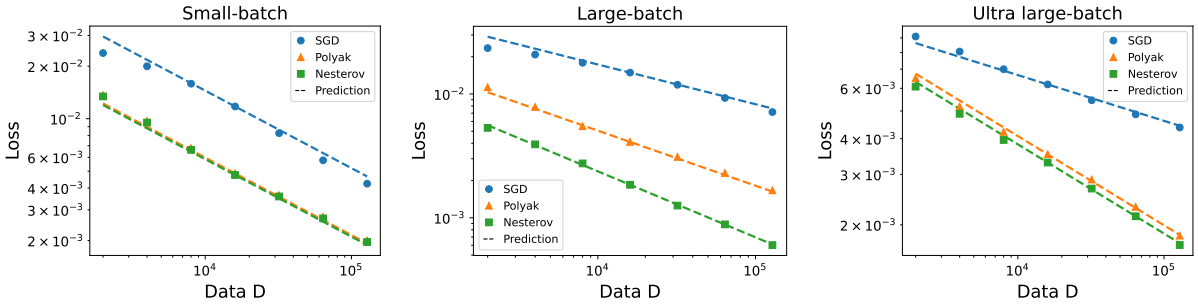


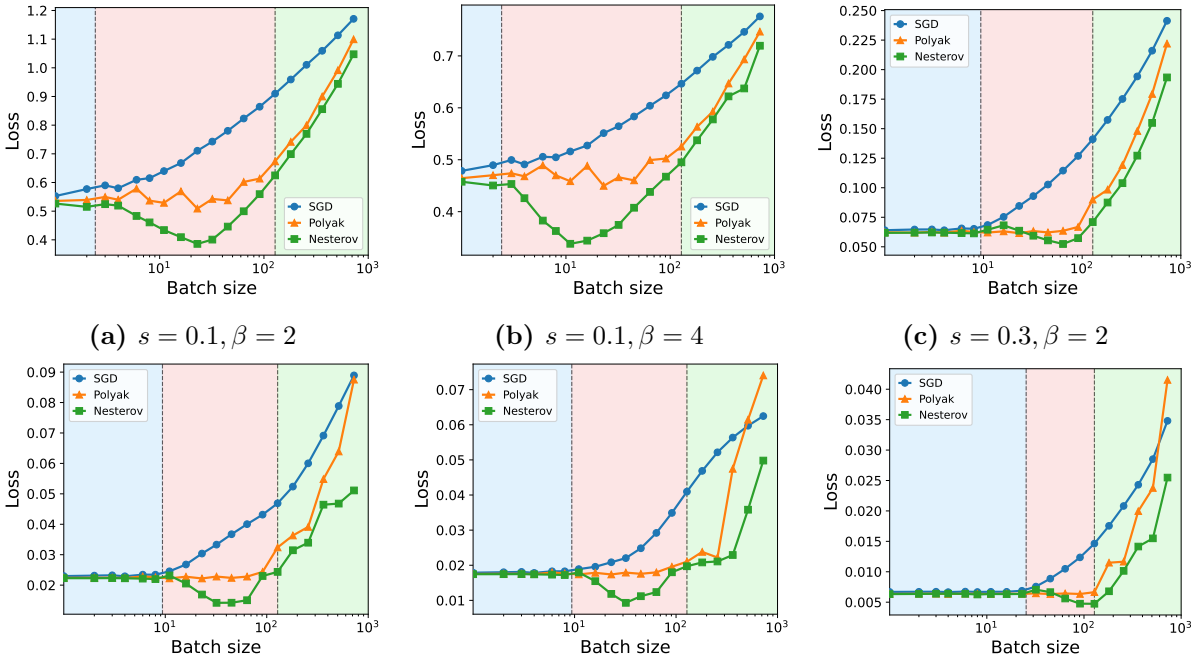
Figure 9: Data scaling with $(s, \beta) = (0.8, 4)$.

The candidate score is the arithmetic mean

$$\overline{\mathcal{E}} = \frac{1}{50} \sum_{r=1}^{50} \overline{\mathcal{E}}^{(r)}.$$

For each algorithm and batch size, Figure 1(right) reports the smallest value of $\overline{\mathcal{E}}$ over the corresponding search grid.

We show more results for critical batch size with different (s, β) in Figure 10.



(a) $s = 0.1, \beta = 2$ **(b)** $s = 0.1, \beta = 4$ **(c)** $s = 0.3, \beta = 2$

(d) $s = 0.3, \beta = 6$ **(e)** $s = 0.3, \beta = 8$ **(f)** $s = 0.5, \beta = 4$

Figure 10: Additional critical batch size experiments under different choices of s and β.

G.5 Delayed instability under increasing momentum

For the blow-up experiment, we simulate the power-law kernel regression model in a finite-dimensional eigenbasis with $d = 2048$, $s = 0.75$, and $\beta = 2$. We set the learning rate $\eta = 0.05$ and batch size $B = 32$. For simplicity, we set additive noise $\sigma = 0$ to discover the role of multiplicative noise. For Polyak and Nesterov, we use the time-varying schedule

$$\rho_k = 1 - \frac{1}{k+1}.$$

Each method is run for at most 5×10^5 iterations over 50 independent runs, and the figure reports the median population excess risk at every iteration. A simulation is terminated if the median risk becomes non-finite or exceeds 10.